



Alexandria Higher Institute of Engineering & Technology

Program of Electronics and Communications Engineering (ECE)

Final Year Project Report

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Integrated Hardware Engineering Solutions

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Implementation of the Egypt's 2030 sustainable Development goals (sdgs) through aiet's projects

SUSTAINABLE DEVELOPMENT GOALS
17 GOALS TO TRANSFORM OUR WORLD



***Alexandria Higher Institute of Engineering
&Technology (AIET)***

Professor kassim kadry (Dean of AIET)

What is Sustainable Development?

“Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”

In other words, Sustainable Development is the criteria for achieving social and economic progress in ways that will not exhaust the earth’s finite resources and not exploit or impoverish one grouping of people for the enrichment of another.



What are the world’s 2030 Sustainable Development Goals?

On September 25, 2015, after three years of debate and negotiations, all 193 member-nations of the UN, voted unanimously to adopt these global Sustainable Development Goals (SDG’s), stating:

“On behalf of the peoples we serve, we have adopted a historic decision on a comprehensive, far-reaching, and people-centered set of universal and transformative goals and targets. We commit ourselves to working tirelessly for the full implementation of this agenda by 2030.”

How are the SDGs supposed to be implemented?

- “...all member states to develop as soon as practicable, ambitious national responses to the overall implementation of this Agenda. These can...build on existing planning instruments, such as national development and sustainable development strategies, as appropriate.”
- “...the essential role of national parliaments through their enactment of legislation and adoption of budgets and their role in insuring accountability for the effective implementation of our commitments.”
- “...governments and public institutions...work closely on implementation with regional and local authorities, sub-regional institutions, international institutions, academia, philanthropic organizations, volunteer groups and others.”

How will the progress or lack of progress be monitored?

A. Voluntary National Reviews at the UN

Each country is encouraged to “...conduct regular and inclusive reviews of progress at the national and sub-national levels, which are country-led and country-driven.” “Regular reviews at the HLPF are to be voluntary, state-led, undertaken by both developed and developing countries, and involve multiple stakeholders.”

B. Stakeholders. That’s YOU! We, the people affected most by the need for these Goals must keep our government accountable to the 2030 Sustainable Development Goals!

To do that, we must organize!

Egypt Vision 2030

Egypt Vision 2030 focuses on improving the life quality and living standard of the Egyptian citizen in various aspects of life through emphasizing the consolidation of justice principles and social integration, and the participation of all citizens in the political and social life. This comes hand in hand with achieving high, inclusive and sustainable economic growth, promoting investment in people and building their creative capabilities by encouraging the increase of knowledge, innovation and scientific research in all fields. The Vision also focuses on the governance of the State's institutions and society through the administrative reform, consolidating transparency, supporting monitoring and evaluation systems, and empowering local administrations. All of these desired goals come within the framework of ensuring the peace and security of Egypt and promoting the Egyptian leadership regionally and internationally.



INTRODUCTION

Inspired by the ancient Egyptian Civilization, linking the present to future, the Sustainable Development Strategy (SDS): Egypt Vision 2030 represents a foothold on the way towards inclusive development. Thus, cultivating a prosperity path through, economic and social justice, and reviving the role of Egypt in regional leadership. SDS represents a roadmap for maximizing competitive advantage to achieve the dreams and aspirations of Egyptians in a dignified and decent life.

It also represents an embodiment of the new constitution's spirit, setting welfare and prosperity as the main economic objectives, to be achieved via sustainable development, social justice, and balanced, geographical and sectoral growth. Therefore, SDS has been developed according to a participatory strategic planning approach; as various civil society representatives, national and international development partners and government agencies have collaborated to set comprehensive objectives for all pillars and sectors of the country.

The current local, regional and global circumstances give the SDS a comparative advantage and importance; for revisiting the strategic vision to cope with international updates and developments. Thus, helping Egypt recover and achieve specific objectives.

SDS has followed the sustainable development principle as a general framework for improving the quality of lives and welfare, taking into consideration the rights of new generations in a prosperous life; thus, dealing with three main dimensions; economic, social, and environmental dimensions.

In addition, SDS is based upon the principles of “inclusive sustainable development” and “balanced regional development”, emphasizing the full participation in development, and ensuring its yields to all parties. The strategy, as well, considers equal opportunities for all, closing development gaps, and efficient use of resources to ensure the rights of future generations.

PARTICIPATORY PLANNING APPROACH TO PREPARE THE SUSTAINABLE DEVELOPMENT STRATEGY (SDS): EGYPT VISION 2030

- Many sessions and workshops have been held, with the active participation of experts, academics, private sector representatives, civil society, government officials and international development institutions.
- The SDS team included representatives from all groups including youth, woman, and disabled.
- All development partners participated in the SDS workshops, which witnessed conflicts, different views, and debates. At the end, partners reached a unified vision and agreed on objectives, KPIs, programs and projects.
- SDS team periodically communicated the main results of each stage of the SDS preparation plan with the media. In addition, SDS updates were continuously published on MoP website and social media pages.
- The strategy was presented for community dialogue in many different local and international forums and seminars. The outcomes of working groups have been revised by all relevant ministries and agencies, and the final document included all comments and feedback.
- An electronic portal and social media pages were launched to present the strategy and allow for interaction with all citizens.

By 2030, the new Egypt will achieve a competitive, balanced, diversified and knowledge-based economy, characterized by justice, social integration, and participation, with a balanced and diversified ecosystem, benefiting from its strategic location and human capital to achieve sustainable development for a better life to all Egyptians.

What are EGYPT's 2030 SDGs?

STRATEGIC VISIONS OF SDS PILLARS

THE ECONOMIC DIMENSION

First pillar: Economic development

By 2030, the Egyptian economy is a balanced, knowledge-based, competitive, diversified, market economy, characterized by a stable macroeconomic environment, capable of achieving sustainable inclusive growth. An active global player responding to international developments, maximizing value added, generating decent and productive jobs, and a real GDP per capita reaching a high-middle income country's level.

Second pillar: Energy

An energy sector meeting national sustainable development requirement and maximizing the efficient use of various traditional and renewable resources contributing to economic growth, competitiveness, achieving social justice, and preserving the environment. A renewable energy and efficient resource management leader, and an innovative sector capable of forecasting and adapting to local, regional and international developments and complying with SDGs.

Third pillar: Knowledge, innovation and scientific research

A creative and innovative society producing science, technology and knowledge, within a comprehensive system ensuring the developmental value of knowledge and innovation using their outputs to face challenges and meet national objectives.

Fourth pillar: Transparency and efficient government institutions

An efficient and effective public administration sector managing State resources with transparency, fairness, and flexibility. Subject to accountability, maximizing citizens, satisfaction and responding to their needs.

THE SOCIAL DIMENSION

Fifth pillar: Social justice

By 2030, Egypt is a fair interdependent society characterized by equal economic, social, political rights and opportunities realizing social inclusion. A society that supports citizens, the right in participation based on efficiency and according to law, encouraging social mobility based on skills. A society that provides protection, and support to marginalized and vulnerable groups.

Sixth pillar: Health

All Egyptians enjoy a healthy, safe, and secure life through an integrated, accessible, high quality, and universal healthcare system capable of improving health conditions through early intervention, and preventive coverage. Ensuring protection for the vulnerable and achieving satisfaction of citizens and health sector employees. This will lead to prosperity, welfare, happiness, as well as social and economic development, which will qualify Egypt to become a leader in the field of healthcare services and research in the Arab world and Africa.

Seventh pillar: Education and training

A high-quality education and training system is available to all, without discrimination within an efficient, just, sustainable, and flexible institutional framework. Providing the necessary skills for students and trainees to think creatively and empower them technically and technologically.

Contributing to the development of a proud, creative, responsible, and competitive citizen

who accepts diversity and differences and is proud of his country's history. **Eighth pillar: Culture**

A system of positive cultural values respecting diversity and differences. Enabling citizens to access knowledge, building their capacity to interact with modern developments, while recognizing their history and cultural heritage. Giving them the wisdom of freedom of choice and of cultural creativity. Adding value to the national economy, representing Egypt's soft power at regional and international levels.

THE ENVIRONMENT DIMENSION

Ninth Pillar: Environment

The environment is integrated in all economic sectors to preserve natural resources and support their efficient use and investment, while ensuring next generations' rights. A clean, safe, and healthy environment leads to diversified production resources and economic activities, supporting competitiveness, providing new jobs, eliminating poverty and achieving social justice.

Tenth pillar: Urban development

A balanced spatial development management of land and resources to accommodate population and improve the quality of their lives.

What are the main Engineering Applications that will be considered for Projects (2025-2026) aiming at SDGs?



Ensure availability and sustainable management of water and sanitation for all.



Ensure access to affordable, reliable sustainable and modern energy for all.



Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.



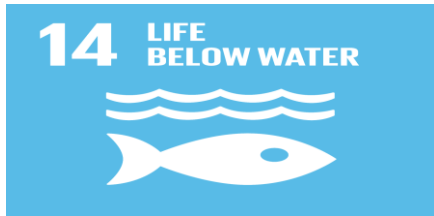
Make cities and human settlements inclusive safe, resilient and sustainable



Ensure sustainable consumption and production patterns.



Take urgent action to combat climate change and its impacts.



Conserve and sustainably use the oceans, seas and marine resources for sustainable development.



Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

Implementation of Egypt's 2030 Sustainable Development Goals (SDGs) through AIET Projects

This graduation project is aligned with Egypt Vision 2030 and the Sustainable Development Goals by demonstrating how digital engineering solutions can support healthcare, infrastructure, innovation, and safer communities. The software part of the project focuses on a smart hospital case study where network infrastructure, web systems, mobile applications, and cloud services work together to improve the quality, accessibility, reliability, and management of healthcare services.

Expected SDGs Achieved by the Project

The project supports sustainable development by improving safety, energy awareness, healthcare monitoring, and emergency response inside a hospital environment. The integration between electrical distribution, fire protection, and smart hardware makes the project aligned with the following goals:

Goal	Project Contribution
SDG 3: Good Health and Well-being	Patient monitoring, fire safety, emergency access, and safer hospital operation.
SDG 7: Affordable and Clean Energy	Lighting control and efficient electrical distribution reduce unnecessary energy usage.
SDG 9: Industry, Innovation and Infrastructure	Combines building services design with embedded systems and cloud connectivity.
SDG 11: Sustainable Cities and Communities	Supports safer and smarter public healthcare buildings.
SDG 12: Responsible Consumption	Automation and monitoring reduce waste of water, electricity, and operational resources.
SDG 13: Climate Action	Energy-aware control and optimized infrastructure help reduce environmental impact.

Mission of Electronics and Communication Engineering Program

Vision:

The department seeks to graduate scientifically distinguished engineers who have the ability to participate in the fields of electronics and communications engineering and scientific research based on high-quality education compatible with regional and international standards and develop their professional capabilities for community service and environmental development.

رؤية قسم الإلكترونيات والاتصالات:

يسعى القسم إلى تخريج مهندسين متميزين علمياً لهم القدرة على المشاركة في مجالات هندسة الإلكترونيات والاتصالات والبحث العلمي اعتماداً على تعليم ذات جودة عالية متوافقة مع المعايير الإقليمية والدولية وتنمية قدراتهم المهنية لخدمة المجتمع وتنمية البيئة.

Mission:

The department aims to graduate generations of engineers in the field of electronics and communications engineering who have the spirit of innovation, technological creativity and the ability to compete in the labor market through the application of the latest teaching and learning systems that keep pace with scientific and research progress in line with the developments of the times to achieve sustainable development and contribute to community service and environmental development.

رسالة قسم الإلكترونيات والاتصالات:

يهدف القسم إلى تخريج أجيال من المهندسين في مجال هندسة الإلكترونيات والاتصالات لديهم روح الابتكار والإبداع التكنولوجي والقدرة على المنافسة في سوق العمل من خلال تطبيق أحدث نظم التعليم والتعلم والتي تواكب التقدم العلمي والبحثي بما يتماشى مع مستجدات العصر لتحقيق التنمية المستدامة والمساهمة في خدمة المجتمع وتنمية البيئة.

Declaration

We hereby declare that this graduation project report, entitled Integrated Hardware Engineering Solutions, was prepared by our project team as part of the requirements for the final year project. The work presented in this report was completed under the guidance of our supervisors and represents our understanding, implementation, and documentation of the project scope.

Student signatures:

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.		
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Date: 22 / 6 / 2026

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Abstract

Integrated Hardware Engineering Solutions is a multidisciplinary graduation project designed to provide a complete engineering solution for a smart hospital building. The project combines electrical distribution design, fire fighting system design, and an embedded hardware system to improve safety, reliability, automation, and operational efficiency inside the hospital environment.

The hardware system presents a smart hospital prototype using ESP32 and Arduino controllers. It includes gas and fire detection, access control, patient health monitoring, temperature and humidity monitoring, mist maker control, smart lighting, and a hospital supply delivery robot. The system integrates several sensors and actuators to support monitoring, safety, and smart response.

The electrical distribution part focuses on designing the hospital power infrastructure, including lighting, power sockets, HVAC loads, panel schedules, circuit breaker selection, cable sizing, equipment sizing, voltage drop calculations, short circuit calculations, single line diagrams, shop drawings, cable tray sizing, and light current systems.

The fire fighting part focuses on protecting the hospital against fire hazards using wet sprinkler systems, pre-action systems for operation rooms, FM200 gas suppression for sensitive rooms, Fire Search systems for electrical panels, fire extinguishers, fire hose cabinets, zone control valves, hydraulic calculations, pipe sizing, fire pump selection, and fire water storage tanks. Together, these tracks demonstrate a complete engineering approach for a safer, smarter, and more reliable hospital environment.

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Project Specification

Item	Description
Project Title	Integrated Hardware Engineering Solutions
Project Type	Multidisciplinary hospital engineering design and prototype implementation
Main Tracks	Hardware System, Electrical Distribution System, Fire Fighting System
Main Objective	To develop a practical engineering solution that improves hospital safety, reliability, automation, and emergency response.
Target Building	Two-floor hospital building with electrical, fire protection, and smart hardware systems.
Hardware Controllers	ESP32 and three Arduino Uno boards.
Electrical Scope	Lighting, power, HVAC, panel schedules, SLD, calculations, shop drawings, and light current systems.
Fire Scope	Sprinklers, pre-action, FM200, Fire Search, FHC, extinguishers, pump room, tank, and hydraulic calculations.

Project Overview

Integrated Hardware Engineering Solutions is a multidisciplinary graduation project that presents a complete engineering solution for a smart hospital building. The project combines three main engineering tracks: Electrical Distribution, Fire Fighting System, and Hardware System. These tracks were integrated together to improve safety, reliability, automation, and operational efficiency inside the hospital environment.

The Electrical Distribution track focuses on designing the main electrical infrastructure of the hospital. It includes lighting design, power socket distribution, HVAC power supply, panel schedules, circuit breaker selection, cable sizing, equipment sizing, single line diagram, voltage drop and short circuit calculations, shop drawings, cable tray sizing, ceiling coordination, wall composite drawings, and light current systems.

The Fire Fighting System track focuses on protecting the hospital from fire hazards by selecting suitable fire protection systems according to the function and risk level of each area. It includes wet sprinkler systems, pre-action systems for operation rooms, FM200 gas suppression for sensitive rooms, Fire Search systems for electrical panels, fire extinguishers, fire hose cabinets, zone control valves, hydraulic calculations, pipe sizing, fire pump selection, and fire water storage tanks.

The Hardware System track presents a smart hospital prototype using ESP32 and Arduino controllers. It includes gas and fire detection, access control, patient health monitoring, temperature and humidity monitoring, mist maker control, smart lighting, and a hospital supply delivery robot. Different sensors and actuators were integrated to create an automated prototype that supports monitoring, safety, and smart response inside the hospital.

Chapter 1: Introduction

Project Background

Modern hospital buildings require reliable electrical infrastructure, safe fire protection systems, and intelligent monitoring solutions. In critical environments such as hospitals, failure in power supply, fire response, access control, or monitoring can affect patient safety and building operation. Therefore, this project was designed as an integrated engineering solution that combines building services design with embedded hardware implementation.

Problem Statement

Traditional building systems are often designed as separate systems. Electrical distribution, fire protection, and hardware automation may work independently without clear integration. This creates challenges in monitoring, safety response, and operational efficiency. The project addresses this problem by presenting a complete hospital-oriented solution that connects design, protection, automation, and monitoring.

Project Objectives

The main objective is to design and demonstrate an integrated smart hospital engineering system. The project aims to provide safe electrical distribution, effective fire protection, patient and environment monitoring, controlled access, emergency response, and automated delivery support.

Project Scope

The project scope includes a hardware prototype, electrical distribution design, and fire fighting design. The hardware prototype represents the smart automation layer, the electrical distribution part represents the building power layer, and the fire fighting part represents the life safety protection layer.

Project Methodology

The methodology started by defining hospital requirements, dividing the project into three tracks, selecting suitable components and systems, preparing drawings and calculations, implementing the prototype, testing the subsystems, and documenting the final design and results.

Track	Main Output
Hardware System	Smart hospital prototype using ESP32, Arduino boards, sensors, actuators, displays, and cloud upload.
Electrical Distribution	Lighting, power, HVAC, panel schedules, sizing, SLD, shop drawings, and light current systems.
Fire Fighting	Sprinkler, pre-action, FM200, Fire Search, pump room, tank, extinguishers, and hydraulic calculations.

Chapter 2: Hardware System

2.1 Hardware System Overview

The hardware system was designed as a prototype layer that simulates how smart monitoring and automatic response can be added to a hospital environment. The design depends on dividing the control tasks between one ESP32 master controller and three Arduino Uno local controllers. This distribution reduces the load on a single controller and makes each subsystem easier to test, troubleshoot, and explain.

The hardware system is the smart automation part of the project. It was implemented as a hospital prototype using one ESP32 controller and three Arduino Uno boards. Each controller was assigned to a clear function to make the system easier to test, debug, and explain. The ESP32 acts as the master controller, while the Arduino boards control protection, access control, and robot movement.

2.2 Integrated Hardware System Architecture

The hardware prototype is organized as a group of embedded subsystems connected together to simulate a smart hospital environment. The ESP32 acts as the main controller for monitoring, display, Wi-Fi/cloud connection, Bluetooth control, and emergency communication. The Arduino Uno boards are used as local controllers for protection, lighting, access control, and robot movement.

Integrated Smart Hospital System Diagram

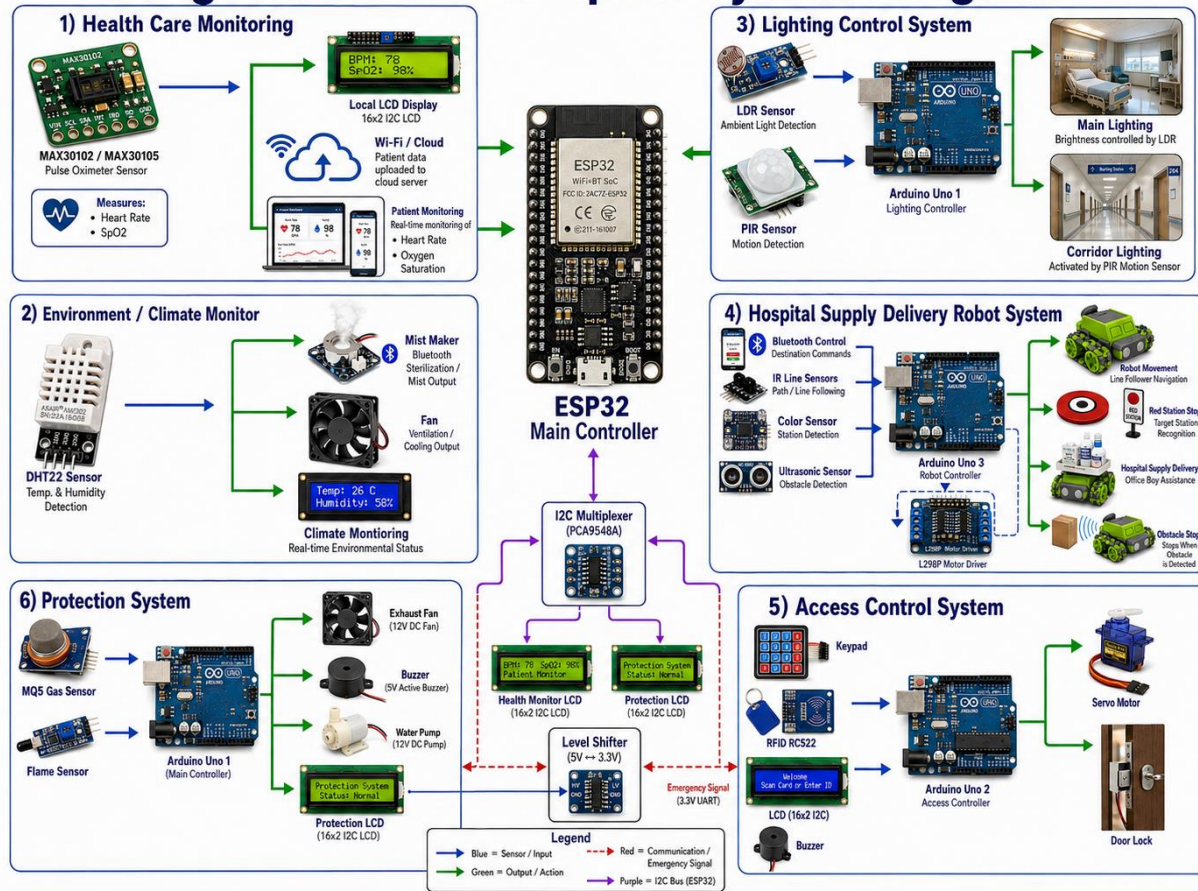


Figure 1: Integrated smart hospital system block diagram

Explanation: This block diagram shows the complete hardware architecture used in the project. It connects the health care monitoring system, environment and climate monitor, lighting control, protection system, access control system, and hospital supply delivery robot. The diagram also clarifies the role of the ESP32, Arduino controllers, I2C multiplexer, level shifter, LCD displays, sensors, and outputs in one integrated smart hospital prototype.

2.3 Health Care Monitoring System

The health care monitoring subsystem is based on the ESP32 and the MAX30102/MAX30105 pulse oximeter sensor. The system measures the patient heart rate and oxygen saturation, displays the readings locally on the LCD, and supports optional cloud upload for remote patient monitoring.

Health Care System Diagram

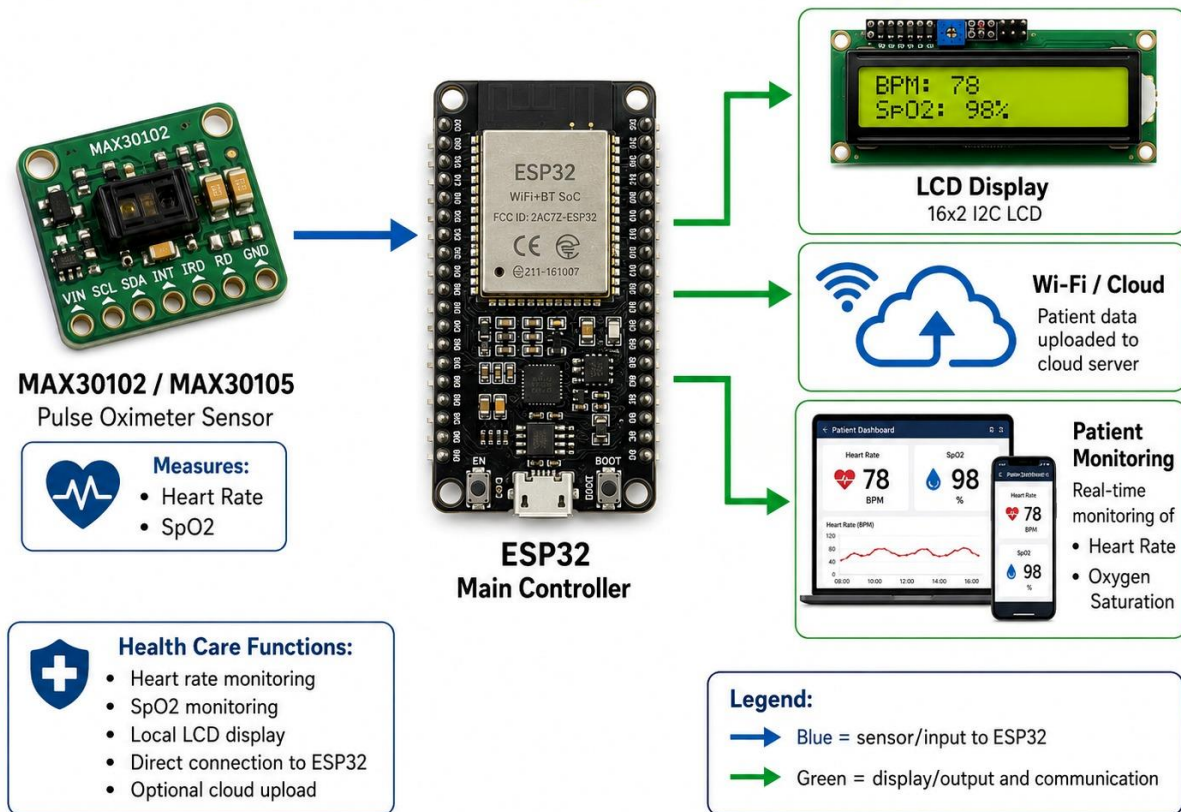


Figure 2: Health care monitoring system diagram

Explanation: This diagram explains how the MAX30102/MAX30105 sensor sends pulse oximeter readings to the ESP32. The ESP32 processes the readings and displays BPM and SpO2 values on the local LCD. The same controller can also upload patient data through Wi-Fi to support real-time monitoring.

2.4 Environment Synthesis and Climate Monitor

The environment subsystem monitors room temperature and humidity using a DHT22 sensor. According to the measured temperature, the ESP32 controls the ventilation fan. The mist maker is controlled through the built-in Bluetooth feature of the ESP32 to represent room sterilization or humidity support.

Environment Synthesis & Climate Monitor

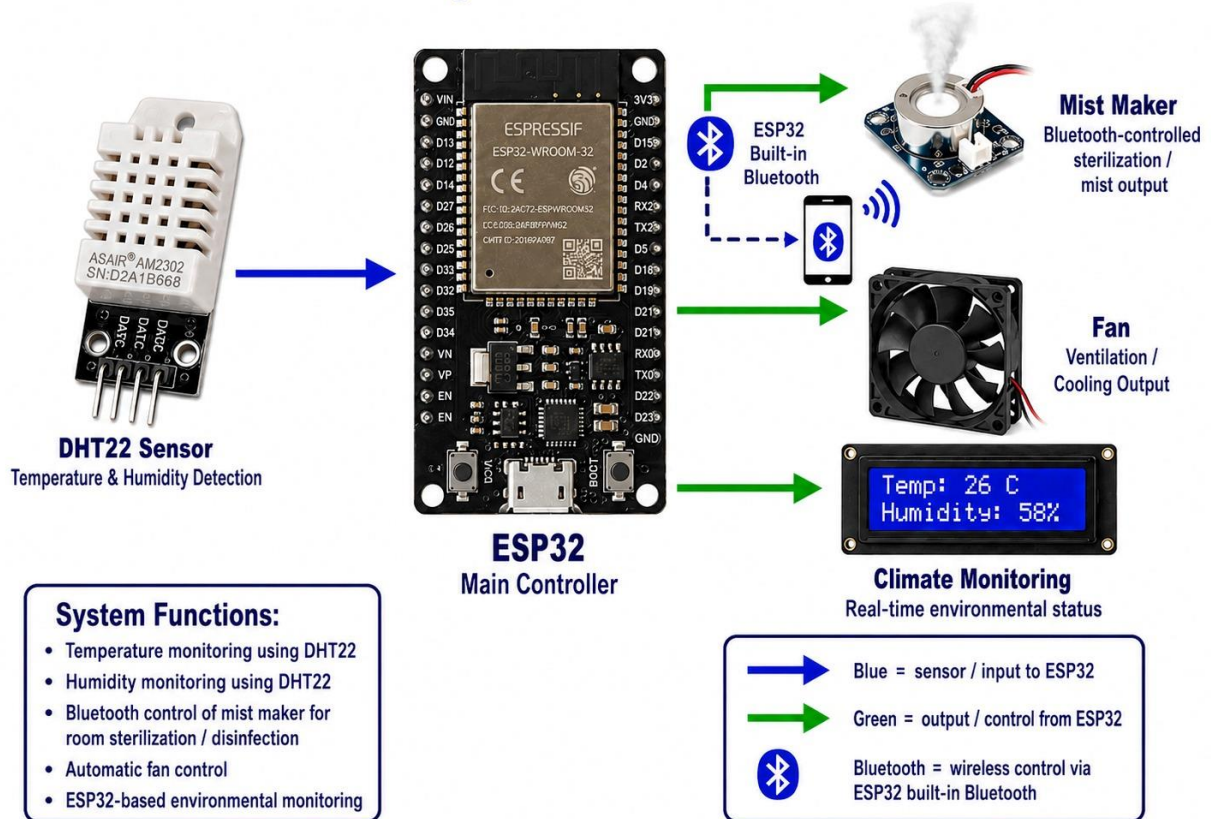


Figure 3: Environment synthesis and climate monitor diagram

Explanation: This diagram shows the DHT22 sensor as the environmental input to the ESP32. The output side includes a fan for cooling/ventilation, a mist maker controlled by Bluetooth, and an LCD for displaying real-time temperature and humidity. The blue arrows indicate sensor input, while the green arrows indicate output/control signals.

2.5 Protection System

The protection system is controlled locally by Arduino Uno 1. It detects gas leakage using the MQ5 sensor and detects fire using the flame sensor. According to the detected condition, it operates the exhaust fan, buzzer, and 5V water pump, then sends the emergency status to the ESP32 through the level shifter.

Protection System Diagram

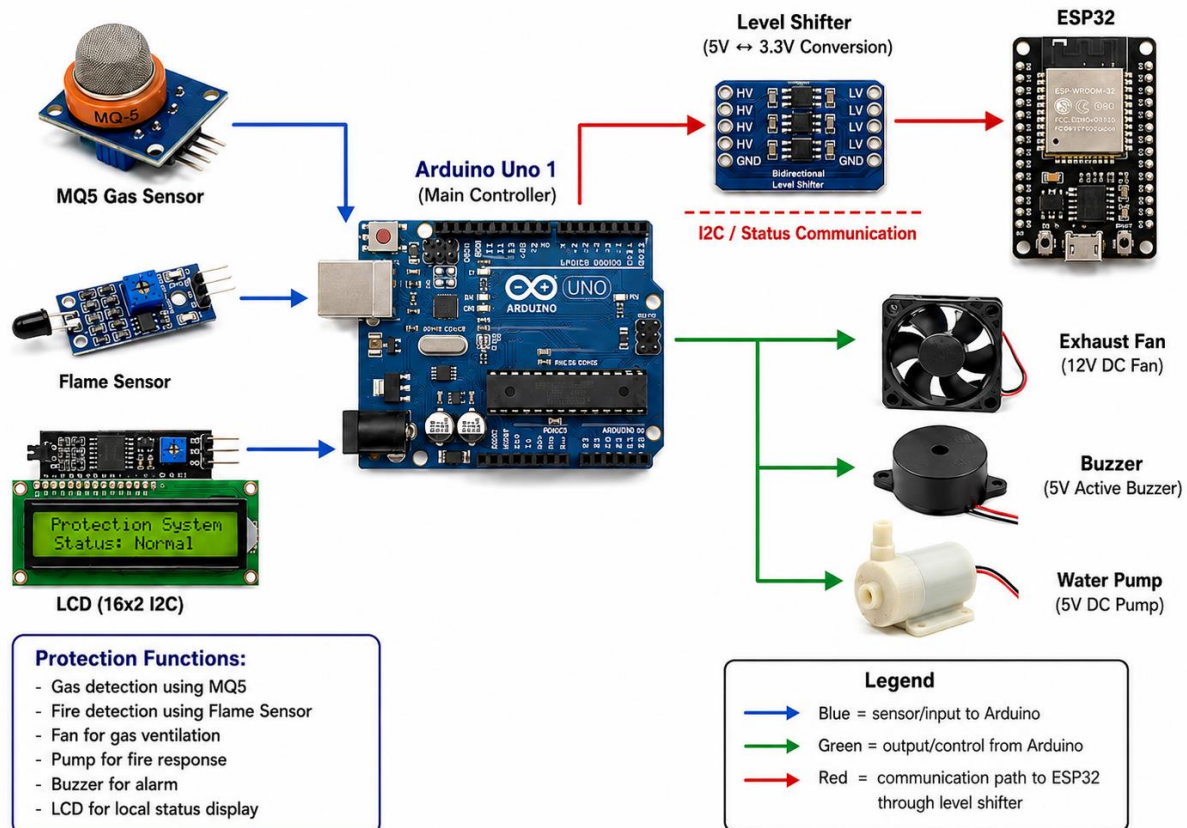


Figure 4: Protection system diagram

Explanation: This diagram presents the gas and fire protection node. Arduino Uno 1 reads the MQ5 gas sensor, flame sensor, and local LCD status. During a gas alarm, the exhaust fan and buzzer can be activated. During a fire alarm, the water pump and buzzer can be activated. The emergency status is transferred to the ESP32 through a level shifter to protect the 3.3V ESP32 logic.

2.6 Lighting Control System

The lighting control subsystem uses Arduino Uno 1 together with an LDR sensor and a PIR motion sensor. The LDR controls the main lighting level according to ambient light, while the PIR sensor supports motion-based corridor lighting.

Lighting Control System Diagram

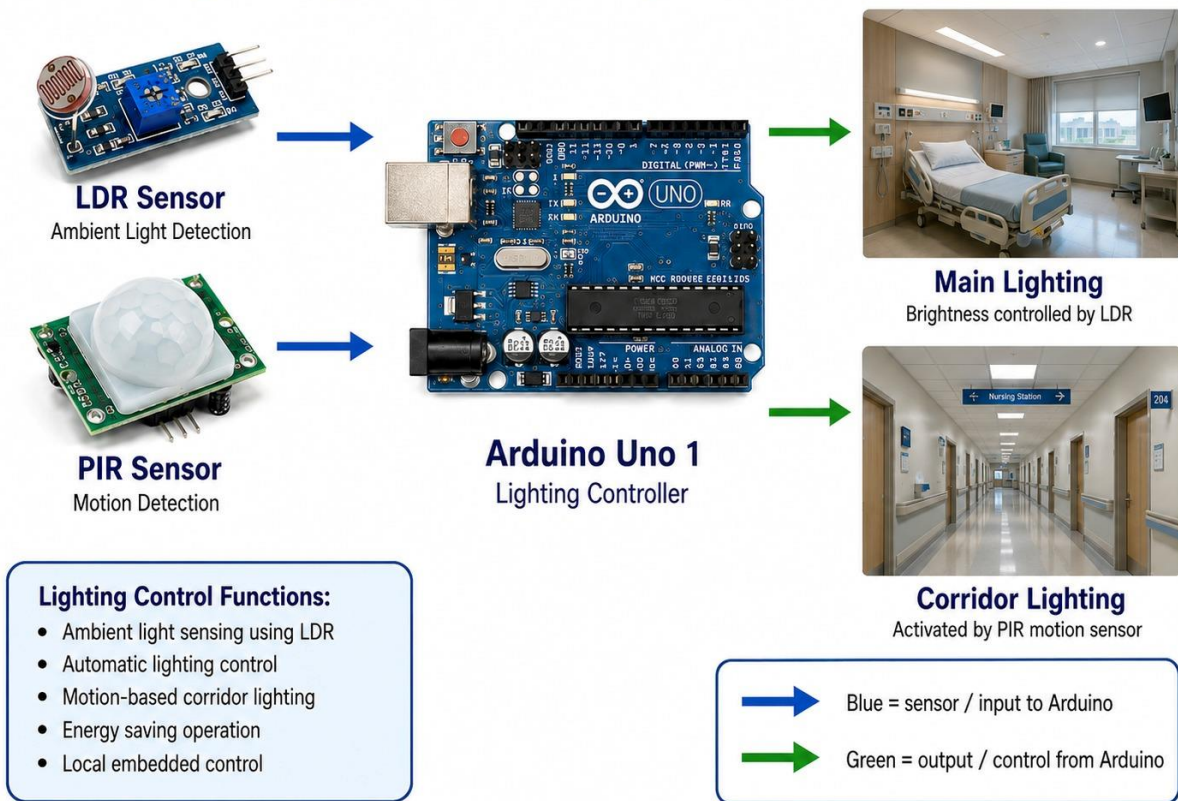


Figure 5: Lighting control system diagram

Explanation: This diagram shows how the LDR sensor and PIR sensor are connected as inputs to Arduino Uno 1. The LDR is used for ambient light detection and automatic brightness control, while the PIR sensor detects motion and activates corridor lighting. This improves energy saving and supports automatic lighting operation inside the hospital prototype.

2.7 Access Control System

The access control subsystem is controlled by Arduino Uno 2. In normal operation, the door can be opened using RFID authentication or keypad password entry. During emergency operation, the ESP32 sends an emergency signal through a level shifter so the Arduino can open the door automatically.

Access Control System Diagram

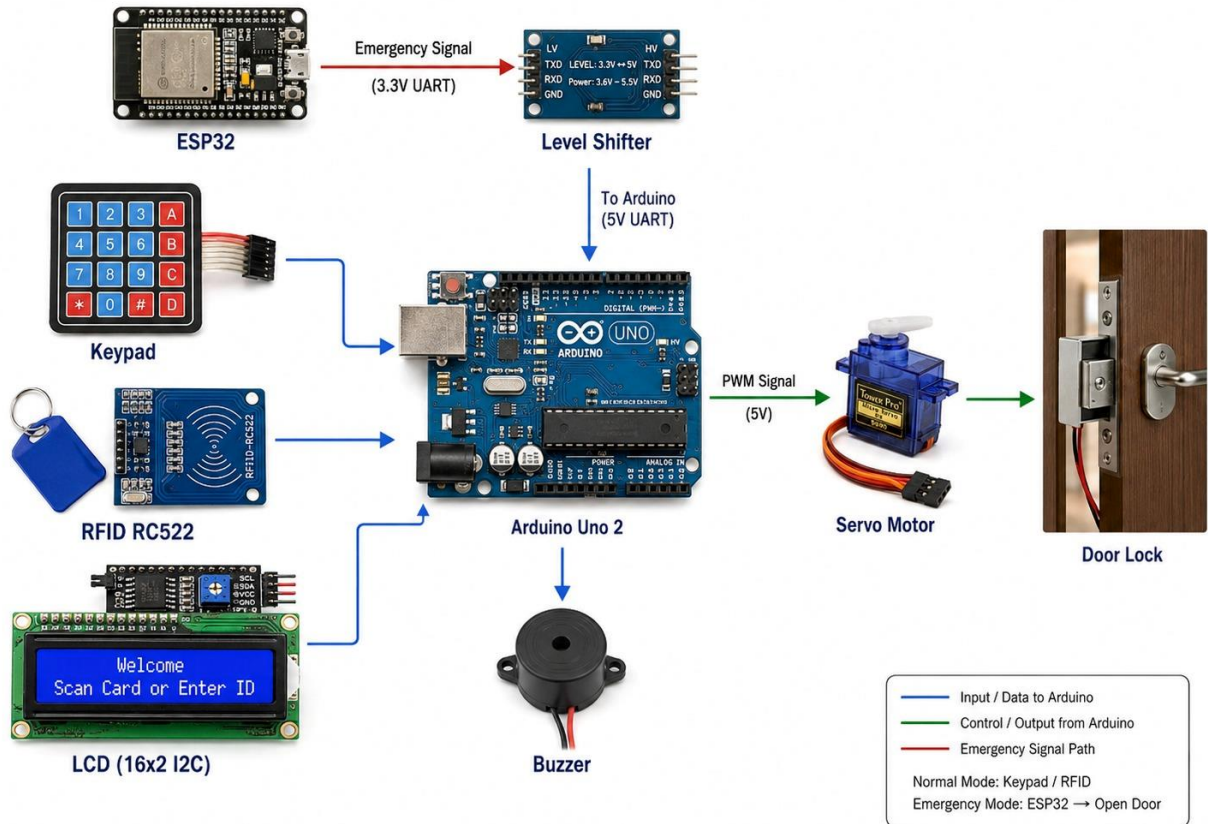


Figure 6: Access control system diagram

Explanation: This diagram explains the complete access control node. The keypad, RFID RC522 module, LCD, buzzer, and servo motor are connected to Arduino Uno 2. The servo motor represents the door lock mechanism. The red path shows the emergency signal coming from the ESP32 through the level shifter, allowing automatic door opening during gas or fire emergency conditions.

2.8 Hospital Supply Delivery Robot System

The hospital supply delivery robot is controlled by Arduino Uno 3 and a motor driver/motor shield. The robot follows a line using IR sensors, receives destination commands through Bluetooth, detects target stations using a color sensor, and stops when an obstacle is detected using the ultrasonic sensor.

Hospital Supply Delivery Robot System

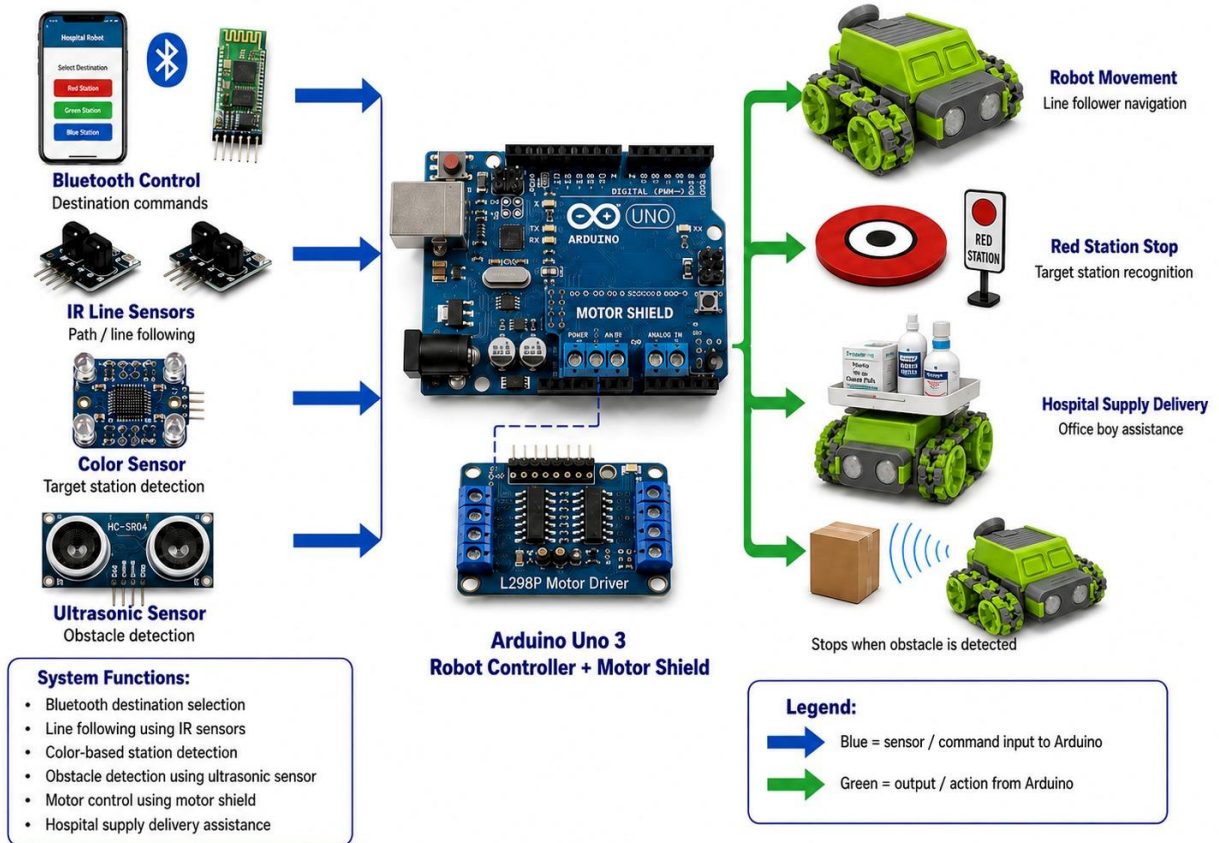


Figure 7: Hospital supply delivery robot system diagram

Explanation: This diagram shows the delivery robot subsystem. Bluetooth is used to select the destination, IR sensors are used for line following, the color sensor is used for station detection, and the ultrasonic sensor is used for obstacle detection. The Arduino Uno 3 and motor driver control the robot movement to represent hospital supply delivery assistance.

Controller	Subsystem	Main Responsibility
ESP32	Main Controller	Health monitoring, DHT22, LCDs, cloud upload, emergency communication, fan and mist control.
Arduino Uno 1	Protection and Lighting	Gas detection, flame detection, pump, fan, buzzers, PIR lighting, LDR lighting, I2C status.
Arduino Uno 2	Access Control	RFID, keypad, servo door, LCD, buzzer, emergency door opening.
Arduino Uno 3	Delivery Robot	Line follower, color station detection, ultrasonic obstacle detection, Bluetooth command control.

Component	ESP32 Pin / Address
I2C SDA	GPIO 21
I2C SCL	GPIO 22
Signal to Access Arduino	GPIO 17
DHT22	GPIO 13
Fan Relay	GPIO 26
Mist Relay	GPIO 27
Mist Fan Relay	GPIO 16
LDR Sensor	GPIO 35
LDR LED 1	GPIO 18
LDR LED 2	GPIO 19
PCA9548A Address	0x70
LCD Address	0x27
Arduino Uno 1 I2C Address	0x08
Component	Arduino Uno 1 Pin
MQ5 Gas Sensor	A0
Gas Buzzer	D4
Gas Fan Relay	D5
Flame Sensor	D3
Fire Buzzer	D6
Fire Pump Relay	D8
PIR Sensor	D12
PIR LED	D9
LDR Sensor	A1
LDR LED 1	D10
LDR LED 2	D11
I2C Address	0x08
Status	Meaning
0	Normal
1	Gas Leak
2	Fire

3	Gas + Fire	
Component		Arduino Uno 2 Pin
Emergency RX from ESP32	A3	
LCD I2C Address	0x27	
RFID SS	D10	
RFID RST	D9	
Buzzer	A1	
Servo Motor	A2	
Keypad Rows	D2, D3, D4, A0	
Keypad Columns	D6, D7, D8, D5	
Emergency Status	Pulse Width	Action
0 Normal	100 ms	Door locked / system normal
1 Gas Leak	200 ms	Door opens and gas warning appears
2 Fire	300 ms	Door opens and fire warning appears
3 Gas + Fire	400 ms	Door opens and combined warning appears
Component		Arduino Uno 3 Pin
Left Motor	Motor Shield M1	
Right Motor	Motor Shield M2	
HC-05 TX	A4	
HC-05 RX	A5 through voltage divider	
Left IR Sensor	D9	
Right IR Sensor	D10	
TCS3200 S0	A0	
TCS3200 S1	A1	
TCS3200 S2	A2	
TCS3200 S3	A3	
TCS3200 OUT	D2	
Ultrasonic TRIG	D11	
Ultrasonic ECHO	D12	
Bluetooth Command		Function
1	Go to red station	

2	Go to white station
3	Go to station / black crossing line
S / s	Stop robot

2.9 Prototype Implementation

The following figures present the physical implementation of the smart hospital hardware prototype. Only the actual implementation photos provided by the team are used in this section. The photos show the overall prototype, the ESP32 monitoring node, the Arduino Uno 1 protection and lighting node, and the Arduino Uno 2 access control node. The implementation photos are arranged from the overall prototype view, then subsystem wiring and practical installation details.



Figure 8: Overall prototype top view

Explanation: This figure shows the complete hospital prototype from the top view. The model layout represents the rooms and corridors where the hardware subsystems are distributed. It shows that the project was implemented as a complete smart hospital model, not only as separate circuits. The different areas include the ESP32 monitoring node, protection and lighting control, access control, and other installed hardware units.

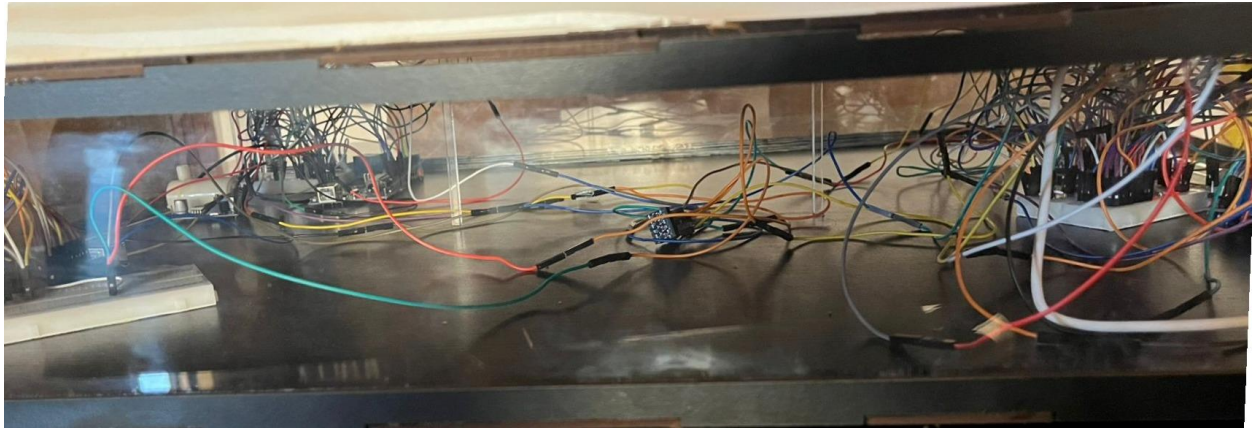


Figure 9: Overall prototype wiring integration

Explanation: This figure shows the full internal wiring of the prototype. It represents the practical integration between the controllers, sensors, relays, displays, and output devices inside the model. The wiring connects the ESP32 and Arduino Uno boards with the different sensors and actuators, proving that the subsystems were physically connected together as one integrated hardware project.

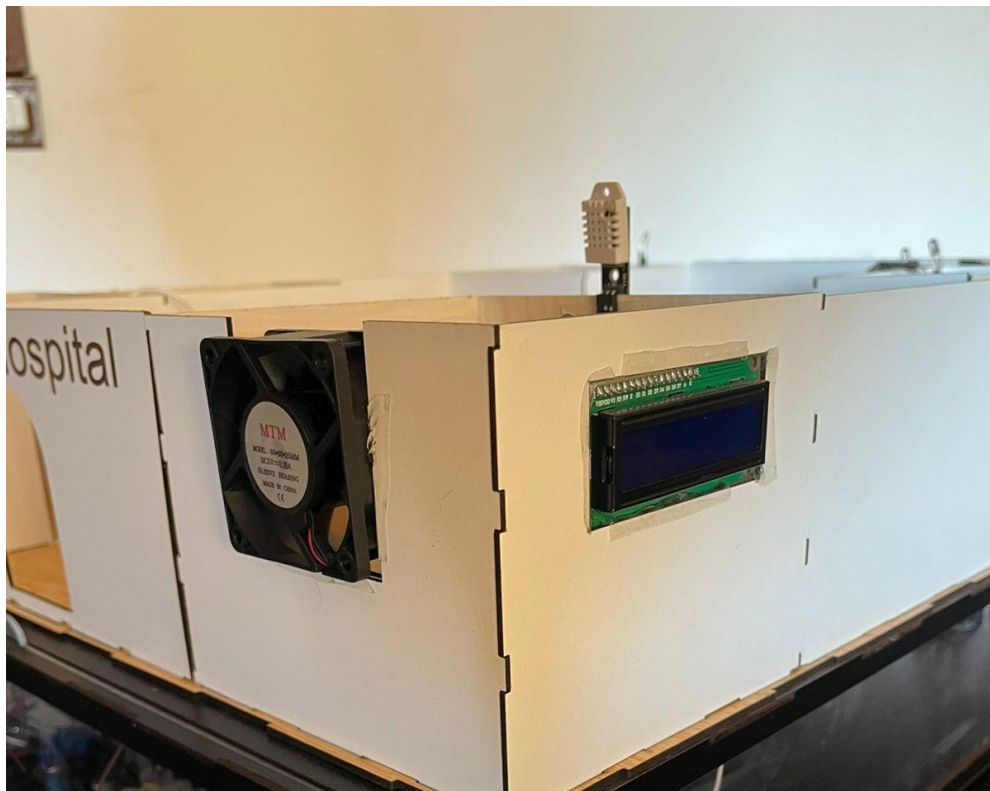
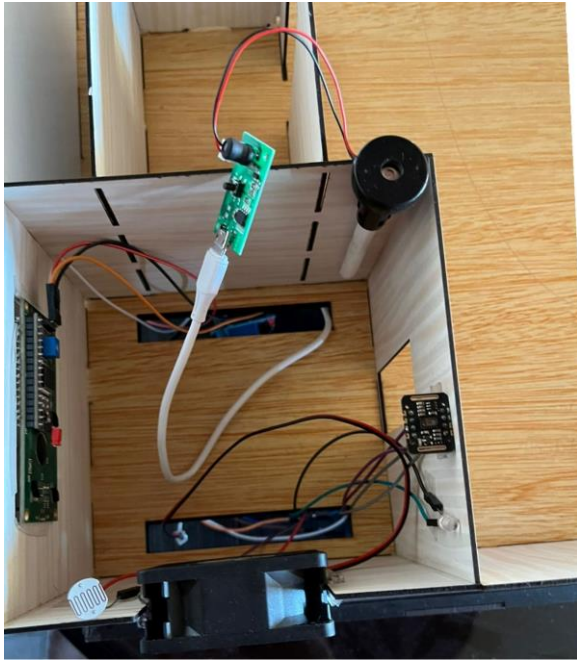


Figure 10: ESP32 climate monitoring implementation

Explanation: This figure shows the ESP32 climate monitoring section. The DHT22 sensor is installed to measure temperature and humidity, while the fan represents the ventilation or cooling output. The LCD display is used to show the real-time readings and system status. This part is controlled by the ESP32 and is used to demonstrate automatic environmental monitoring inside the prototype.



MAX/LDR and sensor area



LCD, fan, mist output area

Figure 11: ESP32 health care and environment sensors implementation

Explanation: This figure shows the ESP32-related sensors and outputs used for health care and environmental functions. The photos include the MAX30102/MAX30105 sensor for patient readings such as BPM and SpO2, the LDR sensor for light detection, the LCD display, the fan, and the mist maker output. These components are controlled by the ESP32 to combine patient monitoring, light sensing, and climate support in the same node.



Figure 12: ESP32 complete wiring implementation

Explanation: This figure shows the complete wiring of the ESP32 subsystem. The wiring connects the ESP32 with the DHT22 sensor, MAX30102/MAX30105 pulse oximeter, LDR sensor, LCD display, fan, mist maker control, relays, and related modules. It represents the internal implementation of the ESP32 node and shows how the monitoring and output-control devices are practically connected inside the prototype.

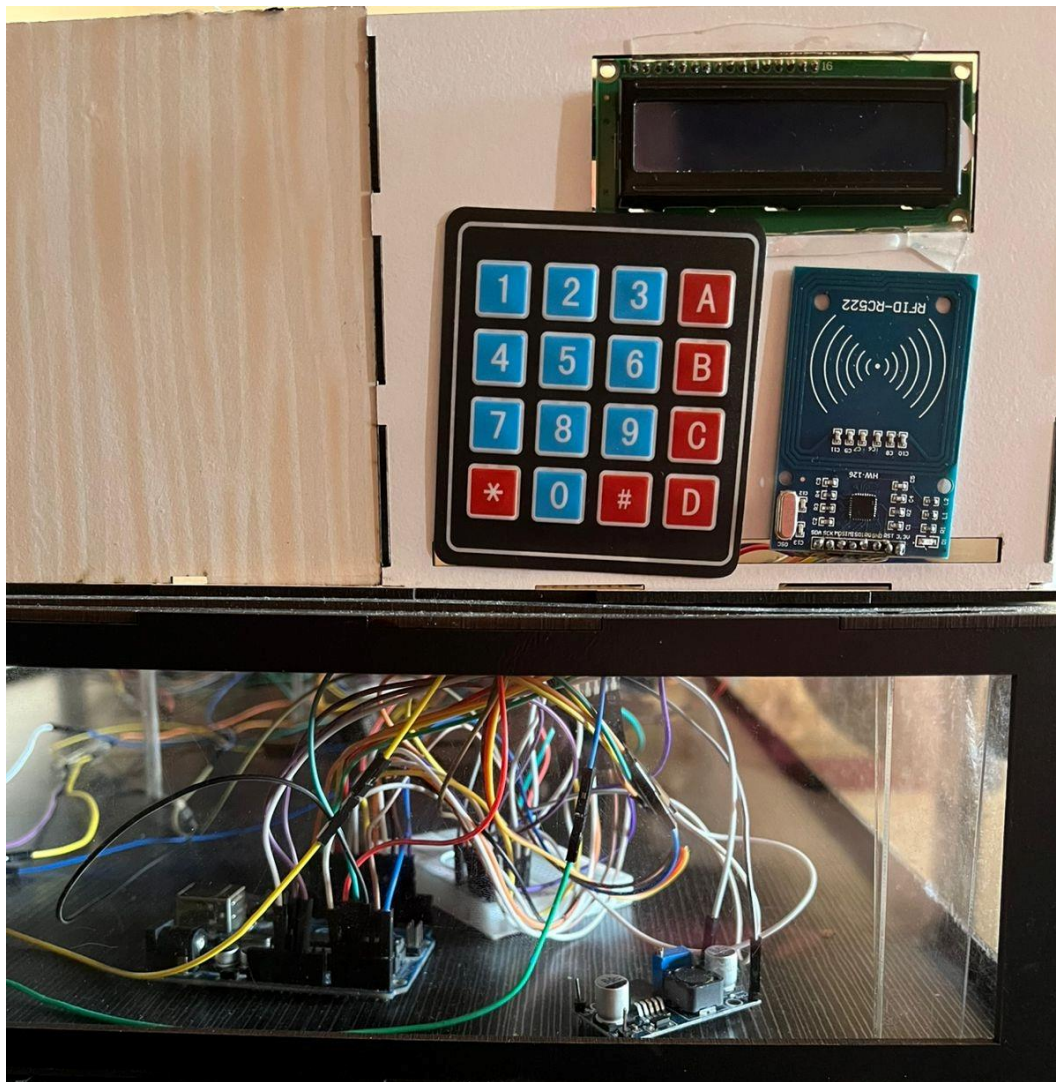


Figure 13: Access control physical implementation

Explanation: This figure shows the access control system implemented inside the prototype. The system includes a 4x4 keypad for password entry, an RFID RC522 reader for card access, and a 16x2 LCD for displaying instructions and access messages. The lower part shows the internal wiring connected to Arduino Uno 2, which controls the door mechanism and also receives the emergency signal from the ESP32 to open the door automatically during gas or fire conditions.

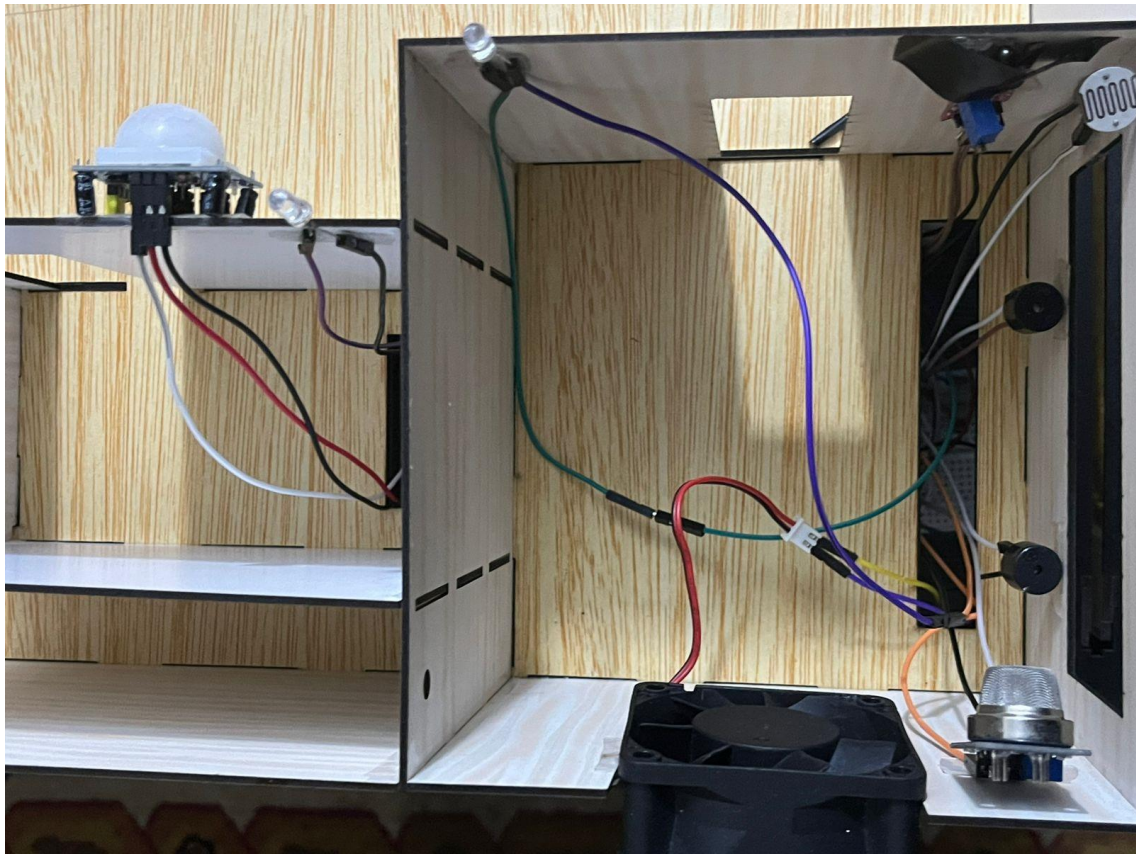


Figure 14: Protection and lighting physical implementation

Explanation: This figure shows the physical implementation of the protection and lighting node. The system includes the MQ5 gas sensor for gas leakage detection, the flame sensor for fire detection, the PIR sensor for motion detection, the LDR sensor for ambient light detection, and output devices such as the fan and buzzer. Arduino Uno 1 is responsible for reading these sensors and controlling the protection and lighting outputs.

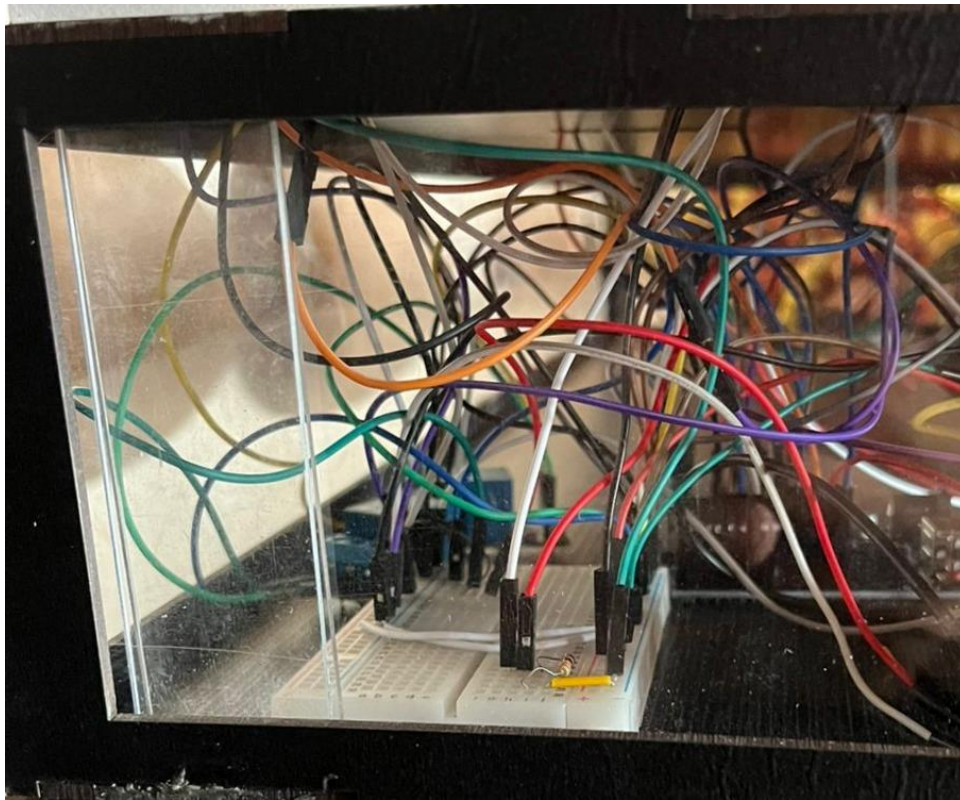


Figure 15: Arduino Uno 1 protection and lighting wiring

Explanation: This figure shows the internal wiring of the Arduino Uno 1 protection and lighting control node. The wiring belongs to the Protection and Lighting subsystem. It connects the MQ5 gas sensor, flame sensor, PIR sensor, LDR sensor, buzzer, fan, pump/output, and lighting outputs. This node detects gas and fire conditions, controls alarm and ventilation outputs, and manages lighting based on motion and ambient light level.

The delivery robot implementation is described in the robot subsystem diagram and code section. A separate physical implementation photo can be added later if a clear robot photo is provided by the team.

2.10 System Communication

The communication strategy was selected according to the function of each subsystem. I2C communication is used between Arduino Uno 1 and the ESP32 for status transfer, while pulse-width signaling is used from the ESP32 to Arduino Uno 2 for emergency access control. Bluetooth is used for manual control of the mist maker and robot, while Wi-Fi is used for cloud upload of patient data.

The hardware system uses more than one communication method. I2C is used between Arduino Uno 1 and the ESP32 for emergency status transfer. Digital pulse width signaling is used from the ESP32 to Arduino Uno 2 for emergency access control. Bluetooth is used for mist maker control and robot commands. Wi-Fi and HTTPS are used by the ESP32 to upload patient readings to the cloud.

Communication	Between	Purpose
I2C	Arduino Uno 1 -> ESP32	Send emergency status
Pulse signal	ESP32 -> Arduino Uno 2	Open door during emergency
Bluetooth	Mobile -> ESP32	Mist maker control
Bluetooth	Mobile -> Robot Arduino	Robot command selection
Wi-Fi / HTTPS	ESP32 -> Cloud API	Patient data upload

2.11 Hardware Problems and Solutions

Problem	Solution
Two LCDs had the same I2C address	PCA9548A I2C multiplexer was used to control each LCD on a separate channel.
Arduino Uno 5V logic and ESP32 3.3V logic are not directly compatible	Logic level shifter was used between ESP32 and Arduino signal lines.
Relay modules were Active LOW	The code initializes relays HIGH to keep them OFF and writes LOW to turn them ON.
Pump and motors can introduce electrical noise	Diode protection and stable wiring were considered to reduce back EMF effects.
Power stability is important for sensors and relays	Buck converter and separated supply lines were used for stable operation.
Wi-Fi connection may be unavailable	ESP32 continues local operation and retries connection without stopping the system.
Robot color readings can vary due to lighting	Color thresholds were calibrated experimentally for red and white stations.

2.12 Hardware System Summary

The hardware system demonstrates how embedded controllers can be used to improve hospital monitoring and emergency response. By using distributed controllers, the prototype became easier to control and debug. The system integrates patient monitoring, climate control, gas and fire detection, lighting control, access control, and robot delivery into one practical prototype.

Hospital General Layout and Project Information

Before starting the design stages such as lighting, power, HVAC, and light current systems, the general architectural layout and the main project data of the hospital must be identified. This initial overview provides a clear understanding of the building shape, the total hospital area, and the number of floors. These data are considered the starting point for all subsequent design calculations and distribution layouts.

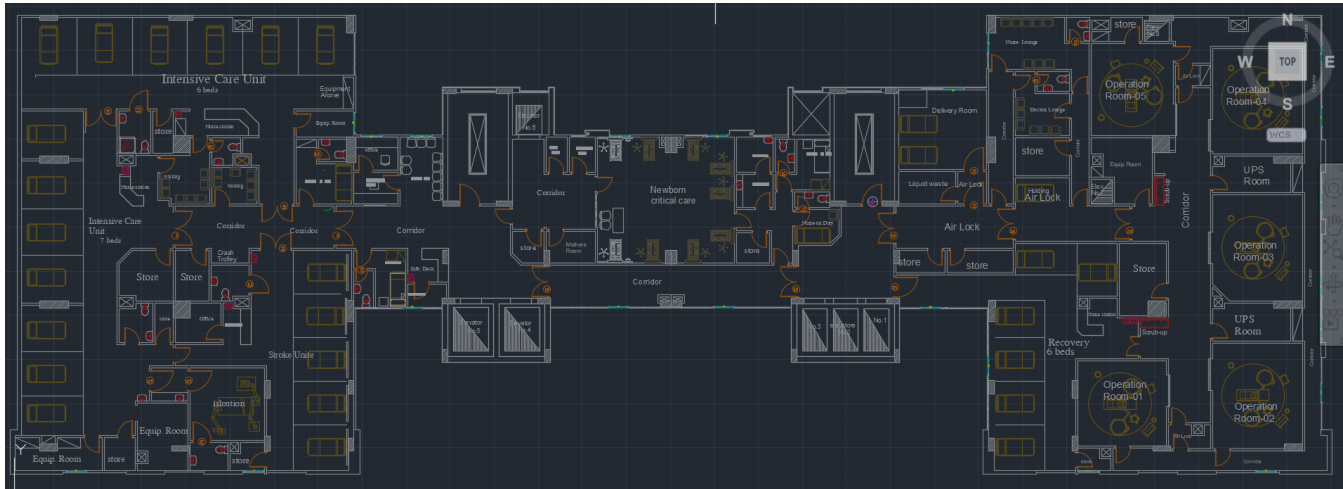


Figure 1. General architectural layout of the hospital.

The hospital building is located in Cairo and consists of two floors. The typical floor area is 1436.49 m², while the total built-up hospital area is 3269.29 m². Knowing these values is important because they directly affect the required number of fixtures, the load estimation, cable routing lengths, HVAC requirements, and the distribution of light current systems.

Accordingly, this page is used as a project introduction before entering the detailed design sections. It serves as the base reference for the following stages of the work: lighting design, power distribution, HVAC design, panel schedules, sizing calculations, shop drawings, cable tray sizing, and the light current systems.

- Place : cairo
- Floor Area : 1436.49 m²
- No Of Floor : 2
- Total Area of Hospital : 3269.29 m²

Project data summary

Electrical Distribution - DIALux Section

Preview only: this sample shows the proposed layout before applying it to the full book.

3.2 DIALux Lighting Simulation

DIALux was used to verify the lighting performance of the hospital spaces before preparing the final lighting drawings. The simulation helped check whether the selected luminaires could provide suitable illumination levels according to the room function and the required hospital lighting criteria.

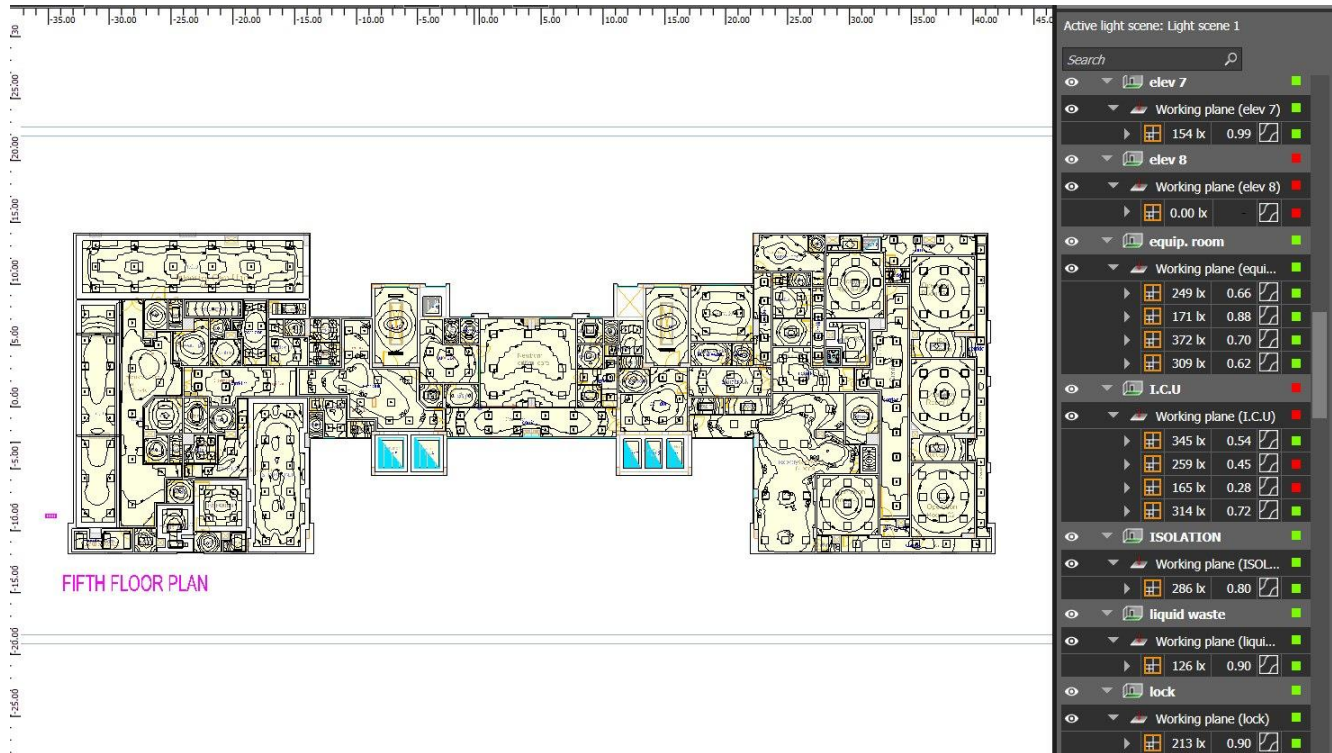


Figure 3.1: DIALux calculation plan and lighting distribution view.

Explanation: This figure shows the complete hospital plan imported into DIALux, including the active light scene and the calculated working planes for the different rooms. The plan was used to check the lighting distribution before transferring the final arrangement to the AutoCAD lighting design.

3.2.1 Lighting Code Requirement and Lamp Selection

٣٠٠	غرفة فحص	
٣٠٠	الإستقبال :	
٥٠٠	عام	
٣٠٠٠	صينية الأدوات	
	فجوة الدم	
٣٠٠	معمل الإستقبال :	
٥٠٠	عام	
١٠٠٠	منضدة العمل	
	إضاءة قطعة الشغل	
٥٠٠	المعامل :	
١٠٠٠	أخذ العينات	
٣٠٠	معمل الأنسجة	
٥٠٠٠	حجرة الفحص المجهرى	المستشفيات
١٠٠٠	التحاليل الكيميائية	
٥٠٠	التحاليل البكتريولوجية	
	تحليل الدم	
٥٠٠	غرفة العمليات :	
١٠٠٠	التخدير والتخدير	
٢٥٠٠٠	عام بغرف العمليات	
١٠٠٠٠٠	مائدة العمليات	
	غرف المرضى :	
٦٠	عام	
٣٠٠	قراءة	
	الإشعة :	
	قسم التشخيص	
٣٠	حجرة الأشعة	
١٠٠٠	فرز الأفلام	
٥٠٠	تحضير الباريوم	

Figure 3.2: Egyptian code lighting requirements for hospital spaces.

Explanation: This reference was used to identify the recommended illuminance level for different hospital spaces such as patient rooms, operation rooms, laboratories, examination rooms, and service areas. It acts as the design target used when checking the DIALux results.

Lamp	Luminous Efficacy (lumens/watt)	CCT (K)	CRI	Burning Hours (hours)	Applications
Incandescent	8-20	2800	100	1000	Indoor lighting specially domestic and also as decorative lighting
Halogen	20-25	2800	100	1200-1500	Flood lighting applications and theatre lighting
Fluorescent	50-70	2700; 4500; 6500	60-90	6000-10000	Indoor lighting specially domestic and small commercial applications
CFL	60-80	2700; 4500; 6500	60-90	6000-15000	Indoor lighting specially domestic and small commercial applications
Mercury Vapour	35-65	2500-4000	>80	20000-24000	At present used vary rarely in outdoor lighting applications
Metal-Halide	75-100	>6500	80-90	6000-15000	Used in wide area overhead lighting of commercial, industrial, and public spaces, such as parking lots, sports arenas, factories, and retail stores, as well as residential security lighting and automotive lighting
HPSV	100-150	<3000	65-85	10000-15000	Street Lighting applications, where good colour rendering is required
LPSV	150-200	<2000	negative	18000	Mostly used in Street Lighting applications where safety is priority and not the colour of the object. Mostly used during foggy weathers
LED	30-80; But nowadays as high as 200	2700; 4500; 6500	60-90	50000	LEDs were used as indicators and signals in earlier days but, now LEDs are used in almost all lighting applications ranging from indoor to road lighting to flood lighting.

Figure 3.3: Lamp type comparison used before luminaire selection.

Explanation: This comparison summarizes the main lamp technologies, including incandescent, halogen, fluorescent, CFL, metal-halide, and LED lamps. LED luminaires were preferred for the final design because they provide higher efficiency, long lifetime, good controllability, and suitable performance for hospital applications.

3.2.2 Luminaire List and Calculation Results

Building 1 - Storey 1 Luminaire list

Φtotal	Power	Luminous efficacy				
1131205 lm	12754.6 W	88.7 lm/W				
pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
60	Philips		CR350B LED60S/840 W60L60	55.0 W	5997 lm	109.0 lm/W
9	Philips		CR350B LED60S/840 W60L60	55.0 W	2998 lm	54.5 lm/W
61	Philips		RC132V W60L60 WIA 1 xLED34S/830 OC	29.0 W	2600 lm	89.7 lm/W
113	Philips		RC132V W60L60 WIA 1 xLED34S/830 OC	29.0 W	2000 lm	69.0 lm/W
41	Philips		RC132V W60L60 WIA 1 xLED34S/830 OC	29.0 W	2500 lm	86.2 lm/W
4	Philips		RC132V W60L60 WIA 1 xLED34S/830 OC	29.0 W	3400 lm	117.2 lm/W
4	Philips		SM136V PSD W20L120 1 x28S/830 OC	22.0 W	2788 lm	126.7 lm/W
4	Philips		SM150C L602 1xLED24S/840	20.0 W	2400 lm	120.0 lm/W
26	Philips		RC430X LED43 930 W60L120 WPA C OC	38.1 W	3799 lm	99.7 lm/W
23	Philips	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1	DN610B 40S-UE/840 PSU-E C	25.0 W	2999 lm	120.0 lm/W
34	Philips	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1	DN610B 40S-UE/840 PSU-E C	25.0 W	1500 lm	60.0 lm/W
1	Philips	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1	DN610B 40S-UE/840 PSU-E C	25.0 W	4200 lm	168.0 lm/W

Luminaire list

Building 1 - Storey 1 (Light scene 1) Calculation objects

Working plane (elev. room) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.000 m	188 lx (≥ 150 lx)	175 lx	197 lx	0.93 (≥ 0.60)	0.89	WP30
Working plane (elev 1) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.210 m	149 lx (≥ 100 lx)	144 lx	152 lx	0.97 (≥ 0.40)	0.95	WP80
Working plane (elev 2) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.210 m	150 lx (≥ 100 lx)	143 lx	152 lx	0.95 (≥ 0.40)	0.94	WP81
Working plane (elev 3) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.210 m	148 lx (≥ 100 lx)	142 lx	150 lx	0.96 (≥ 0.40)	0.95	WP82
Working plane (elev 4) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.393 m	133 lx (≥ 100 lx)	124 lx	138 lx	0.93 (≥ 0.40)	0.90	WP78
Working plane (elev 5) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.393 m	133 lx (≥ 100 lx)	124 lx	138 lx	0.93 (≥ 0.40)	0.90	WP79
Working plane (elev 6) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.225 m	154 lx (≥ 100 lx)	153 lx	154 lx	0.99 (≥ 0.40)	0.99	WP83
Working plane (elev 7) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.180 m	149 lx (≥ 100 lx)	148 lx	149 lx	0.99 (≥ 0.40)	0.99	WP84
Working plane (elev 8) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.180 m	152 lx (≥ 100 lx)	152 lx	153 lx	1.00 (≥ 0.40)	0.99	WP85
Working plane (equip. room) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.400 m	374 lx (≥ 300 lx)	267 lx	421 lx	0.71 (≥ 0.60)	0.63	WP7
Working plane (equip. room) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.200 m	313 lx (≥ 300 lx)	203 lx	366 lx	0.65 (≥ 0.60)	0.55	WP9
Working plane (equip. room) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.300 m	175 lx (≥ 150 lx)	155 lx	186 lx	0.89 (≥ 0.60)	0.83	WP21

Calculation objects and result check

Figure 3.4: DIALux luminaire list and calculation object results.

Explanation: The luminaire list summarizes the selected lighting products, quantities, power ratings, luminous flux, and luminous efficacy. The calculation object sheet shows the lighting result checks for the working planes. The green check marks indicate that the lighting values were verified against the required room criteria.

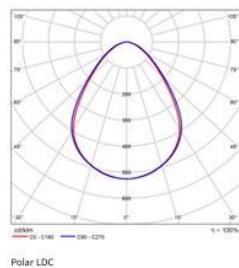
3.2.3 Product Data Sheets

The product data sheets were grouped together in compact tables to show the real photometric data used in the DIALux model without wasting many pages. Each sheet includes important product information such as power, luminous flux, luminous efficacy, color temperature, CRI, and polar light distribution curve.

Product data sheet

Philips - CR350B LED60S/840 W60L60

	
P	55.0 W
Φ _{Lamp}	3000 lm
Φ _{Luminaire}	2998 lm
η	99.95 %
Luminous efficacy	54.5 lm/W
CCT	4000 K
CRI	80

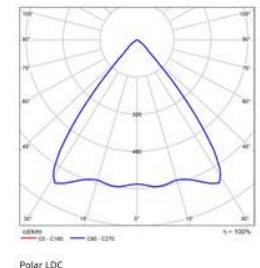


Product data sheet 1

Product data sheet

Philips - DN610B 40S-UE/840 PSU-E C

	
Article No.	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1
P	25.0 W
Φ _{Lamp}	3000 lm
Φ _{Luminaire}	2999 lm
η	99.97 %
Luminous efficacy	120.0 lm/W
CCT	3000 K
CRI	100

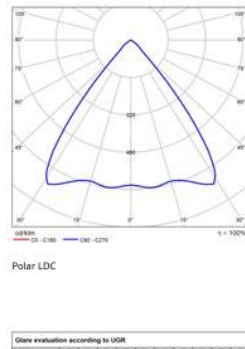


Product data sheet 2

Product data sheet

Philips - DN610B 40S-UE/840 PSU-E C

	
Article No.	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1
P	25.0 W
Φ_{Lamp}	1500 lm
$\Phi_{Luminaire}$	1500 lm
η	99.97 %
Luminous efficacy	60.0 lm/W
CCT	3000 K
CRI	100

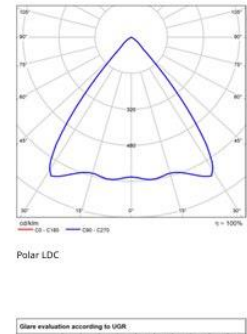


Product data sheet 3

Product data sheet

Philips - DN610B 40S-UE/840 PSU-E C

	
Article No.	DN610B1-5e3e1331-2b2a-42e0-b1fb-7194e0f473b1
P	25.0 W
Φ_{Lamp}	4201 lm
$\Phi_{Luminaire}$	4200 lm
η	99.97 %
Luminous efficacy	168.0 lm/W
CCT	3000 K
CRI	100



Product data sheet 4

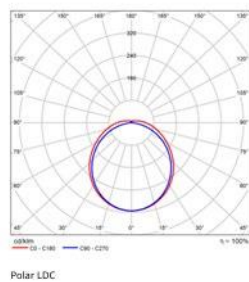
Figure 3.5: DIALux product data sheets - group 1.

Explanation: The first group of product data sheets shows the selected Philips luminaire references used in the lighting simulation. These sheets support the design by showing that the luminaire selection was based on real manufacturer data and polar distribution curves.

Product data sheet

Philips - SM150C L602 1xLED24S/840

	
	
P	20.0 W
Φ_{Lamp}	2400 lm
$\Phi_{Luminaire}$	2400 lm
η	99.98 %
Luminous efficacy	120.0 lm/W
CCT	4000 K
CRI	80

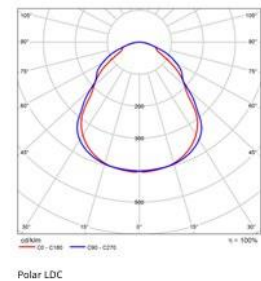


Product data sheet 5

Product data sheet

Philips - RC430X LED43 930 W60L120 WIA C OC

	
P	38.1 W
Φ_{Lamp}	3800 lm
$\Phi_{Luminaire}$	3799 lm
η	99.98 %
Luminous efficacy	99.7 lm/W
CCT	3000 K
CRI	100

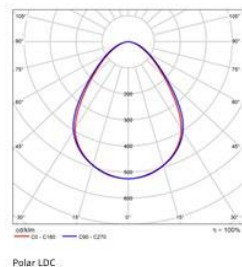


Product data sheet 6

Product data sheet

Philips - CR350B LED60S/840 W60L60

	
	
P	55.0 W
Φ_{Lamp}	6000 lm
$\Phi_{Luminaire}$	5997 lm
η	99.95 %
Luminous efficacy	109.0 lm/W
CCT	4000 K
CRI	80



Product data sheet 7

Figure 3.6: DIALux product data sheets - group 2.

Explanation: The second group continues the product data sheets used in the DIALux model. Grouping the sheets in a compact layout keeps the section readable while still documenting the technical source of the selected luminaires.

Lighting Design Section

This prototype shows the proposed layout for the Lighting Design section only. The figures are placed in the same order received: lighting layout, lighting legend, wiring concept, ceiling/lighting applications, luminaire types, and switch types.

1. Lighting Layout

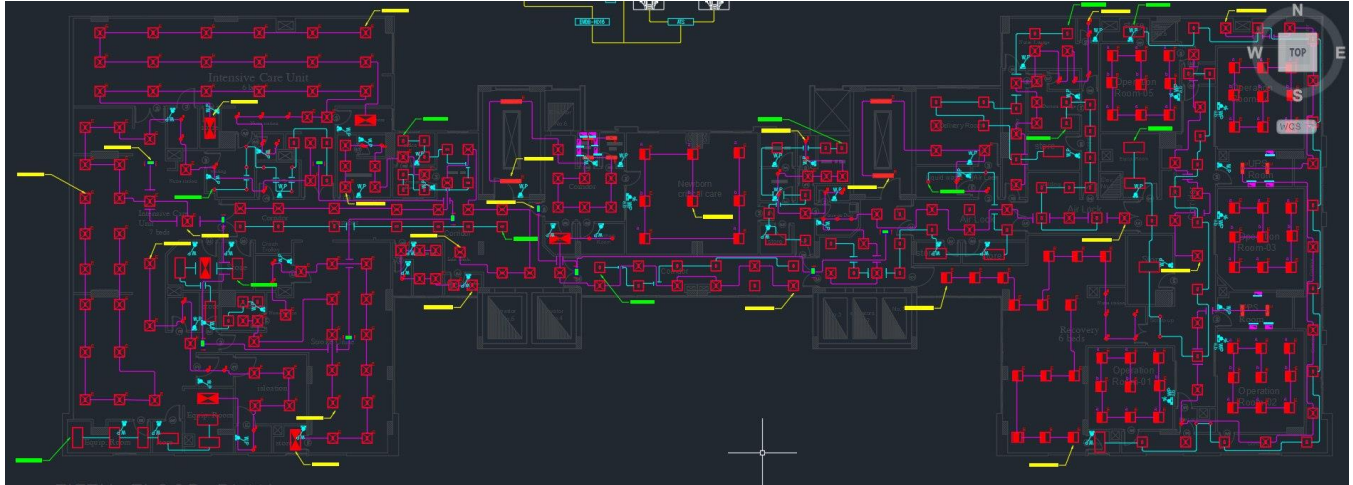


Figure L.1: Lighting layout drawing.

Explanation: This drawing shows the lighting distribution for the hospital floor. The luminaires are distributed according to the room function, ceiling layout, required illumination level, and circuit grouping. The drawing includes normal, emergency, and critical lighting points according to the selected power source for each circuit.

2. Lighting Legend

LIGHTING SWITCHES			
	SW1	LIGHTING SWITCH, ONE WAY, ONE GANG, 50HZ, 240V.	
	SW2	SAME AS ABOVE BUT WEATHER PROOF.	
	SW3	LIGHTING SWITCH, ONE WAY, TWO GANGS, 50HZ, 240V. (ALSO CONTROLS FAN/CEILING FAN ETC.)	
	SW4	LIGHTING SWITCH, TWO WAY, ONE GANG, 50HZ, 240V.	
	SW5	SAME AS SW3 BUT WEATHER PROOF.	

CODE		SYMBOL	DESCRIPTION
F1			RECESSED LUMINAIRES (RCL32V WOL 80WMA 1xLED3x5 830/CC) LED LAMP 30 WATT LUMINOUS FLUX = 2800 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F2			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery
F3			RECESSED Downlight (DW100 400-LE 840/PSU-E C) LED LAMP 36 WATT LUMINOUS FLUX = 1800 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F4			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery
F5			RECESSED LUMINAIRES (RCL430X LED 830 43WBL 120WMA C DC) LED LAMP 36.1 WATT LUMINOUS FLUX = 3800 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F6			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery
F7			RECESSED LUMINAIRES ANTI BACTERIAL (RCL300 LED 840/WOL80) LED LAMP 85 WATT LUMINOUS FLUX = 3800 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F8			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery

F8			SURFACE MOUNTED LUMINAIRES (SM130C L1 602xLED240W/0) LED LAMP 28 WATT LUMINOUS FLUX = 2400 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F10			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery
F11			SURFACE MOUNTED LUMINAIRES (SM138V PSD WOL 1 120x285 830/40) LED LAMP 30 WATT LUMINOUS FLUX = 2400 LUMEN
			SAME AS ABOVE BUT CONNECTED TO DIESEL GENERATOR
F12			SAME AS ABOVE BUT EQUIPPED WITH 3 HRS NI-CD Battery
EX			EMERGENCY EXIT LIGHTING SIGN W/ LED LIGHTING LAMP = 6W CONNECTED TO NI-CD Battery

Figure L.2: Lighting legend and lighting symbols.

Explanation: The lighting legend defines the symbols used in the lighting drawings. It links each symbol to its corresponding luminaire, switch, or lighting type so the drawing can be read correctly during design review and site execution.

3. Wiring of Lighting Loads

WIRING OF LIGHTING

- 1) Normal Loads
- 2) Emergency Loads
- 3) Critical Loads

Figure L.3: Wiring classification of lighting loads.

Explanation: Lighting circuits were divided into three categories. Normal loads are supplied from the transformer through normal distribution panels. Emergency loads are supplied from the generator to keep essential lighting working during power failure. Critical loads are supplied from the UPS, and the UPS itself is backed up by the generator to maintain power for highly sensitive areas.

Load Type	Power Source	Purpose
Normal lighting	Transformer	General lighting operation during normal supply.
Emergency lighting	Generator	Maintains essential lighting during utility power failure.
Critical lighting	UPS backed by generator	Maintains sensitive and critical lighting circuits without interruption.

4. Ceiling Lighting Applications

	
Ceiling applications reference	External / landscape lighting reference

Figure L.4: Lighting applications in ceiling and external areas.

Explanation: The lighting design considers different application areas. Ceiling lighting is used for indoor rooms and corridors according to ceiling type and required lux level, while external lighting can be used for outdoor or landscape areas to improve visibility and safety around the building.

5. Luminaire Types Used



Figure L.5: Recessed, downlight, and ceiling luminaire examples.

Explanation: The design uses different luminaire types according to the function of the space. Recessed panel lights are suitable for offices, corridors, and rooms with suspended ceilings. Downlights can be used in smaller areas or spaces requiring focused lighting. Louver or linear ceiling luminaires can be used where wider light distribution is required.



Figure L.6: High bay luminaire example.

Explanation: High bay luminaires are used in high-ceiling areas or service spaces where a stronger light output and wider coverage are required. They are not used everywhere, but only where the room height and application need this type of fixture.

6. Switch Types

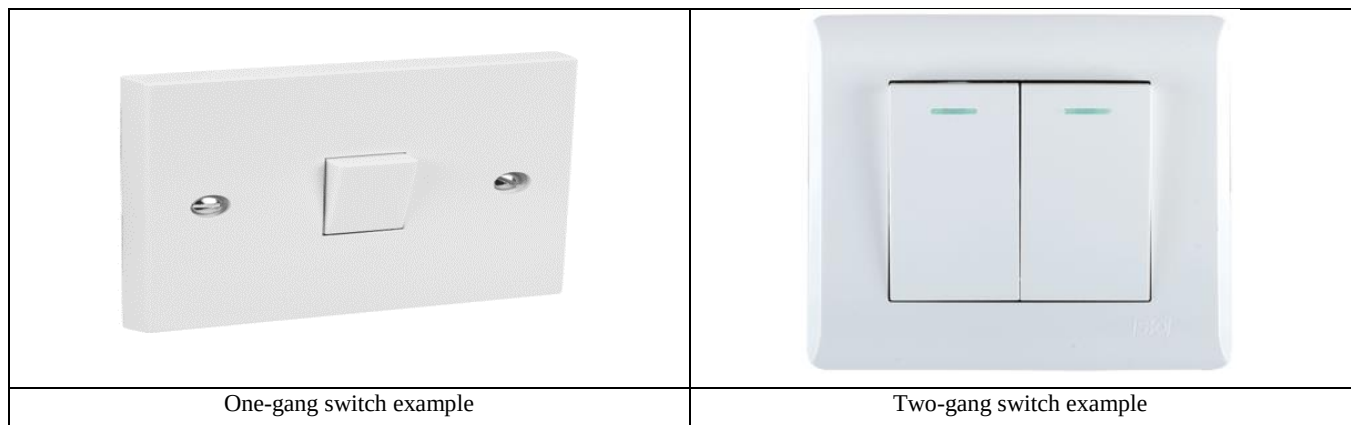


Figure L.7: One-gang and two-gang switch examples.

Explanation: Switches were selected according to the required control method and number of lighting groups. A one-gang switch controls one lighting group, while a two-gang switch can control two separate lighting groups from the same plate.

TYPES OF SWITCHES

- 1-ONE GANG SWITCH , ONE WAY
- 2-ONE GANG SWITCH , TWO WAY
- 3-TWO GANG SWITCH , ONE WAY
- 4-TWO GANG SWITCH , TWO WAY

Figure L.8: Types of switches used in the lighting design.

Explanation: The switch types include one-gang one-way, one-gang two-way, two-gang one-way, and two-gang two-way switches. One-way switching is used for simple local control, while two-way switching is used when a lighting circuit must be controlled from two different locations, such as corridors or room entrances.

3.4 Power Socket and Distribution

The power socket and distribution system defines the location, symbol, and supply type of socket outlets inside the hospital. The system includes normal sockets, UPS sockets, waterproof sockets, bed head unit points, floor boxes, TV/data service points, and dedicated power outlets according to the function of each room.

In the final report, the power section will be placed after the lighting design section. The images below are arranged exactly as: full power layout, small power legend, code and installation references, socket types, and socket zoom-in detail.

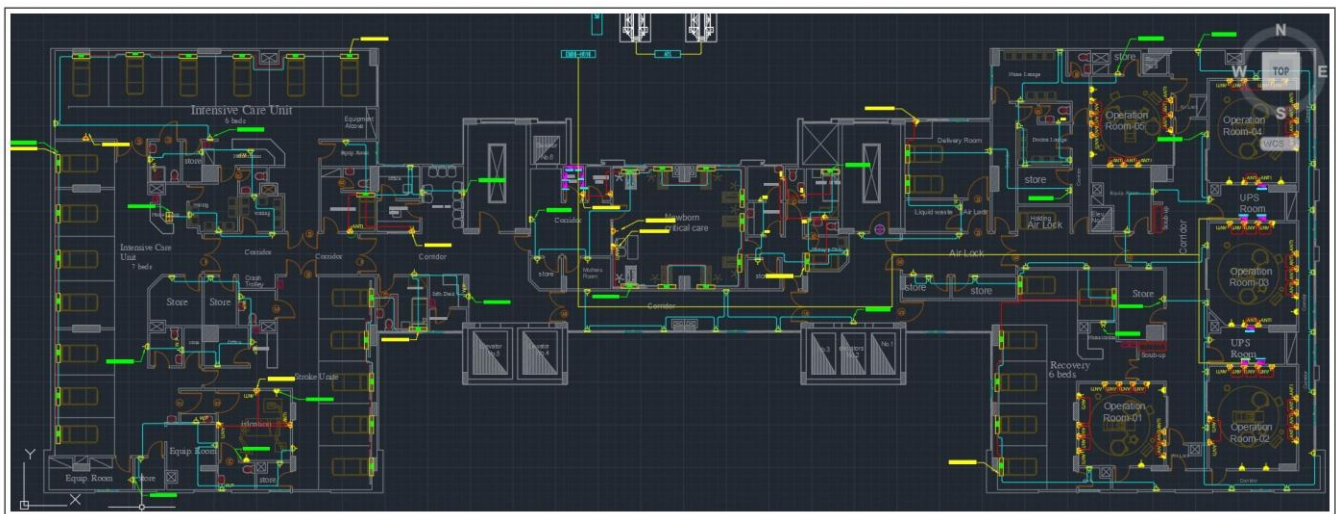


Figure P.1: Complete small power and socket layout.

Explanation: This figure shows the complete power socket layout for the hospital floor. It presents the distribution of socket outlets, bed head unit points, UPS outlets, TV/data related service points, and dedicated power connection points. The layout is used as the main reference for understanding the socket locations before moving to the legend and the socket type explanation.

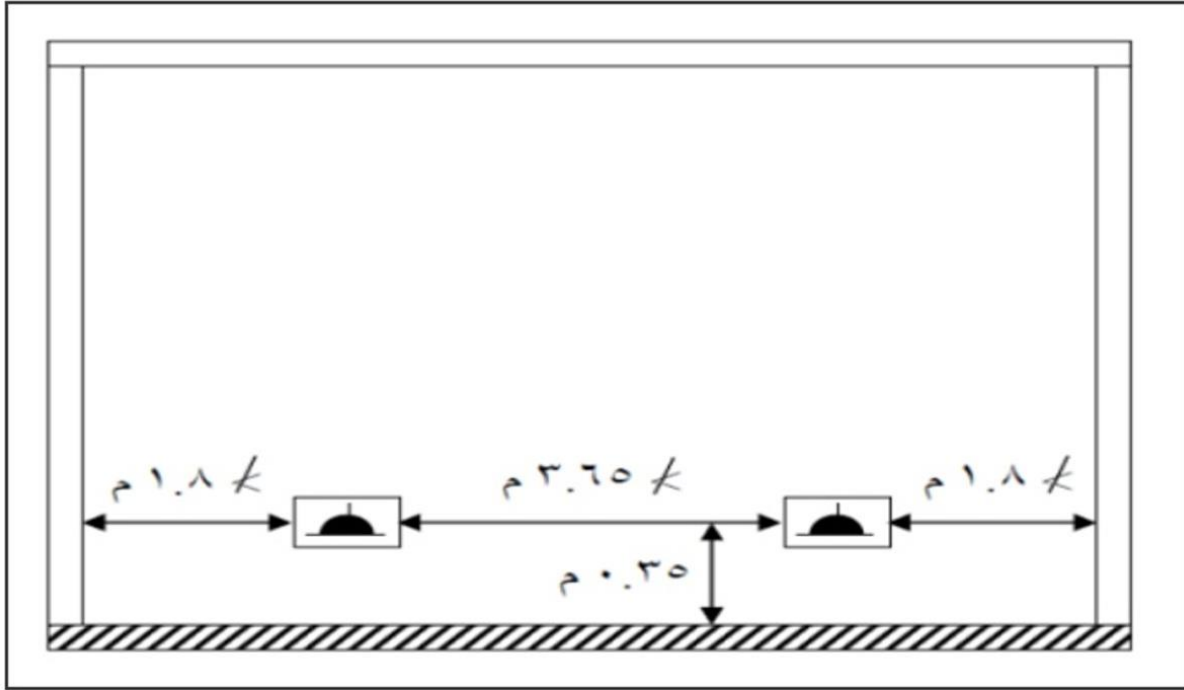
Power Legend and Symbol Interpretation

SMALL POWER LEGEND:		
TYPE	SYMBOLS	DESCRIPTION
SO1		230V- 16A, CONVENTIONAL SOCKET OUTLET, 2P+E, IP20 FLUSH MOUNTED WITH SAFETY SHUTTERS AND SIDE EARTH CONTACT (SCHUKO TYPE)
		SAME AS SO1 BUT CONNECTED TO UNINTERRUPTIBLE POWER SUPPLY (MARKED BY RED SOCKET LUGS)
SO2		SAME AS SO1 BUT DUPLEX SOCKET OUTLETS, WITH GROUND
		SAME AS SO2 BUT CONNECTED TO UNINTERRUPTIBLE POWER SUPPLY (MARKED BY RED SOCKET LUGS)
SO3		POWER SOCKET WATER PROOF TYPE IP54, MOUNTED AT HEIGHT 120 CM FROM F.F.L
SO4		SAME AS SO1 BUT WATER PROOF IP65
SO5		SAME AS SO4 BUT CONNECTED TO UPS
SO6		SAME AS SO2 BUT CONNECTED TO UNINTERRUPTIBLE POWER SUPPLY ANTI BACTERIAL (MARKED BY RED SOCKET LUGS)
SO7		SAME AS SO2 BUT DUPLEX SOCKET OUTLETS FOR TV.
BHU		(ONE SINGLE SOCKET NORMAL + ONE SINGLE SOCKET (UPS)+TEL+DATA)
FB1		FLOOR BOX CONTAINS :- (DOUBLE SOCKET(NORMAL)+TEL.+DATA)
FB2		FLOOR BOX CONTAINS :- (DOUBLE SOCKET(NORMAL)+(DOUBLE SOCKET(UPS)+TEL.+DATA)
SO8		SAME AS SO3 BUT CONNECTED TO UPS
SMALL POWER LEGEND:		
TYPE	SYMBOLS	DESCRIPTION
IS1		220V, 50Hz, WITH 2P+E SURFACE FLUSH MOUNTED LOAD BREAK SWITCH - RATING AS SHOWN ON DWG
IS3		380V, 50Hz, 3P+N+E SWITCH DISCONNECTOR, RATING AS SHOWN ON DWG

Figure P.2: Small power legend used in the power socket layout.

Explanation: This legend defines the symbols used in the power drawing. It links each AutoCAD symbol to its real function, such as conventional socket outlets, UPS socket outlets, waterproof sockets, antibacterial sockets, bed head unit points, floor boxes, TV socket points, and disconnect switches. The legend must be placed before the socket type images so the reader can understand the drawing symbols correctly.

Code and Installation References



Typical socket spacing / mounting reference

جدول رقم (٢-٥): أنواع المقابس المستخدمة

م	النوع	العدد	السعة (أمبير)	الجهد (فولت)	عدد الأطراف	الاستخدام
١	مقبس بريزة أحادي	١	١٠	٢٥٠	٣ 1Ph + N+E	الأحمال العادية فياجورة + مكثفة - راديو - تليفزيون - ألخ
٢	مقبس بريزة أحادي	١	١٦	٢٥٠	٣ (1Ph + N+E)	أحمال القوى كالسخان
٣	مقبس بريزة مزدوج	٢	١٠	٢٥٠	٣ (1Ph + N+E)	أحمال الخدمة العادية والتي تتطلب أكثر من مصدر متجاوز كالفيديو و التليفزيون
٤	مقبس بريزة ثلاثي	١	٢٥ . ١٦ ٦٣ . ٥٠ ٢٥٠ . ١٢٥	٤٠٠	٥ (3Ph + N+E)	أحمال القوى الخاصة بالصيانة كمنشآر كهربى، مثقاب محمول، .. ألخ

Egyptian code table for socket types and applications

Figure P.3: Egyptian code and socket installation references.

Explanation: This figure combines the socket installation spacing reference and the Egyptian code table used for power outlets. The spacing reference helps define the practical socket mounting and separation requirements, while the code table explains the required socket types, number of phases, voltage level, and typical use. These references support the power layout decisions in hospital rooms and service spaces.

Socket Types Used in the Design

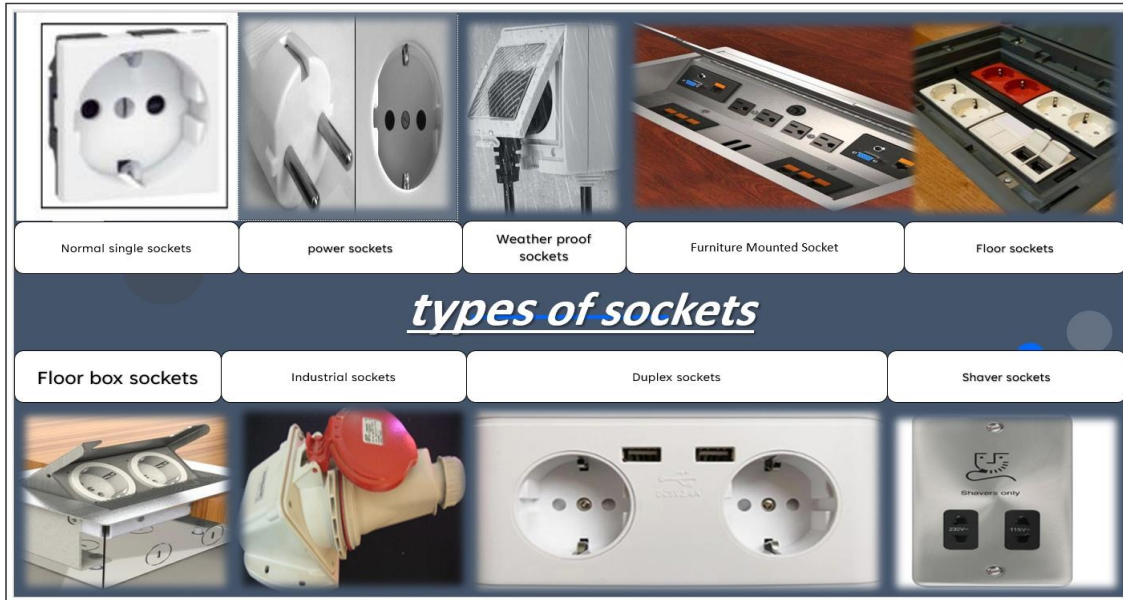


Figure P.4: Types of sockets used in the hospital power design.

Explanation: This figure presents the main socket types used in the design, including normal single sockets, power sockets, weatherproof sockets, furniture mounted sockets, floor sockets, floor box sockets, industrial sockets, duplex sockets, and shaver sockets. These types are not used randomly; each type is selected according to the room function, environment, load type, and required safety level.

Examples of usage: normal sockets are used for general loads, weatherproof sockets are used in wet or service areas, floor boxes are used for flexible room layouts, industrial sockets are used for high-power equipment, and duplex or combined sockets can be used where more than one service is required in the same location.

Socket Detail Inside the Drawing



Figure P.5: Zoom-in view of socket and TV/service point arrangement.

Explanation: This zoom-in view shows how socket symbols appear inside the AutoCAD drawing at room level. It helps connect the legend symbols with their actual locations in the layout. The highlighted area includes socket outlets and TV/service points, showing how the power design is coordinated with the architectural room arrangement and furniture locations.

Prototype note: When this section is inserted into the final book, the same order should be preserved: main power layout, power legend, code/installation references, socket types, and zoom-in detail. No image from this system should be moved into HVAC, panel schedule, or shop drawing sections.

3.5 HVAC Power Design

HVAC power design defines the electrical supply required for air conditioning and ventilation units in the hospital. The electrical design depends on the mechanical load data, unit location, voltage level, disconnect switch rating, circuit breaker selection, and cable size. In this section, the HVAC drawings and mechanical load reference are kept together so the design sequence remains clear.

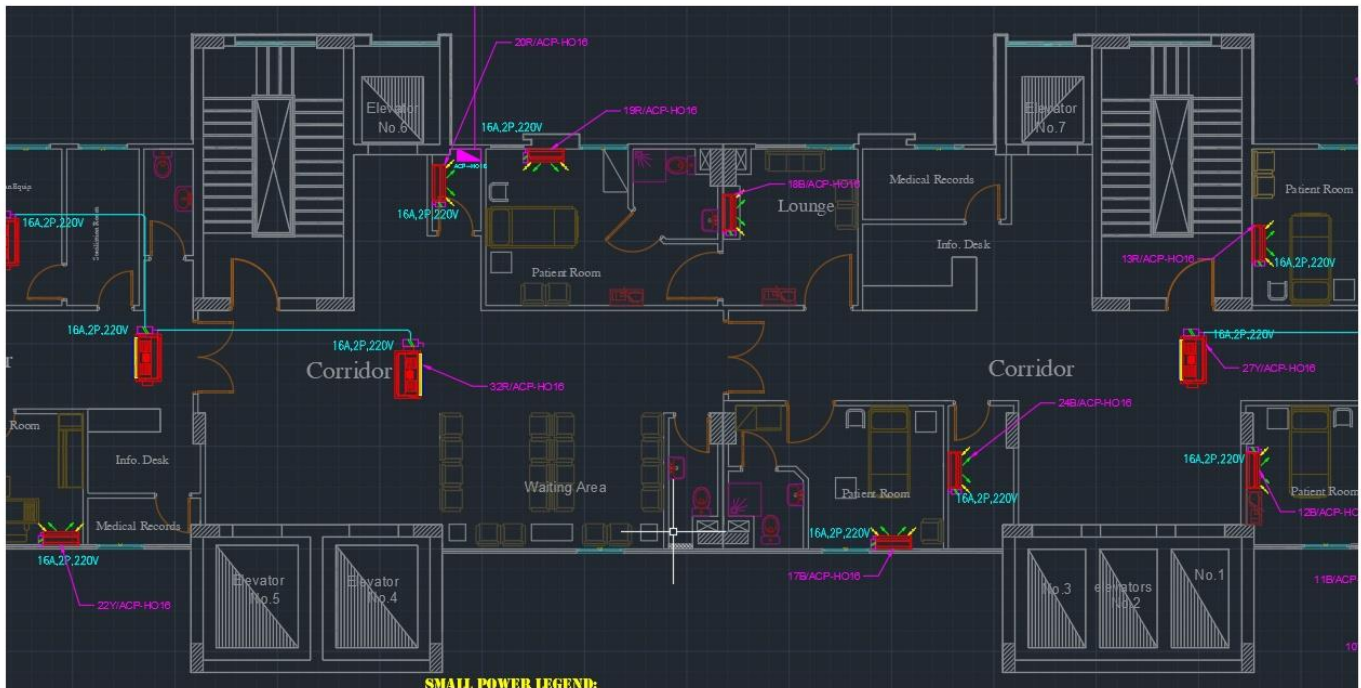


Figure HVAC-1: HVAC power layout reference.

Explanation: This drawing shows the HVAC electrical power points distributed on the hospital floor. The labels beside the units identify the circuit reference and supply information used later in panel schedules, circuit breaker selection, cable sizing, and shop drawing routing. This figure is kept as the main HVAC plan reference, not as a power legend or panel schedule.

HVAC Unit References

The HVAC system contains different types of air-conditioning equipment. The two references below clarify the difference between a Fan Coil Unit (FCU) and a split unit before discussing the electrical sizing method.



Figure HVAC-2: FCU concept and split unit reference.

Explanation: An FCU is normally connected to a central chilled-water system. Air is supplied through ducts and diffusers, and the unit contains a fan and coil to condition the air. A split unit is usually a direct expansion air-conditioning system with an indoor unit and an outdoor unit. Electrically, both are treated as mechanical loads, but their locations, ratings, control requirements, and supply points may be different.

Item	FCU	Split Unit
System concept	Connected to central chilled-water / air-side distribution concept.	Direct expansion unit with indoor and outdoor sections.
Air distribution	Air can be supplied through ducts and diffusers.	Air is supplied directly from the indoor unit.
Electrical design need	Requires unit power data, control point, circuit breaker, cable and disconnect switch.	Requires indoor/outdoor electrical supply data, breaker, cable and isolator as per unit rating.
Typical use	Hospital rooms, corridors and spaces served by central HVAC systems.	Small rooms or isolated spaces that need individual air-conditioning.

HVAC Mechanical Load Sizing

Mechanical load sizing starts by converting the HVAC unit power from horsepower to electrical current. After calculating the base current, a design factor is applied according to the project calculation reference, and then a suitable disconnect switch, circuit breaker, and cable size are selected.

MECHANICAL LOADS:

MECHANICAL LOADS:

Sizing for Disconnect switch :

$$\bullet \text{ } I_b = (HP \times 746) / (V \times P.F \times EFF.)$$

In mechanical loads :

$$I = I_b \times 2.5$$

In heater loads : $I = I_b \times 1.25$

AREA M**2	HP
12	1.5 HP
18	2.25 HP
24	3 HP
30	4 HP
40	5 HP

Figure HVAC-3: Mechanical load sizing and disconnect switch reference.

Explanation: The sizing reference shows the relationship between room area and approximate HVAC capacity in horsepower. It also shows the current calculation formula used as the base for selecting the disconnect switch. The design current is then increased by a factor to provide a safe margin for the mechanical load before selecting the standard protective device and cable size.

Step	Description
1. Collect mechanical data	Read the HVAC unit capacity, voltage, power factor and efficiency from the mechanical schedule or equipment reference.
2. Calculate base current	Use the horsepower formula: $I_b = (HP \times 746) / (V \times PF \times Eff.)$.
3. Apply design factor	Multiply the base current by the project design factor shown in the HVAC calculation reference to obtain the design current.
4. Select disconnect switch	Choose a standard disconnect switch rating higher than the design current.
5. Select breaker and cable	Select the circuit breaker and cable so that the cable current-carrying capacity is suitable for the selected breaker and load current.

HVAC Power Design Sequence

Design stage	Purpose
HVAC layout	Identify HVAC unit locations and circuit labels on the plan.
Unit type	Differentiate between FCU and split units because the electrical supply method may be different.
Mechanical load	Use the room area or mechanical schedule to estimate the unit capacity in HP.
Current calculation	Calculate base current using HP, voltage, power factor and efficiency.
Protection and cable	Select disconnect switch, circuit breaker and cable size according to design current and standard ratings.
Panel schedule	Transfer the selected load and breaker values to the HVAC panel schedule or related distribution panel.

3.6 Panel Schedule Design

3.6.1 Three-Phase Distribution Panel Schedule

Panel Name:	LP2-HO16	load current	5.798319136
FED FROM:	SMDG-HO16	s125	7.24789852
Panel Location:	16		
Power System:	LIGHTING		
Panel type :	Distribution		

AIN C.B. RATIO	20	CABLE SIZE:	W(4+1)mm ² CU/XLPE/PVC
NO.OF VAYS :	4 WAY	MOUNTING:	WALL MOUNTED

		Ground			
--	--	--------	--	--	--

CT NO.	TYPE OF LOAD	NOTE	WIRE	PHASE LOAD			BREAKER TYPE	BREAKER RATING	R Y B		PHASE LOAD			WIRE	NOTE	TYPE OF LOAD	CT NO.
				R	Y	B					R	Y	B				
1	LIGHTING		3x(T2.5)mm ²	470.3			MCB	16				310.1		3x(T2.5)mm ²		LIGHTING	2
3	LIGHTING		3x(T2.5)mm ²		375.0		MCB	16				347.4		3x(T2.5)mm ²		LIGHTING	4
5	LIGHTING		3x(T2.5)mm ²			392.8	MCB	16					303.9	3x(T2.5)mm ²		LIGHTING	6
7	LIGHTING		3x(T2.5)mm ²	187.5			MCB	16				300.0		3x(T2.5)mm ²		LIGHTING	8
9	LIGHTING		3x(T2.5)mm ²		262.5		MCB	16				331.3		3x(T2.5)mm ²		LIGHTING	10
11	LIGHTING		3x(T2.5)mm ²			255.4	MCB	16					285.8	3x(T2.5)mm ²		LIGHTING	12
Sub. Total :				657.8	637.5	639.1											
Total Power :				610.1	670.7	695.6											
				1267.3	1316.2	1277.8	3.812	kVA									

Avag.Power =	1270.6	VA	99.8%	103.6%	96.6%
Un balance percentage			7%		
THE LOAD TO BE MANAGED			44.22		

NOTE:	TRANSFER THE LOAD FORM MAX PHASE TO MINIMUM PHASE
NOTE:	الرجاء تعديل توزيع الأحمال (إذا كانت أحمال صناعية تحتوي على أحمال أخرى Main تشغيل الدارة diversity factor)

Load Type :	Lighting(left)	Lighting(right)	Power(left)	Power(right)	A/C(left)	A/C(right)	Others(left)	Others(right)	Total	Max. Demand
Control Load (VA)	1933.375	1878.440	0.000	0.000	0.000	0.000	0.000	0.000	3811.82	
Demand Factor (VA)	100.0%	100.0%	96.00%	96.00%	100.0%	100.0%	65.0%	65.0%		3811.815
Demand Load (VA)	1933.375	1878.440	0.000	0.000	0.000	0.000	0.000	0.000	3811.815	
Diversity Load (VA)									3849.452	

Figure 3.17: Three-phase panel schedule for a normal/emergency distribution panel.

Explanation: This schedule is used for ordinary three-phase panels such as lighting panels, power panels, normal distribution panels, and emergency panels. It shows the panel name, the upstream source, the location, the power system, the panel type, the load current, the safety factor, the main circuit breaker, the cable size, the number of ways, and the wall-mounted installation type.

3.6.2 Main Data Read from the Panel Schedule

Item	Meaning
Panel Name	The unique tag of the panel, for example LP2-HO16.
Fed From	The upstream panel or board that supplies this panel.
Panel Location	The room or area where the panel is installed.
Power System	The category of the panel load, such as lighting, power, normal, emergency, or isolated system.
Panel Type	Defines whether the panel is distribution, main, isolated, or another special type.
Load Current	The current calculated from the connected or demand load.
x 1.25	A safety factor applied to the current before selecting the main breaker.
Main C.B. Rating	The selected main circuit breaker rating after applying the safety factor.
Cable Size	The selected feeder cable connected to the panel.
No. of Ways	The number of outgoing circuits available in the panel.
Mounting	Shows whether the panel is wall mounted or installed using another method.

Normal Distribution Panel Board Reference



The schedule is the calculation and documentation sheet, while the distribution panel board is the real installed enclosure. The breaker arrangement, outgoing circuits, cable sizes, and panel labels must match the schedule during installation. The normal distribution panel can serve normal or emergency circuits depending on its source and load category.

Figure 3.18: Normal distribution panel board reference.

3.6.3 Load Balance, Max Demand, and Diversity Load

[illegible]

Figure 3.19: Load balance, demand load, diversity load, and max demand area in the panel schedule.

Explanation: For three-phase panels, the loads must be distributed across R, Y, and B phases. The unbalance percentage should be less than 10%. If one phase carries a higher load, some circuits should be transferred from the maximum phase to the minimum phase until the balance becomes acceptable.

Term	Explanation
Connected Load	The total installed load before applying demand or diversity.
Demand Factor	A factor that represents the expected operating percentage of each load type.
Diversity Load	The load after applying demand/diversity factors. It gives a realistic design value.
Max Demand	The maximum expected demand used to select the main breaker and feeder cable.
Load Balance	A check between R, Y, and B phase loads. The design target is less than 10% unbalance.

3.6.4 Single-Phase Isolated Panel Schedule

Panel Type:	SURFACE TYPE	load current	27.64176136	MAIN C.B. RATING :	40
FED FROM.:	EMDB	x1.25	34.5522017	CABLE SIZE:	4 x 25 mm² Cu/PVC/PVC
Panel Location:	UPS ROOM			NO.OF WAYS :	
Power System :	ISOLATION PANEL ONE			MOUNTING:	WALL MOUNTED

Notes	line no.	Description	C.B.	C.s.a	Lighting load	EMERGENCY socket load	disconnect switch HVAC	disconnect switch HEATER	Power socket	R	
	1R	LIGHTING ROOM	16	3x(1x2.5)mm²	618.75					618.75	
	2R	SMALL POWER	16	3x(1x4)mm²		1980				1980	
	3R	POWER	32	3x(1x10)mm²					3000	3000	
	4R	HVAC	32	3x(1x25)mm²			2098.125			2098.125	
					Total lighting load	Normal socket load	disconnect switch HVAC	disconnect switch HEATER	Power socket	R	
					Total conncteld load	618.75	1980	2098.125	0	3000	7696.875
								Balance percentage		0	
					Demand factor	0.9	0.7	0.9	0.5	0.75	
					Total Demand load	556.875	1386	1888.3125	0	2250	6081.2
											Total demand load (VA)

Note :-

1- SELECT ONE PHASE AND DELETE OTHERS

2- ENTER THE LOAD WITH VA

6.0812

Figure 3.20: Single-phase isolated panel schedule used for operation-room related loads.

Explanation: The isolated panel schedule is used for critical medical areas, especially operation-room related circuits.

Unlike the normal three-phase panel, this isolated panel is treated as a single-phase panel in the shown schedule. The R phase column contains the connected loads, and the schedule focuses on safe medical supply, correct circuit identification, suitable breaker rating, and feeder cable size.

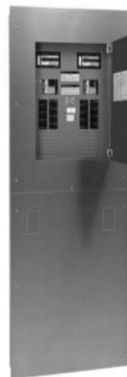


Figure 3.21: Isolated panel board reference.

3.6.5 Reading and Comparing Panel Schedule Types

Panel Type:	SURFACE TYPE	load current	27.64176136	MAIN C.B. RATING :	40
FED FROM.:	EMDB	x1.25	34.5522017	CABLE SIZE:	4 x 25 mm ² CuPVC/PVC
Panel Location:	UPS ROOM			NO.OF WAYS :	
Power System :	ISOLATION PANEL ONE			MOUNTING:	WALL MOUNTED

Notes	line no.	Description	C.B.	C.s.a	Lighting load	EMERGENCY socket load	disconnect switch HVAC	disconnect switch HEATER	Power socket	R
	1R	LIGHTING ROOM	16	3x(1x2.5)mm ²	618.75					618.75
	2R	SMALL POWER	16	3x(1x4)mm ²		1980				1980
	3R	POWER	32	3x(1x10)mm ²					3000	3000
	4R	HVAC	32	3x(1x25)mm ²			2098.125			2098.125
					Total lighting load	Normal socket load	disconnect switch HVAC	disconnect switch HEATER	Power socket	R

Figure 3.22: Isolated panel main data, breaker rating, cable size, and load categories.

Explanation: The isolated panel schedule includes the panel source, panel location, isolated power system, load current, safety factor, main circuit breaker rating, cable size, and mounting type. This information is important because the isolated panel supplies sensitive medical loads and must be clearly documented before installation.

Point	Three-phase panel	Single-phase isolated panel
Typical use	Normal, emergency, lighting, power, and distribution panels.	Operation room related critical medical loads.
Phase arrangement	Loads are distributed across R, Y, and B.	Load appears on one phase in the shown schedule.
Main check	Load balance must be less than 10%.	Correct isolated supply and protection are the main concern.
Selection output	Main C.B., cable size, number of ways, phase balance, and max demand.	Main C.B., cable size, load categories, and single-phase demand.

Panel Schedule Summary

The panel schedule section confirms how each panel is documented before installation. The three-phase schedules are used for normal and emergency distribution panels, while the isolated schedule is used for special operation-room related loads. The engineer checks source, panel type, load current, safety factor, main breaker, cable size, number of ways, mounting, max demand, diversity load, and phase balance before moving to circuit breaker and cable selection.

3.X Circuit Breaker and Cable Selection

This subsection explains the method used to select circuit breakers and cables after preparing the panel schedules. The selection depends on load current, safety factor, standard circuit breaker ratings, cable current-carrying capacity, installation method, and catalogue data.

Multicore Cables, with Stranded, Copper Conductors, XLPE Insulated and PVC Sheathed



Product Code	Nominal Cross sectional area	Maximum Conductor Resistance		Current Rating			Approx. Overall Diameter	Approx. Overall Diameter
		DC at 20 °C	AC at 90 °C	Laid in ground	Laid in duct	Laid in free air (Shaded)		
	mm²	Ω/Km	Ω/Km	A	A	A	mm	Kg/Km
4 core cables - Cu/XLPE/PVC								
CX1-T104-U04	1.5 RM	12.1	14.6	32	26	24	10.3	155
CX1-T104-U06	2.5 RM	7.41	8.87	43	33	35	11.4	210
CX1-T104-U08	4 RM	4.61	5.54	55	40	45	12.6	280
CX1-T104-U09	6 RM	3.08	3.69	70	50	55	14	385
CX1-T104-U10	10 RM	1.83	2.19	92	67	75	15.5	535
CX1-T104-U11	16 RM	1.15	1.39	118	84	98	17.9	790
CX1-T104-U12	25 RM	0.727	0.9273	155	112	131	22.1	1170
CX1-T104-U13	35 RM	0.524	0.6686	186	136	161	24.8	1545
CX1-T104-U14	50 SM	0.387	0.494	225	162	197	26.3	1970
CX1-T104-U15	70 SM	0.268	0.3425	276	204	249	30.4	2825
CX1-T104-U16	95 SM	0.193	0.2471	330	243	303	33.1	3825
CX1-T104-U17	120 SM	0.153	0.1964	374	282	352	37.2	4785
CX1-T104-U18	150 SM	0.124	0.1597	421	321	405	41.7	5875
CX1-T104-U19	185 SM	0.0991	0.1284	475	369	467	46.7	7395
CX1-T104-U20	240 SM	0.0754	0.0988	551	431	554	52.5	9620
CX1-T104-U30	300 SM	0.0601	0.0799	621	493	636	58.1	11995
CX1-T104-U40	400 SM	0.047	0.0641	706	571	741	66.7	15480
CX1-T104-U50	500 SM	0.0366	0.0518	797	653	851	74.2	19815

CB & CABLE SIZING

Load Current

The beginning of the calculations is depend on the load current, load current can calculated by $I_b = \frac{kVA}{\sqrt{3} \times V_{LL}}$

CB rating

after calculate the load current ,calculation of the I_n (nominal current of circuit breaker) by $I_n = I_b \times 1.25$
then choose the standard rating of the circuit breaker as shown:

Circuit Breaker Ratings																										
10	16	20	25	32	40	50	63	80	100	125	160	200	250	400	630	800	1000	1250	1600	2000	2500	3200	4000	5000	6300	C (A)
MCB											ACB															
MCCB																										

Figure CB-1: Elsewedy cable catalogue reference and circuit breaker standard rating method.

Explanation: Cable sizes are selected from the Elsewedy cable catalogue according to current-carrying capacity and installation conditions. After calculating the load current, a safety factor of 1.25 is applied to select the breaker nominal current. The nearest suitable standard rating is then selected as MCB, MCCB or ACB depending on the feeder size.

Selection item	Design meaning
Load current	Calculated from the connected kVA and system voltage.
Breaker rating	$I_n = I_b \times 1.25$, then the nearest standard CB rating is selected.
Cable selection	Cable capacity must coordinate with the protection device and the design current.
Selection rule	Cable capacity > breaker rating > design load current.

Equipment Sizing Prototype

Transformer, Generator and UPS sizing section to be inserted before the Single Line Diagram

Equipment Sizing Overview

Equipment sizing is prepared after load estimation, panel schedules, circuit breaker selection, cable selection and cutting list preparation. This section confirms the main electrical equipment ratings before preparing the Single Line Diagram. The selected ratings are based on the calculated kVA load, design margin, standard equipment sizes, main circuit breaker rating and feeder cable size.

Equipment	Design role	Selected size	Purpose
Transformer	Normal supply source	500 kVA	Supplies MDB-HO16 and normal distribution panels
Generator	Emergency supply source	250 kVA	Supplies emergency loads through ATS
UPS-1	Critical supply source	20 kVA	Supplies selected critical/isolated loads
UPS-2	Critical supply source	36 kVA	Supplies additional critical/isolated loads

Transformer Sizing

	Transformer reference: the transformer is the normal supply source feeding the MDB and downstream normal distribution panels. Its rating is selected from the calculated demand, loading criterion, standard size, main ACB and feeder cable.
Item	Value
Calculated load	387.9525 kVA
Design loading criterion	80% loading
Required transformer size	484.940625 kVA
Selected standard transformer	500 kVA
Main CB calculation	$I_b = 757.58 \text{ A}$, $1.25 \times I_b = 946.97 \text{ A}$
Selected main CB	1000 AT ACB
Selected cable	3 runs $[(3 \times 240 \text{ mm}^2 + 120 \text{ mm}^2) + 120 \text{ mm}^2]$, Cu XLPE/PVC

Transformer (Dry Type) - A1	
MDB IN KVA	387.9525
Elec fire pump(KVA)	0
ACCORDING TO NEMA 80%	484.940625
Standard(KVA)	500
T.R (C.B) ACB	
I_b	757.5757576
$1.25 \times I_b/c.b$	946.969697
C.B Ratings AIR	1000 AT
Cable	3 Runs of $[(3 \times 240 \text{ mm}^2 + 120 \text{ mm}^2) + 120 \text{ mm}^2]$, Cu, XLPE/PVC

Figure ES.1: Transformer reference and sizing calculation.

Explanation: The transformer was sized from the total MDB demand load. The calculated demand was divided by the 80% loading criterion to avoid operating the transformer at full capacity continuously. The nearest standard rating above the required value was selected as 500 kVA. The main ACB and feeder cable were then selected according to the calculated current and the project cable schedule.

Generator Sizing



Generator is selected to supply emergency loads during normal power failure. It feeds emergency distribution boards through the ATS, keeping essential hospital services operating until the main supply returns.

Item	Value
Static load	68.16585 kVA
Miscellaneous load	33.75 kVA
Total calculated load	169.5823125 kVA
Design loading criterion	80% loading
Required generator size	211.9778906 kVA
Selected standard generator	250 kVA
Main CB calculation	$I_b = 378.79 \text{ A}$, $1.25 \times I_b = 473.48 \text{ A}$
Selected main CB	500 AT MCCB
Selected cable	$(3 \times 300 \text{ mm}^2 + 150 \text{ mm}^2) + 150 \text{ mm}^2$, Cu XLPE/PVC

Generator (OPEN Type)	
Static Loads (KVA)	68.16585
Miscellaneous Loads (KVA)	33.75
Total	169.5823125
80%	211.9778906
Standard (KVA)	250KVA
GEN (C.B)	
I_b	378.7878788
$1.25 \times I_b/c.b$	473.4848485
C.B Ratings AIR	500 AT
Cable	$(3 \times 300 \text{ mm}^2 + 150 \text{ mm}^2) + 150 \text{ mm}^2$, Cu, XLPE/PVC

Figure ES.2: Generator sizing calculation reference.

Explanation: The generator rating was selected from the total emergency demand load. After applying the 80% loading criterion, the required value was increased to the nearest standard size. The selected generator is 250 kVA, and the feeder breaker and cable are selected to coordinate with the generator output current and emergency load requirement.

UPS Sizing



The UPS system supplies critical loads that require continuous power without interruption. In this project, two UPS units are used instead of one UPS per operating room. The UPS units support critical/isolated loads and are backed by the generator during long utility outages.

Item	Value	
UPS-1 calculated load	12.6652 kVA	
UPS-1 required size at 80%	15.8315 kVA	
UPS-1 selected standard size	20 kVA	
UPS-1 breaker and cable	40 A MCCB, (3 x 10 + 10) + 10 mm ²	
UPS-2 calculated load	18.2436 kVA	
UPS-2 required size at 80%	22.8045 kVA	
UPS-2 selected standard size	36 kVA	
UPS-2 breaker and cable	80 A MCCB, (3 x 25 + 16) + 16 mm ²	

UPS-1 (C.B)		UPS-1	
Ib	30.3030303	Load in KVA	12.6652
1.25 * Ib/c.b	37.87878788	80%	15.8315
C.B Ratings	40 A MCCB	Standard(KVA)	20
Cable	(3×10 mm ² + 10 mm ²) + 10 mm ² , Cu, XLPE/PVC		
UPS -2 (C.B)		UPS-2	
Ib	54.54545455	Load in KVA	18.2436
1.25 * Ib/c.b	68.18181818	80%	22.8045
C.B Ratings	80 A MCCB	Standard(KVA)	36
Cable	(3×25 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC		

Figure ES.3: UPS sizing calculation reference.

Explanation: The UPS units were sized from the critical load groups. Each UPS calculated load was divided by the 80% loading criterion, then the nearest standard UPS rating was selected. This arrangement keeps sensitive critical loads supplied during power interruption and allows the generator to support the UPS system for longer operation.

Equipment sizing position in the design track: This section must be placed after shop drawing design checks and before the Single Line Diagram. The selected transformer, generator and UPS ratings become the main supply values that appear later in the SLD, voltage drop checks and short-circuit verification.

3.X Cutting List and SLD Preparation

This subsection is placed after equipment sizing because the feeder list, cable schedule and source hierarchy are used directly before preparing the Single Line Diagram.

Load Draft Classification

HO16											
NORMAL				EMERGENCY				UPS			
Location	SORCE	Name	Load(KVA)	LOCATION	SOURCE	Name	Load(KVA)	LOCATION	SOURCE	NAME	LOAD(KVA)
HO16	16/(TRANS)	MDb-HO16	387.9525 KVA	HO16	GENERATOR	EMDB-HO16	127.394KVA	UPS-ROOM	UMDB-HO15	ISO-H015-O1	6.0812
HO16	16/(MDB-HO16)	SMDB-HO16	331.7025 KVA	HO16	EMDB-HO16	ESMDB-HO16	37.25825KVA	UPS-ROOM	UMDB-HO15	ISO-H015-O2	6.584
HO16	SMDB-HO16	DSP-HO16	55KVA	HO16	ESMDB-HO16	UPP-HO16	9.216KVA	UPS-ROOM	UMDB-HO15	ISO-H015-O3	6.0712
HO16	SMDB-HO16	ACP-HO16	48.697KVA	HO16	ESMDB-HO16	ELP-HO16	6.172675KVA	UPS-ROOM	UMDB-HO15	ISO-H015-O4	6.0612
HO16	SMDB-HO16	PP-HO16	33.648KVA	HO16	EMDB-HO16	EMCC-HO16	33.75KVA	UPS-ROOM	UMDB-HO15	ISO-H015-O5	6.1
HO16	SMDB-HO16	LP-HO16	3.6KVA	HO15	ESMDB-HO16	EPP-H015	10.608KVA				
HO16	16/(MDB-HO16)	MCC-HO16	45KVA	HO15	ESMDB-HO16	ELP-HO15	3.81KVA				
HO15	SMDB-HO16	PP-HO15	19.536KVA	HO16	EMCC-HO16	ELV-HO16-7	9KVA				
HO15	SMDB-HO16	LP-HO15	3.811KVA	HO16	EMCC-HO16	ELV-HO16-5	9KVA				
HO15	SMDB-HO16	ACP-HO15	39.87	HO16	EMCC-HO16	ELV-HO16-3	9KVA				
HO15	SMDB-HO16	DIS-HO15	25.2 KVA	HO16	ESMDB-HO16	UMDB-HO15	30.9076KVA				
HO16	MCC-HO16	ELV-HO16-6	9KVA								
HO16	MCC-HO16	ELV-HO16-4	9KVA								
HO16	MCC-HO16	ELV-HO16-2	9KVA								
HO16	MCC-HO16	ELV-HO16-1	9KVA								

Figure CL-1: Load draft summary for normal, emergency and UPS loads.

Explanation: The load draft classifies hospital panels into normal, emergency and UPS/critical supplies. Normal loads are supplied from the transformer, emergency loads are supplied from the generator, and UPS loads are supplied from the UPS system. This sheet defines the electrical hierarchy before drawing the SLD.

Cutting List and Feeder Schedule

FROM	TO	LOAD	IB	ICB	CB	CABLE	LENGTH			total legnth
							Horizontal	Vertical	Total	
16/(TRANS)	MDb-HO16	387.9525	587.8068182	734.759	630	(3x185 mm ² + 95 mm ²) + 95 mm ² , Cu, XLPE/PVC	2.6	4	6.6	11.6
16/(MDB-HO16)	SMDB-HO16	331.7025	502.5795455	628.224	400	(3x120 mm ² + 50 mm ²) + 50 mm ² PE, Cu, XLPE/PVC	3.62	4	7.62	12.62
SMDB-HO16	DSP-HO16	55	83.33333333	104.167	160	(3x50 mm ² + 25 mm ²) + 25 mm ² , Cu, XLPE/PVC	20.588	4	24.588	29.588
SMDB-HO16	ACP-HO16	48.697	73.78333333	92.2292	160	(3x35 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC	11.46	4	15.46	20.46
SMDB-HO16	PP-HO16	33.648	50.98181818	63.7273	80	(3x25 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC	19.73	4	23.73	28.73
SMDB-HO16	LP-HO16	3.6	5.454545455	6.81818	25	(3x2.5 mm ² + 2.5 mm ²) + 2.5 mm ² , Cu, XLPE/PVC	17.72	4	21.72	26.72
SMDB-HO16	PP-HO15	19.536	29.6	37	40	(3x10 mm ² + 10 mm ²) + 10 mm ² , Cu, XLPE/PVC	14.37	4	18.37	23.37
SMDB-HO16	LP-HO15	3.811	5.774242424	7.2178	16	(3x2.5 mm ² + 2.5 mm ²) + 2.5 mm ² , Cu, XLPE/PVC	15.2	4	19.2	24.2
SMDB-HO16	ACP-HO15	39.87	60.40909091	75.5114	100	(3x35 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC	16.1	4	20.1	25.1
SMDB-HO16	DIS-HO15	25.2	38.18181818	47.7273	50	(3x10 mm ² + 10 mm ²) + 10 mm ² , Cu, XLPE/PVC	17.8	4	21.8	26.8
16/(MDB-HO16)	MCC-HO16	45	68.18181818	85.2273	100	(3x25 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC	3.04	4	7.04	12.04
MCC-HO16	ELV-HO16-6	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	66.54	4	70.54	75.54
MCC-HO16	ELV-HO16-4	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	67.56	4	71.56	76.56
MCC-HO16	ELV-HO16-2	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	67.46	4	71.46	76.46
MCC-HO16	ELV-HO16-1	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	68.64	4	72.64	77.64
GENERATOR	EMDB-HO16	127.394	193.0212121	241.277	160	(3x35 mm ² + 16 mm ²) + 16 mm ² PE, Cu, XLPE/PVC	6.91	4	10.91	15.91
EMDB-HO16	ESMDB-HO16	37.25825	56.45189394	70.5649	80	(3x25 mm ² + 16 mm ²) + 16 mm ² , Cu, XLPE/PVC	6.26	4	10.26	15.26
ESMDB-HO16	UPP-HO16	9.216	13.96363636	17.4545	25	(3x2.5 mm ² + 2.5 mm ²) + 2.5 mm ² , Cu, XLPE/PVC	21.23	4	25.23	30.23
ESMDB-HO16	ELP-HO16	6.172675	9.352537879	11.6907	25	(3x2.5 mm ² + 2.5 mm ²) + 2.5 mm ² , Cu, XLPE/PVC	21.76	4	25.76	30.76
ESMDB-HO16	EPP-HO15	10.608	16.07272727	20.0909	25	(3x2.5 mm ² + 2.5 mm ²) + 2.5 mm ² , Cu, XLPE/PVC	15.1	4	19.1	24.1
ESMDB-HO16	ELP-HO15	3.81	5.772727273	7.21591	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	15.59	4	19.59	24.59
EMDB-HO16	EMCC-HO16	33.75	51.13636364	63.9205	80	(3x16 mm ² + 16 mm ²) + 16 mm ² PE, Cu, XLPE/PVC	4.05	4	8.05	13.05
EMCC-HO16	ELV-HO16-7	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	49.844	4	53.844	58.844
EMCC-HO16	ELV-HO16-5	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	49.234	4	53.234	58.234
EMCC-HO16	ELV-HO16-3	9	13.63636364	17.0455	25	(3 x 25 mm ² + 16 mm ²), Cu, XLPE/PVC	48.484	4	52.484	57.484
SMDB-HO16	UMDB-HO15	30.9076	46.82969697	58.5371	50	(3x10 mm ² + 10 mm ²) + 10 mm ² , Cu, XLPE/PVC	72.26	4	76.26	81.26
UMDB-HO16	ISO-H015-O1	6.0812	9.213939394	11.5174	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	62.845	4	66.845	71.845
UMDB-HO16	ISO-H015-O2	6.584	9.975757576	12.4697	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	64.315	4	68.315	73.315
UMDB-HO16	ISO-H015-O3	6.0712	9.198787879	11.4985	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	60.3	4	64.3	69.3
UMDB-HO16	ISO-H015-O4	6.0612	9.183636364	11.4795	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	61.6	4	65.6	70.6
UMDB-HO16	ISO-H015-O5	6.1	9.242424242	11.553	16	(3x10 mm ² + 10 mm ²) + 10 mm ² PE, Cu, XLPE/PVC	58.5	4	62.5	67.5

Figure CL-2: Cutting list showing feeder source, destination, breaker, cable and length.

Explanation: The cutting list converts the load draft into practical feeders. Each feeder includes the source panel, destination panel, connected load, design current, selected circuit breaker, selected cable size and route length. It is used for execution, quantity estimation, voltage drop checks and final SLD preparation.

Item	Purpose in SLD preparation
FROM / TO	Defines the source and destination panel used in the SLD feeder path.
LOAD / Ib	Shows connected load and calculated load current for each feeder.
CB	Shows the selected standard circuit breaker rating.
CABLE / LENGTH	Shows selected cable type, size and route length for cutting and installation.

Single Line Diagram, Voltage Drop and Short Circuit Verification

This prototype section presents the final electrical distribution verification stage after load estimation, panel schedules, circuit breaker selection, cable sizing, and cutting list preparation.

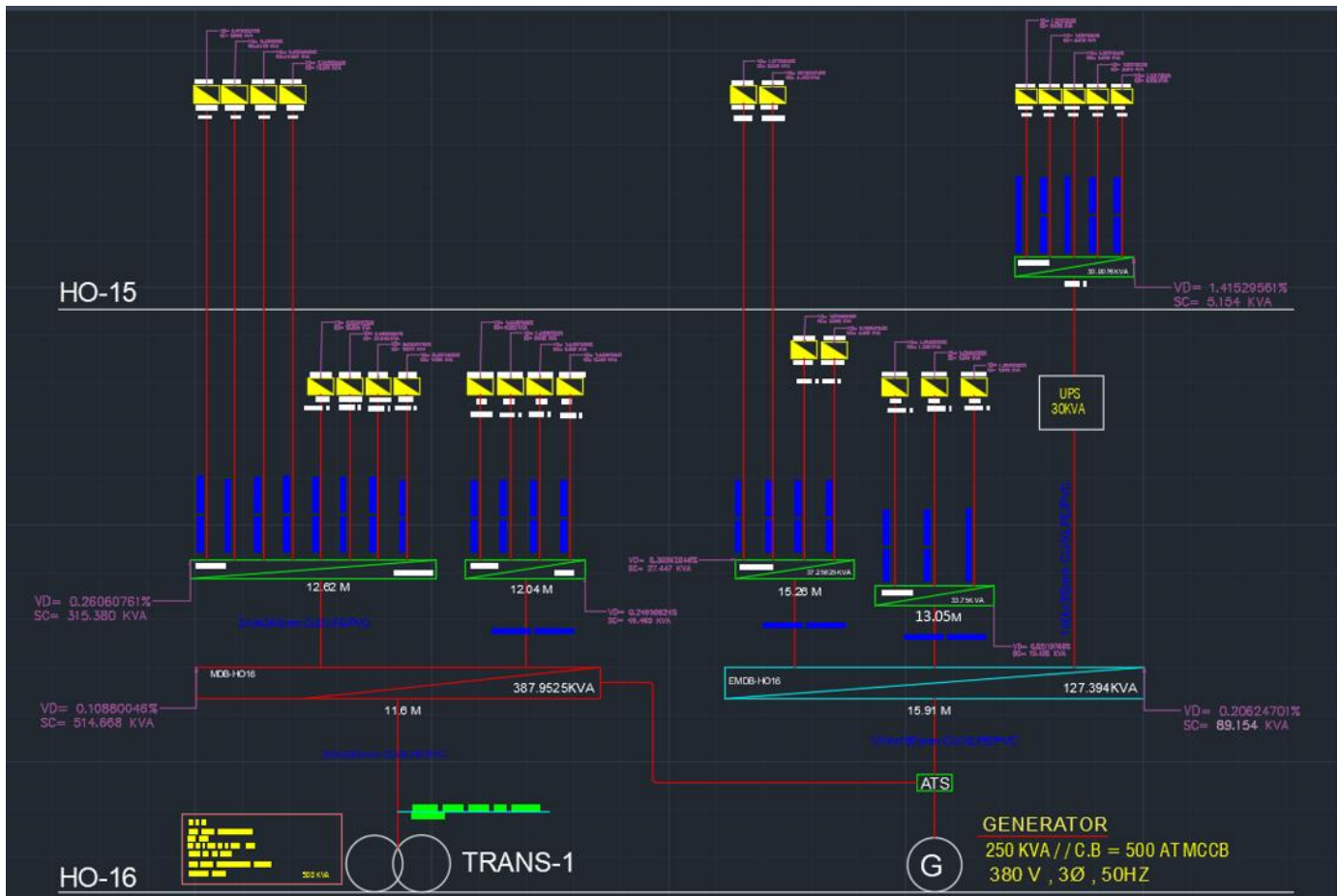


Figure X.1: Single Line Diagram for HO-16 Electrical Distribution

The single line diagram summarizes the power path of the project. It shows the normal supply from the transformer, the emergency supply from the generator through the ATS, and the critical supply through the UPS system. It also shows the main distribution boards, sub-main boards, feeder cables, circuit breaker ratings, cable lengths, and the calculated voltage drop and short-circuit values at different points.

Normal supply	Emergency supply	Critical supply
Transformer feeds normal distribution panels.	Generator feeds emergency loads through ATS.	UPS feeds critical / isolated operation room loads.

Manual Voltage Drop and Short Circuit Calculations

The following Excel calculation sheets were used to verify the selected feeder cables. The calculation checks both voltage drop acceptance and receiving-end short-circuit current based on cable data, length, current, and upstream values.

Individual Voltage Drop									
!	=	mV/A/m	x	I rated	x	L/1000	=	0.497	
		0.155	x	254.295	x	12.62			
	% V.D1	=	0.1244						
Accumulated Voltage Drop									
	=	V.D1	+	V.D on upstream (Feeding Cable)					
	=	0.124357	+	0.0978					
	=	0.22215679	%		V.D Status		Acceptance		
Short Circuit Calculation									
	1.129	Receiving End Symmetrical Fault Ampere in (KA) =						#UNKNOWN!	
	1.129	=	204.489	KA					

Figure X.2: Voltage Drop and Short Circuit Calculation Summary

In this sheet, the individual voltage drop is calculated using the selected cable voltage drop value, rated current, and route length. The accumulated voltage drop also includes the upstream feeder voltage drop. The result is then compared with the accepted voltage drop limit and marked as accepted when it is within the allowed range.

Project Name:		HO 16								
Owner:		AIET								
Consultant:		PRO MEP								
Calculation By:		ENG-KAREEM IBRAHEM								
SMDB-HO16										
Cable Information:				Panel Load Information:		Transformer Information:				
From:	MDB-HO16	Route:	Duct	Rated Voltage (V):	400	Trans. Rating in (KVA):	500			
Length (m):	12.62	Insulation:	XLPE	P.F.:	0.8	% Impedance	6%			
Phase:	3-PH	Conductor:	CU	Load	KVA	S.C Rating in (KA):	12			
Circuit Breaker Sizing:										
#UNKNOWN!				#UNKNOWN!						
				Rated Current = 254.295 A						
Safety Factor:		1.25		According To IEC-60947						
Designed Breaker Ampere:		317.87		A		MCCB				
Nearest Standard Rating C.B.:		400		AT		Selection Status				
Nearest Standard Frame C.B.:		630		AF		Correct				
Recommended Circuit Breaker:		630		AT		MCCB				
Cable Sizing:										
Duct Temp.	40	»	C1 =	0.95	Cable Conditions	De-Rating Factor (C)	=	C1xC2xC3xC4xC5		
Burial Depth	1.5	»	C2 =	0.93			=	0.945345		
Soil Thermal	100	»	C3 =	1.07						
No. of Cables	2	»	C4 =	1	Ground	Cable Ampere	=	I C.B. / C		
Spacing	0.15							=	423.126	A
Formation	Flat									
No. of Cables	2	»	C5 =	1	Air	Cable C.S.A	400	mm	Ic =	472
No. of Trays	2						Neutral Size	Same Size	=	400
Installation	H Ladder Touching					No. of Cables	=	1		
No. of Cables	1	»	C5 =	1	Duct					

Figure X.3: Circuit Breaker and Cable Sizing Calculation Sheet

This detailed sheet combines panel information, transformer data, circuit breaker sizing, derating factors, and cable sizing. The rated current is multiplied by a safety factor, then the nearest standard circuit breaker rating is selected. After that, the cable cross-section is checked according to installation conditions and derating factors.

ETAP Voltage Drop Verification

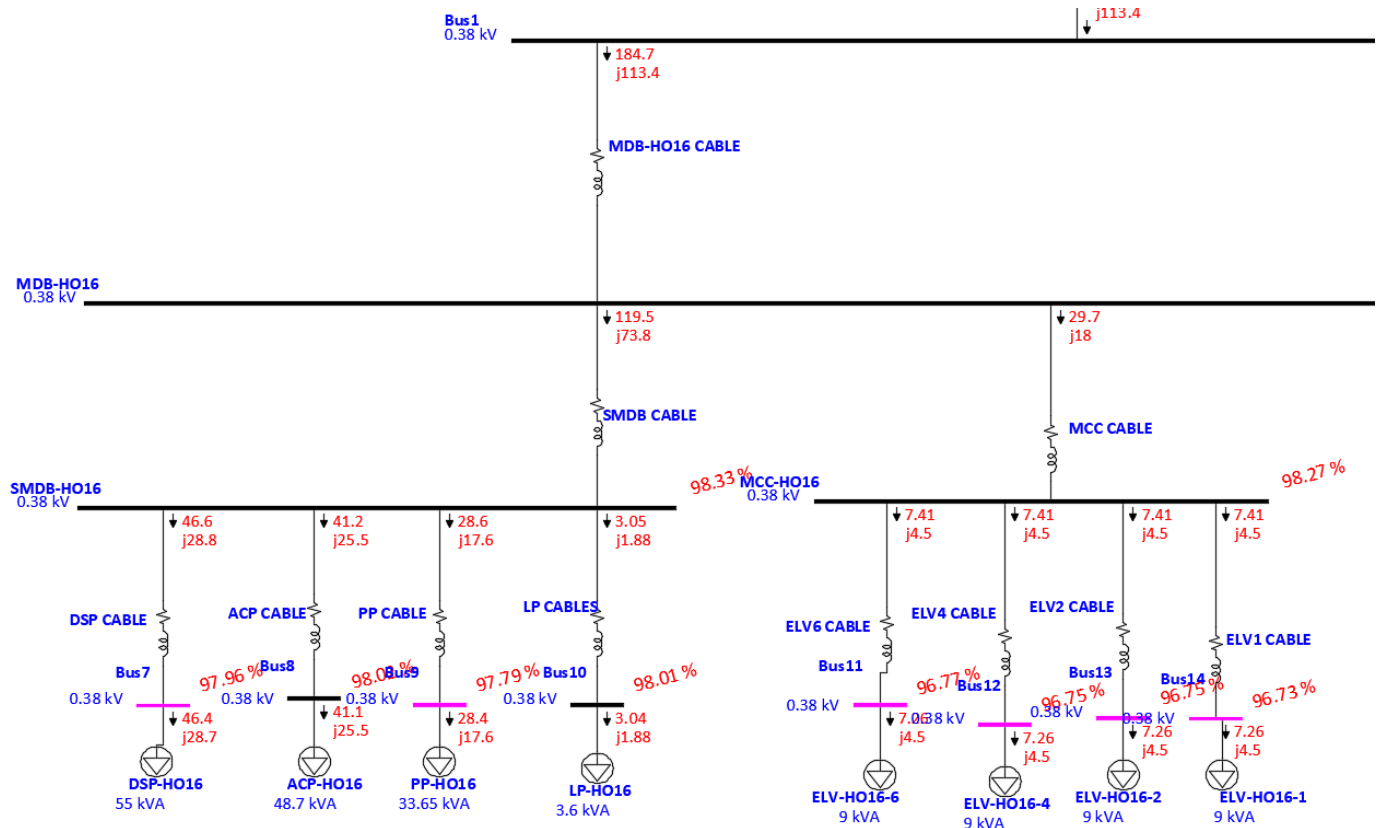


Figure X.4: ETAP Voltage Drop Verification

ETAP was used as a verification tool after completing the manual calculations. This view shows the voltage drop / voltage profile at the downstream buses and load points. The ETAP model helps confirm that the selected cables and feeders keep the voltage level within an acceptable operating range.

The ETAP results are used here as a supporting check only. The main design values were prepared manually in Excel using the selected loads, cable sizes, breaker ratings, and route lengths.

ETAP Short Circuit Verification

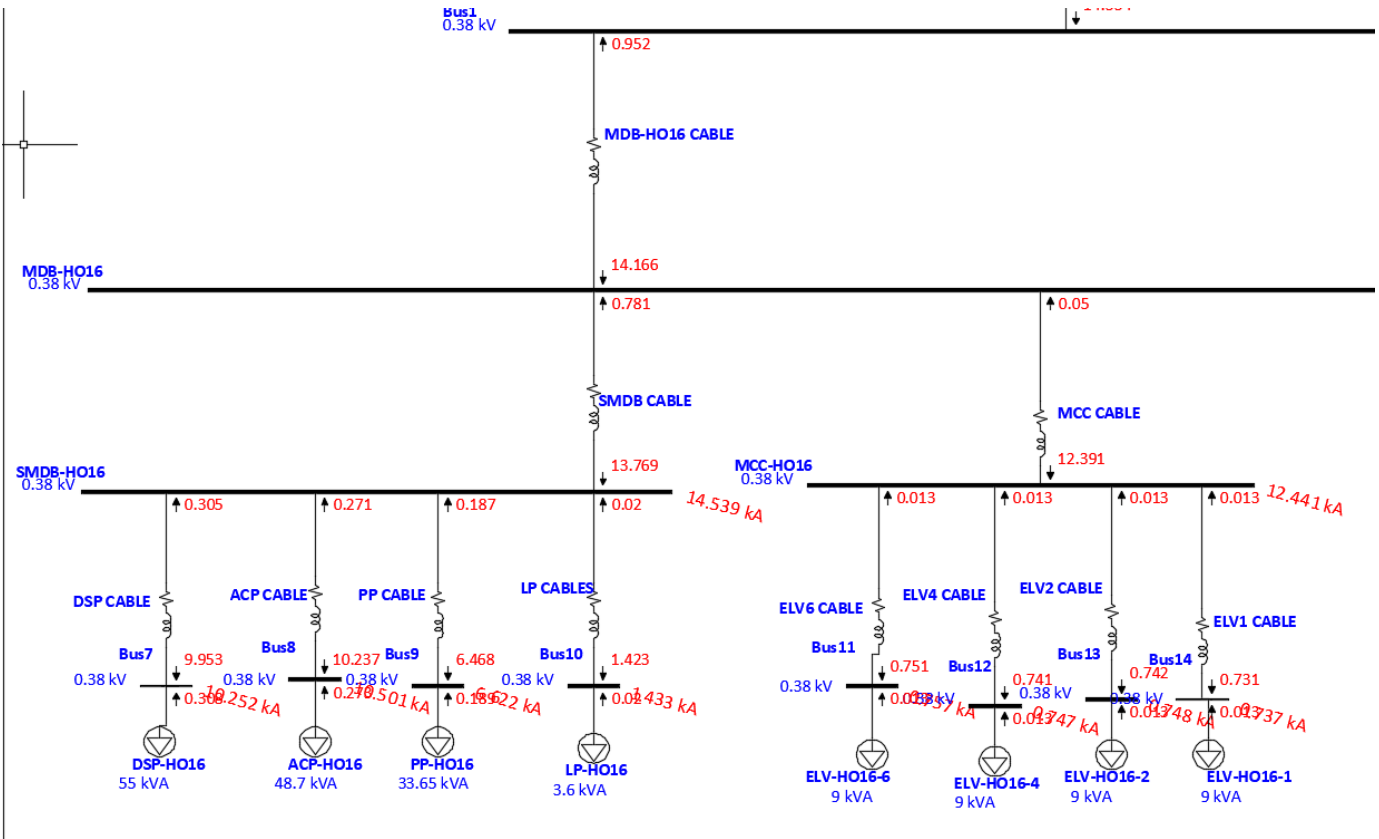


Figure X.5: ETAP Short Circuit Verification

This ETAP view shows the calculated short-circuit current at the main and sub-main buses. These values are used to verify that the selected circuit breakers have suitable breaking capacity and that the system can safely clear faults.

The short-circuit verification is important because the circuit breaker must not only carry the load current during normal operation, but must also interrupt fault current safely. Therefore, the selected breaker type and rating are checked against the expected fault level at each distribution point.

Purpose	Input basis	Final output
Verify voltage drop and short-circuit performance.	Load draft, cutting list, breaker ratings, cable sizes, and lengths.	Confirmed SLD values and accepted feeder selections.

Lighting Shop Drawing Prototype

This prototype shows the proposed layout for the Lighting Shop Drawing subsection only. The section explains the meaning of shop drawings first, then presents the lighting execution drawing and the hatch mark legend used to read the wiring marks on the drawing.

1. What is a Shop Drawing?

Shop drawing is the practical execution drawing prepared after the design stage. It converts the design layout into an installation drawing that can be used on site by engineers and technicians. It shows actual routes, wiring paths, circuit references, installation notes, and coordination with the ceiling, walls, rooms, and cable containment.

For lighting systems, the shop drawing explains how the luminaires, switches, emergency lighting points, and circuit routes will be installed. It is not only a design drawing; it is a site coordination drawing used to connect the lighting symbols with real wiring routes and installation details.

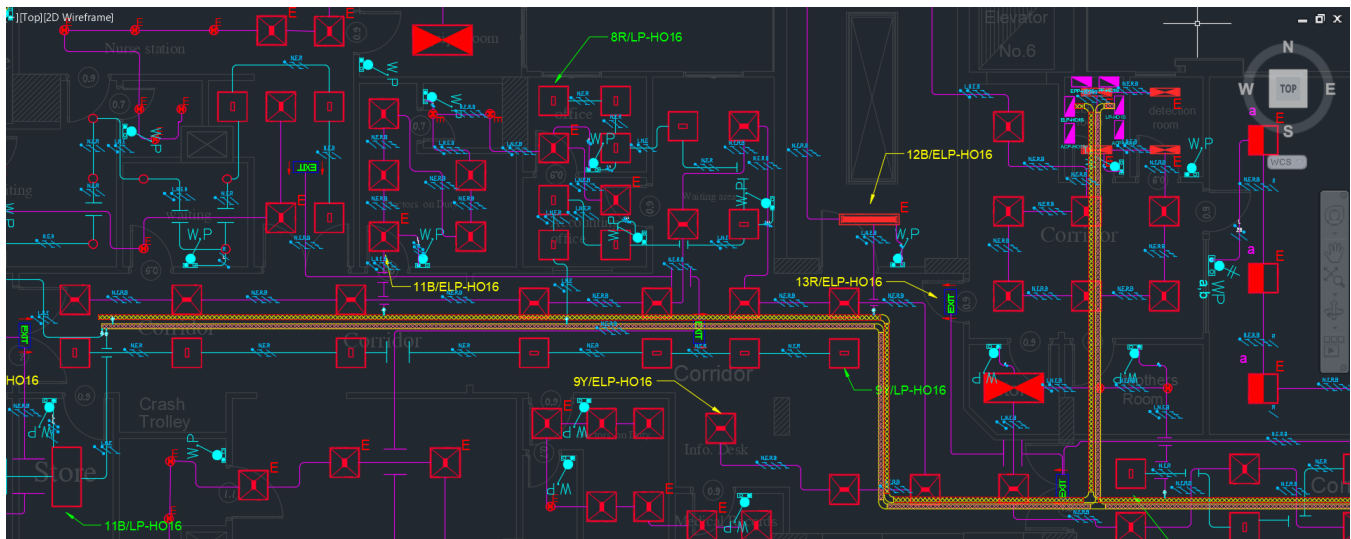


Figure SD-1: Lighting shop drawing full layout.

Explanation: This figure shows the lighting shop drawing layout for the hospital floor. The drawing includes lighting points, switching routes, emergency/critical lighting references, circuit numbers, and installation routes. It is used to guide the site installation and to coordinate the lighting system with architectural spaces and ceiling services.

2. Lighting Hatch Mark Legend

Hatch marks are small symbols placed on the lighting route to identify the wires running in the conduit or path. They make the shop drawing readable without writing long wiring notes on every route. In lighting drawings, hatch marks help the installer know whether the route contains line, neutral, earth, return, or common wires.

LIGHTING HATCH MARK LEGEND			How the hatch mark legend is read
TYPE	SYMBOL	DESCRIPTION	
L		LINE	L = Line: the live supply wire feeding the lighting circuit. N = Neutral: the neutral return path required to complete the circuit. E = Earth: the protective earthing conductor for safety. R = Return: the switched return wire from the switch to the lighting point. C = Common: the common terminal/wire used mainly in switching arrangements.
N		NEUTRAL	
E		EARTH	
R		RETURN	
C		COMMON	

Figure SD-2: Lighting hatch mark legend.

Explanation: The hatch mark legend defines the wire marks used in the lighting shop drawing. These marks are placed along the route to indicate the type of conductor inside the conduit. Using this legend reduces drawing clutter and helps the site team install the correct wiring for each lighting circuit.

3. Lighting Shop Drawing Reading Sequence

Step	Meaning
1. Check lighting symbols	Read the luminaires, switches, emergency points, and circuit references from the legend.
2. Follow the route	Trace the wiring/conduit path from the panel or switch to the lighting points.
3. Read hatch marks	Use L, N, E, R, and C marks to identify the wires included in each route.
4. Coordinate on site	Install according to ceiling layout, walls, rooms, and service coordination.

Prototype note: this subsection should be placed at the beginning of the Shop Drawing track before Power Shop Drawing and HVAC Shop Drawing. The lighting shop drawing is shown first, followed by the hatch mark legend so the reader can understand how the wiring routes are executed on site.

Power Shop Drawing

Shop drawing is the practical execution drawing used on site after finishing the design calculations. It does not only show the location of sockets or electrical loads; it also shows the installation route, conduit/tray path, cable direction, circuit reference, and the practical coordination between the electrical layout and the architectural spaces.

For the power system, the shop drawing explains how socket circuits, UPS outlets, TV/service points, and dedicated power points are routed from the panel to the final load points. It helps the site engineer install the conduits, pull the correct cables, and check that the routing matches the voltage drop and filling ratio calculations.

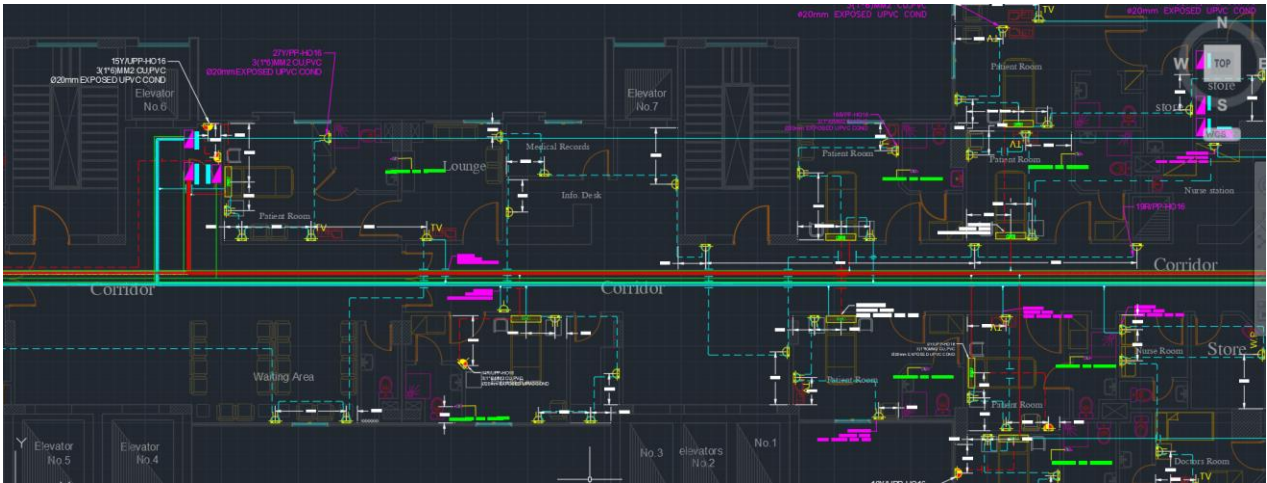


Figure PSD-1: Power shop drawing layout and power routing.

Explanation: This figure shows the AutoCAD power shop drawing. The drawing includes the power circuit routing, socket/service points, conduit or tray paths, and circuit references. It is used during site execution to know how each power point is connected back to its panel and how the route passes through corridors and rooms. The shop drawing is prepared after the design layout and after the main panel/cable selections are known.

Shop drawing item	Purpose in power installation
Circuit route	Shows the practical path from panel to load point.
Circuit reference	Identifies which circuit feeds each socket or power point.
Conduit / tray path	Guides the installation route used by site workers.
Coordination	Avoids conflicts with architecture, false ceiling and other services.

Voltage Drop and Filling Ratio in Power Shop Drawing

After defining the power routing, the route length is transferred to the voltage drop and filling ratio sheet. Voltage drop checks that the cable length and load current will not cause an excessive voltage reduction at the final socket or power load. Filling ratio checks that the selected conduit size can safely contain the required number of cables without overcrowding.

$\%V.D = \frac{(2 \times R \times I \times L) \times 100}{(V^2) \times 1000} \rightarrow ((\leq 2\%))$ In case of single phase										$\%V.D = \frac{(\sqrt{3} \times R \times I \times L) \times 100}{(V^2) \times 1000} \rightarrow ((\leq 3\%))$ In case of three phase										$A(T) = A1 + A2 + A3 + \dots = \left(\frac{\pi}{4}\right) D^2 = 0.79 D^2$ where : A1- 1 cable in pipe A2- 2 cable in pipe A3-3 cable in pipe A(T)- total C.S.A of cable inside pipe D-outside diameter of cable from catalogue max occupancy ratio : 3C=53%/3 3C=31%/3 3C=40%/3									
voltage drop calculation										filling ratio calculation																			
UDBS-4	conductor	load-VA	D.LOAD-VA	voltage-V	current-A	length-m	R-ohm/km	%V.D	acceptance	no of cable in pipe	OD-mm	A1-mm²	A2-mm²	A3-mm²	A(T)-mm²	conduit-mm²	%fill ratio	results	acceptance	conduit size									
1R	3(1X6) MM2	1800.00	1620.00	220.00	7.36	81.00	3.68	2.00	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
2R	3(1X6) MM2	1800.00	1620.00	220.00	7.36	76.51	3.68	1.88	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
3V	3(1X6) MM2	1620.00	1458.00	220.00	6.63	73.47	3.68	1.63	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
4R	3(1X6) MM2	1800.00	1620.00	220.00	7.36	75.38	3.68	1.86	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
5R	3(1X6) MM2	1260.00	1134.00	220.00	5.15	78.20	3.68	1.35	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
6R	3(1X6) MM2	1260.00	1134.00	220.00	5.15	77.92	3.68	1.34	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
7R	3(1X6) MM2	360.00	324.00	220.00	1.47	68.98	3.68	0.34	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
8R	3(1X6) MM2	360.00	324.00	220.00	1.47	62.61	3.68	0.31	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
9V	3(1X6) MM2	360.00	324.00	220.00	1.47	65.60	3.68	0.32	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
10R	3(1X6) MM2	360.00	324.00	220.00	1.47	59.41	3.68	0.29	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
11R	3(1X6) MM2	1500.00	1350.00	220.00	6.14	59.76	3.68	1.23	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
12R	3(1X6) MM2	1500.00	1350.00	220.00	6.14	60.79	3.68	1.25	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
13R	3(1X6) MM2	360.00	324.00	220.00	1.47	63.90	3.68	0.31	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
14R	3(1X6) MM2	1260.00	1134.00	220.00	5.15	77.40	3.68	1.33	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										
15V	3(1X6) MM2	1620.00	1458.00	220.00	6.63	78.28	3.68	1.74	✓	3	4.40	15.29	0.00	0.00	45.88	352.81	13.01	✓	25MM										

Figure PSD-2: Voltage drop and filling ratio calculation sheet for power circuits.

Explanation: The left side of the sheet calculates voltage drop using circuit number, conductor size, load, voltage, current, route length and cable resistance. The result is compared with the accepted limit and marked as accepted when it is within the required range. The right side calculates the filling ratio by comparing the total cable area inside the conduit with the conduit area. The selected conduit size is accepted when the fill ratio is within the allowed limit.

Check	How it is calculated	Why it is needed
Voltage drop	Uses cable resistance, load current and route length.	Confirms that the final load receives acceptable voltage.
Filling ratio	Compares total cable area with conduit internal area.	Confirms that conduit size is suitable for the installed cables.
Acceptance	Both VD and filling ratio must satisfy design limits.	Prevents overheating, installation difficulty and poor performance.

Power shop drawing sequence: Design layout -> panel/circuit selection -> routing on AutoCAD -> route length extraction -> voltage drop check -> filling ratio check -> final approved installation route.

3.X HVAC Shop Drawing

A shop drawing is the practical execution drawing used on site after finishing the design stage. It does not only show the load locations, but also shows the real installation routes, conduit paths, cable routing, cable sizes, circuit references, and coordination with ceilings, corridors, rooms, and architectural elements.

For the HVAC power system, the shop drawing explains how the electrical supply is routed from the panel to each air-conditioning unit or disconnect switch. The drawing must match the panel schedule, circuit breaker selection, cable sizing, voltage drop calculation, and conduit filling ratio calculation.

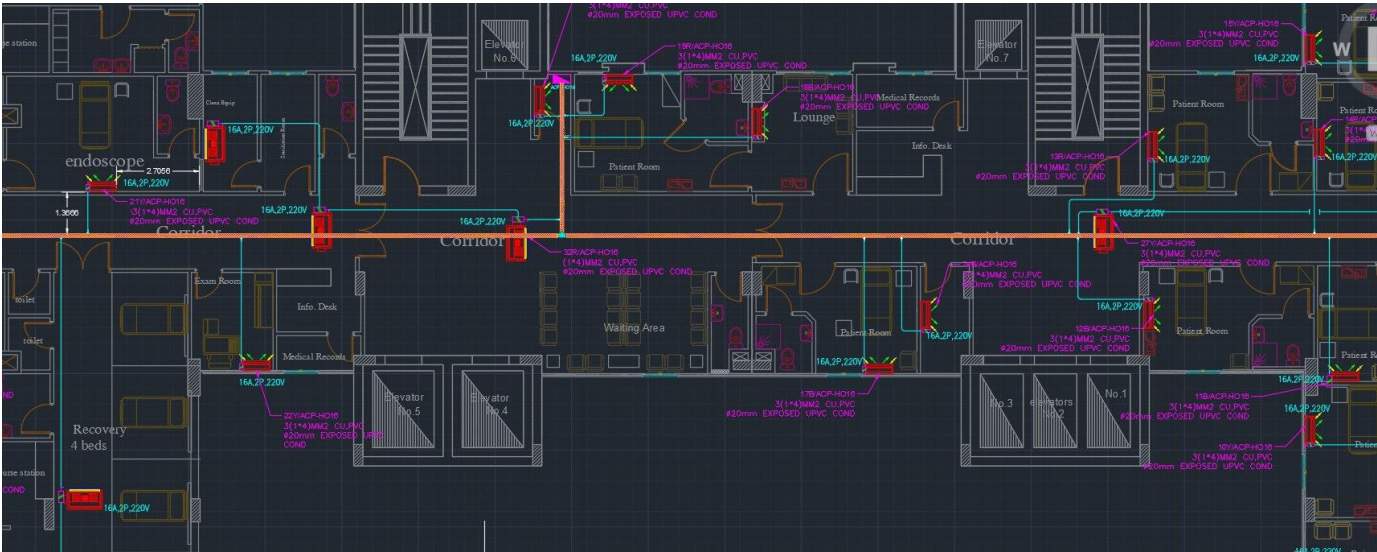


Figure SD-HVAC.1: HVAC power shop drawing layout.

Explanation: This figure shows the HVAC power shop drawing route. The plan includes the locations of HVAC electrical points, disconnect switches, circuit labels, conduit routes, cable references, and the main path of installation. It is used by the site team to understand the actual routing and installation method, not only the design load positions.

Voltage Drop and Filling Ratio Calculations

The HVAC shop drawing is supported by two main checks: voltage drop and filling ratio. Voltage drop confirms that the selected cable size and route length will not cause excessive voltage loss at the end of the circuit. Filling ratio confirms that the selected conduit size is suitable for the number and cross-sectional area of cables inside it.

$\%V.D = \frac{(2 \times R \times I \times L) + 100}{(V) \times 1000} \rightarrow ((< 2\%))$ <i>in case of single phase</i>										$\%V.D = \frac{(\sqrt{3} \times R \times I \times L) \times 100}{(V) \times 1000} \rightarrow ((< 3\%))$ <i>in case of three phase</i>										$A(t) = A1 + A2 + A3 + \dots \quad A = \left(\frac{\pi}{4}\right) D^2 = 0.79 D^2$ where : A1- 1 cable in pipe A2- 2 cable in pipe A3- 3 cable in pipe A(T)= total C.S.A of cable inside pipe D= outside diameter of cable from catalogue max occupancy ratio : 1C=53%fill , 2C=31%fill 3C+=40%fill									
voltage drop calculation										filling ratio calculation																			
circuit		load			cable		results			circuit	c.s.a of cables				c.s.a of pipes		results												
circuit no	conductor	load-VA	D.LOAD -VA	voltage -V	current -A	length -mt	R-ohm /km	%V.D	acceptance	no of cable in pipe	OD-mm	A1-mm²	A2-mm²	A3-mm²	A(t)-mm²	conduit -mm²	%fill ratio	acceptance	conduit size										
UDB-S-4										UDB-S-4																			
1R	3(1*6)MM2	2098.1	1888.29	220.00	8.58	58.87	3.60	1.65	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
2R	3(1*6)MM2	2098.1	1888.29	220.00	8.58	58.39	3.60	1.64	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
3Y	3(1*6)MM2	2098.1	1888.29	220.00	8.58	52.49	3.60	1.47	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
4Y	3(1*6)MM2	2098.1	1888.29	220.00	8.58	59.10	3.60	1.66	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
5B	3(1*6)MM2	2098.1	1888.29	220.00	8.58	40.98	3.60	1.15	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
6B	3(1*6)MM2	2098.1	1888.29	220.00	8.58	48.73	3.60	1.37	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
7R	3(1X4) MM2	1398.80	1258.92	220.00	5.72	60.79	5.51	1.74	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
8R	3(1X4) MM2	1398.80	1258.92	220.00	5.72	59.83	5.51	1.71	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
9Y	3(1X4) MM2	1398.80	1258.92	220.00	5.72	52.60	5.51	1.51	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
10Y	3(1X4) MM2	1398.80	1258.92	220.00	5.72	45.70	5.51	1.31	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
11B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	38.80	5.51	1.11	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
12B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	30.30	5.51	0.87	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
13B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	31.04	5.51	0.89	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
14R	3(1X4) MM2	1398.80	1258.92	220.00	5.72	36.20	5.51	1.04	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
15Y	3(1X4) MM2	1398.80	1258.92	220.00	5.72	53.41	5.51	1.53	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
16Y	3(1X4) MM2	1398.80	1258.92	220.00	5.72	64.95	5.51	1.86	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
17B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	23.23	5.51	0.67	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
18B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	12.30	5.51	0.35	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										
19B	3(1X4) MM2	1398.80	1258.92	220.00	5.72	7.70	5.51	0.22	✓	3	3.90	12.02	0.00	0.00	36.05	211.13	17.07	✓	20mm										

Figure SD-HVAC.2: HVAC voltage drop and filling ratio calculation sheet.

Explanation: The left side of the sheet checks voltage drop for each HVAC circuit by using the load current, cable resistance, cable length, and supply voltage. The result is compared with the accepted limit, and accepted circuits are marked in the acceptance column. The right side checks the filling ratio by comparing the total cable area with the available conduit area, then selecting a suitable conduit size such as 20 mm or 25 mm depending on the circuit.

Check	Purpose
Voltage drop check	Verifies that cable length and current do not reduce the voltage beyond the accepted limit.
Filling ratio check	Verifies that the conduit has enough space for the number and size of cables.
Conduit size	Selected after checking the cable area and the maximum allowed occupancy ratio.

Cable and Conduit Reference

The calculation sheet depends on catalogue data. The cable catalogue provides the cable cross-sectional area, conductor resistance, and current rating. The conduit reference provides the standard conduit sizes and the heavy gauge conduit data used to select the final conduit diameter.

WIRES

Single Core Cables with Solid or Stranded Copper Conductors and PVC Insulated



Description

- Soft annealed solid or stranded Copper conductors insulated with PVC compound rated 70 °C or 85 °C according to IEC 60227 & BS 6004.

Application

- For indoor fixed installations in dry locations, laid in conduits, as well as in steel support brackets.

Product - code	Nominal Cross Sectional Area mm²	Max. Conductor Resistance		Current Rating in Air			Approx. Overall Diameter mm	Approx. Weight kg/km
		DC at 20 °C Ω/km	AC at 70 °C Ω/km	Free A	In Pipes A			
CPD-5001-U04	1.5 mm	12.1000	14.6000	17	13		2.8	20
CPD-1001-U04	1.5 mm	12.1000	14.6000	17	13		3.0	21
CPD-5001-U05	2 mm	9.1300	10.9000	19	15		3.2	27
CPD-1001-U05	2 mm	9.1300	10.9000	19	15		3.4	28
CPD-5001-U06	2.5 mm	7.4100	8.8900	24	19		3.4	31
CPD-1001-U06	2.5 mm	7.4100	8.8900	24	19		3.6	32
CPD-5001-U07	3 mm	6.1000	7.4100	27	21		3.6	37
CPD-1001-U07	3 mm	6.1000	7.4100	27	21		3.8	39
CPD-5001-U08	4 mm	4.6100	5.5100	32	23		3.9	47
CPD-1001-U08	4 mm	4.6100	5.5100	32	23		4.2	50
CPD-5001-U09	6 mm	3.0800	3.6800	40	29		4.4	68
CPD-1001-U09	6 mm	3.0800	3.6800	40	29		4.7	71

Item Code	Size mm	Wall Thickness mm	Packing m
NGCH20	20	1.80	100m
NGCH20	20	2.1	100m
NGCH25	25	1.90	80m
NGCH25	25	2.2	80m
NGCH32	32	2.50	45m
NGCH40	40	2.80	30m
NGCH50	50	3.40	15m

HEAVY GAUGE



$$2+3.6+(0.4*4*2)$$

Figure SD-HVAC.3: Cable catalogue and heavy gauge conduit reference.

Explanation: This reference is used to read the cable properties and select the conduit size. For example, the selected cable size is checked using catalogue values such as conductor resistance and current capacity, while the conduit size is selected from the standard heavy gauge conduit table. These values are then transferred to the voltage drop and filling ratio calculation sheet.

Cable Tray Sizing

Placed after HVAC Shop Drawing and before Light Current Systems

Cable Tray Sizing Method

Cable tray sizing is prepared after completing lighting, power and HVAC shop drawings because the final cable routes and cable quantities become clear at this stage. The tray size is selected according to the total cable cross-sectional area, the allowable filling ratio, spare capacity for future additions, and available standard tray dimensions.

Sizing item	Design meaning
Sum of C.S.A	Total cable cross-sectional area inside the tray.
Filling ratio	Maximum allowed tray occupation; 40% was used in the sizing sheet.
20% spare	Additional reserve for future cables and safer installation.
Selected tray	The calculated width is rounded up to the nearest standard tray size.

PANELS		NORMAL HOMERUN TRAY					
		CABLE	D-CABLE	R-CABLE	C.sa of CABLE	No.of CABLES	Total C.sa
NORMAL WIRING	LIGHTING NORMAL	3x(1*2.5)mm25	11.5	5.75	103.81625	26	2699.2225
SUM OF C.S.A							2699.2225
F.R							0.4
Area of TRAY							6748.05625
Area of TRAY with 20% spare							8435.070313
Height							50
Width							168.7014063
TRAY							200*50*1.5

Figure CT.1: Normal homerun cable tray sizing calculation.

Explanation: This sheet shows how the normal homerun tray size was calculated. The lighting normal cables are counted, their cross-sectional areas are summed, and the filling ratio is applied. After adding 20% spare capacity, the required tray width is rounded to a practical standard size. In this example, the selected tray is 200 x 50 x 1.5 mm.

Standard Tray and Cable References

After calculating the required tray area, the final tray is chosen from the standard cable tray dimensions catalogue. The cable catalogue is also used to confirm the cable outside diameter and resistance values used in the sizing sheet.

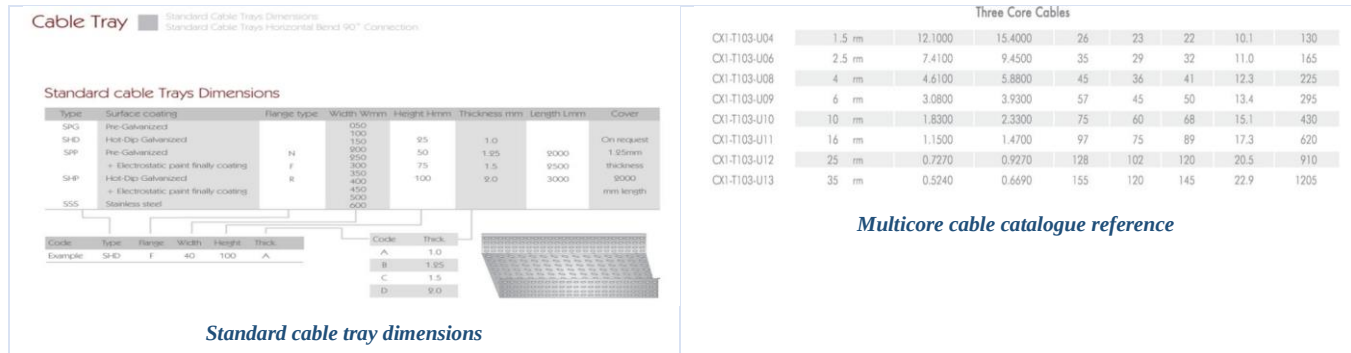


Figure CT.2: Cable tray and cable catalogue references.

Explanation: The standard tray table is used to select the nearest available tray size according to the required width, height and thickness. The cable catalogue provides data such as conductor size, resistance and cable outside diameter, which are needed to calculate occupied tray area and verify cable routing.

Design Decision

Step	What is checked	Result in this design
1	Count cables routed in the tray	Normal lighting homerun cables are collected.
2	Calculate total C.S.A and apply filling ratio	Required tray area is obtained with 40% filling ratio.
3	Add spare capacity	20% spare is added for future expansion.
4	Select nearest standard tray	200 x 50 x 1.5 mm tray is selected.

Location in report: This section should be placed after the HVAC Shop Drawing section and before the Light Current systems, because cable tray sizing depends on the finalized electrical routes and comes before starting the low-current design documentation.

Light Current Systems

Introduction to Light Current Systems

After completing the main electrical distribution design, shop drawings, cable tray sizing, and final power coordination, the report moves to the Light Current Systems section. Light current systems are low-voltage systems used for communication, safety, security, monitoring, control, and hospital support services.

Unlike power systems, which supply electrical energy to loads such as lighting, sockets, HVAC units, panels, transformers, generators, and UPS systems, light current systems mainly transfer signals, data, alarms, audio, video, and control commands. Therefore, they require proper device distribution, cable routing, riser diagrams, rack or panel sizing, and clear shop drawing coordination.

In this project, the light current part includes Fire Alarm, Data and Telephone, CCTV, Access Control, Nurse Call, and Public Address systems. Each system is presented in a clear order: main plan, legend, riser diagram, shop drawing or installation detail, calculations when required, and the main components used in the system.

Light Current Systems Covered in This Section

System	Main Purpose
Fire Alarm	Detects fire conditions and activates alarm notification devices.
Data and Telephone	Provides structured communication outlets, WAP points and rack connections.
CCTV	Provides video surveillance and monitoring coverage.
Access Control	Controls and monitors access to selected rooms and secure areas.
Nurse Call	Allows patients to call nursing staff and indicate the calling room.
Public Address	Distributes announcements and emergency voice messages through loudspeakers.

Fire Alarm System

1.1 System Overview

The fire alarm system is a light-current life-safety system used to detect fire conditions at an early stage and notify occupants using audible and visual alarm devices. In this project, an addressable fire alarm system was used. Each detector, manual call point, module, or notification/control device is identified by a unique address, allowing the Fire Alarm Control Panel (FACP) to locate alarms and faults accurately.

The floor is divided into two fire alarm zones/loops. This zoning is used in the shop drawing to split the floor into two clear areas, reduce routing complexity, and make alarm location, maintenance, and fault tracing easier. Each loop is checked separately for voltage drop to confirm that the devices receive acceptable 24 VDC supply voltage.

1.2 Main Plan



Figure FA.1: Fire alarm main plan.

Explanation: The main plan shows the distribution of fire alarm devices across the hospital floor. It includes smoke detectors, above-ceiling detectors, multi detectors, heat detectors, manual call points, sounders, strobes, door holders, control modules, monitor modules, and the main Fire Alarm Control Panel. The plan is the base drawing used before preparing the riser diagram, shop drawing, and loop calculations.

1.3 Fire Alarm Legend

FIRE ALARM LEGEND		
TYPE	SYMBOLS	DESCRIPTION
		FIRE ALARM CONTROL PANEL
F1		ADDRESSABLE SMOKE SENSOR DETECTOR
F2		ABOVE CEILING ADDRESSABLE SMOKE SENSOR DETECTOR
F3		ADDRESSABLE MULTI DETECTOR
F4		ADDRESSABLE HEAT DETECTOR HEAT RAISE TYPE
F5		addressable sarnie
F6		ADDRESSABLE CONTROL MODULE
F7		ADDRESSABLE MONITOR MODULE
F8		DOOR HOLDER
F9		STROBE LIGHT WITH CANDELA RATING AS INDICATED IN DRAWING.
F9		ADDRESSABLE INTELLIGENT FIRE ALARM MANUAL

1. ALL DIMENSIONS ARE IN METER UNLESS OTHERWISE INDICATED.

2. PULL BOXES ARE PROVIDED EVERY 15M &/OR 2# 90° BENDS

3. LOOP LINE FIRE RETARDANT CABLE 2x2.5 mm² IN Ø25mm²(EMT COND. EXPOSED/PVC COND. EMBEDDED)

4. DC LINE FIRE RETARDANT CABLE 2x2.5 mm² IN Ø25mm²(EMT COND. EXPOSED/PVC COND. EMBEDDED)

- FACP: Fire Alarm Control Panel.
- F1: Addressable smoke detector.
- F2: Above-ceiling addressable smoke detector.
- F3: Addressable multi detector.
- F4: Addressable heat detector.
- F5: Addressable sounder / bell.
- F6: Addressable control module.
- F7: Addressable monitor module.
- F8: Door holder.
- F9: Strobe light.
- F10: Intelligent manual call point.

Figure FA.2: Fire alarm legend and symbol reference.

Explanation: The legend explains the symbols used in the fire alarm drawings. It is placed before the technical drawings so the reader can identify each device symbol on the main plan and shop drawing. The notes also define installation requirements such as loop fire-retardant cable and DC line cable routes in exposed PVC or embedded conduit.

1.4 Riser Diagram

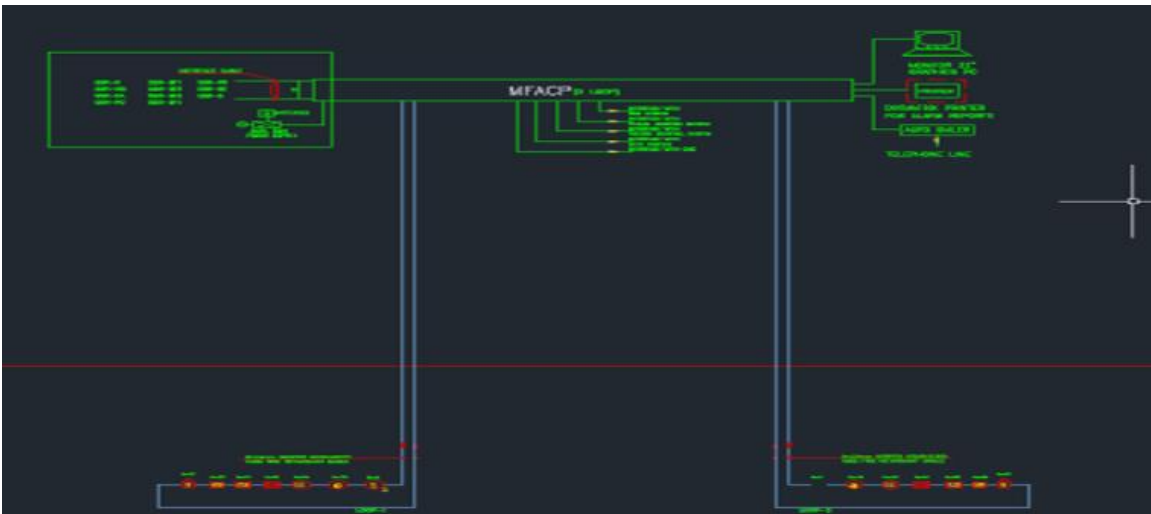


Figure FA.3: Fire alarm riser diagram.

Explanation: The riser diagram shows how the Fire Alarm Control Panel is connected vertically to the field loops, notification appliance circuits, graphic annunciator, and remote annunciator. It clarifies the relationship between the FACP, loop cards, field devices, and monitoring points through the building floors.

1.5 Shop Drawing and Installation Detail

The fire alarm shop drawing is the practical installation drawing used on site. It shows the actual routing of loop cables, device positions, conduit paths, control/monitor module locations, and the relation between the design plan and installation method. The shop drawing is divided into two zones/loops per floor to organize the circuit routes and simplify fault location during maintenance.

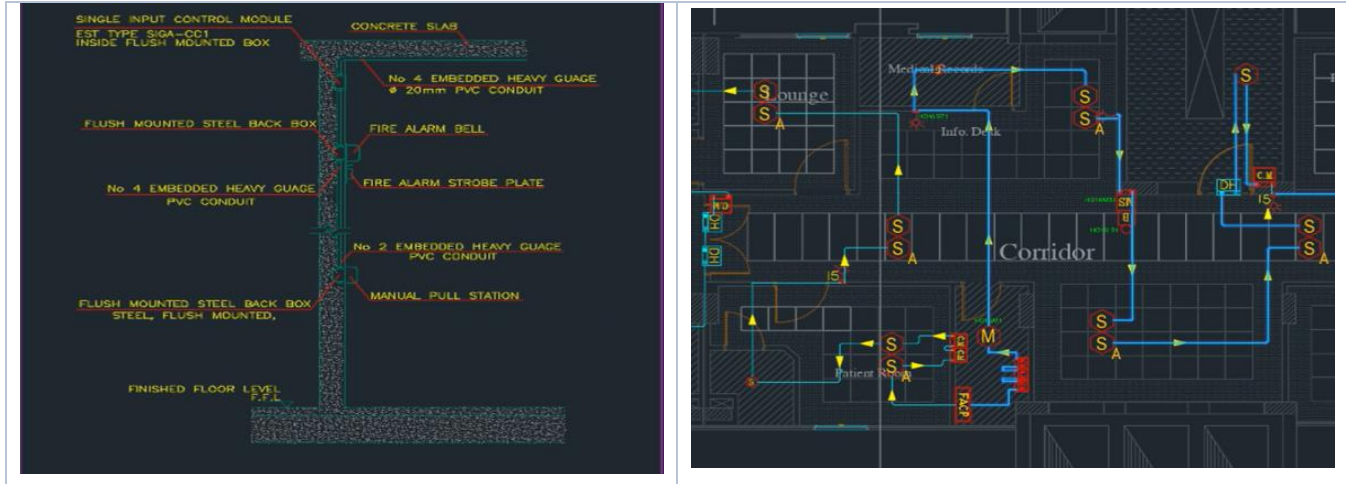


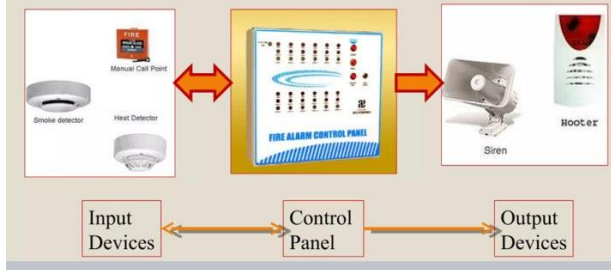
Figure FA.4: Fire alarm installation detail and shop drawing route.

Explanation: The installation detail shows typical mounting and conduit connections for devices such as manual pull station, fire alarm bell, fire alarm strobe plate, and input/control modules. The shop drawing detail shows the practical route of the loop cable between devices, including direction arrows, device locations, and connection back to the FACP. This drawing is used by the site team for installation, coordination, and final approval.

1.6 System Operation and Addressable Concept

The fire alarm system follows a simple operation sequence: input devices detect fire or are manually activated, the signal is sent to the FACP, and the panel activates output devices such as sounders, bells, horns, and strobes. The panel also supervises loops, monitors faults, and identifies alarm locations.

Block diagram of FDA system

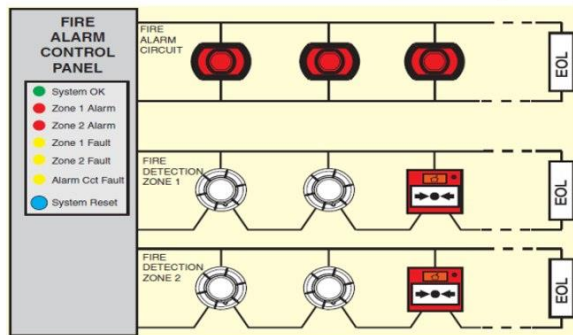


Block diagram of fire alarm system

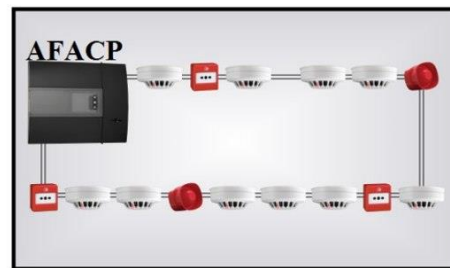


Main fire alarm control panel

من المجموعة إشارة إلى لوحة التحكم للإنذار بوجود حريق دون تحديد رقم أو عنوان الكاشف الذي أكتشف الحريق.



Conventional fire alarm concept



- يتم توصيف لوحة الحريق من النوع المعنون طبقاً لعدد Loop والتي يمكن أن تصل إلى 10 Loop للوحة الواحدة لدى بعض الشركات.

Addressable fire alarm loop concept

Figure FA.5: Fire alarm operation, control panel, conventional system and addressable loop concept.

Conventional System

- Devices are grouped by zones.
- The panel usually identifies the alarm zone only.
- Fault tracing is slower because the exact device is not directly identified.

Addressable System Used in This Project

- Each device has a unique address.
- The panel identifies the exact device or location.
- Maintenance and troubleshooting are faster, and the system is more flexible.

1.7 Voltage Drop Calculation for Fire Alarm Loops

V.D Loop 1							
DEVICES	STANDBY CURRENT FOR DEVICE	NO.	TOTAL STANDBY CURRENT	CABLE SIZE	CABLE RESISTANCE / KM	LOOP LENGTH (KM)	RESISTANCE (OHM)
SMOKE DETECTOR	0.000500	90	0.0450000	2*1.5 mm2	12.1	0.19	2.32
MULTI DETECTOR	0.000050	1	0.0000500				
CONTROL MODULE	0.000800	1	0.0008000				
MONITOR MODULE	0.000550	3	0.0016500				
BELL	0.000000	4	0.0000000				
STROBE LIGHT	0.000000	11	0.0000000				
DOOR HOLDER	0.000020	1	0.0000200				
RESISTANCE (OHM) x 2	TOT STAND BY CURRNT FOR ALL DEVICES	VOLTAGE DROP CALCULATIONS	OPERATING VDC	V.D%			
4.64	0.05	0.22	24.00	0.92%			

V.D Loop 2							
DEVICES	STANDBY CURRENT FOR DEVICE	NO.	TOTAL STANDBY CURRENT	CABLE SIZE	CABLE RESISTANCE / KM	LOOP LENGTH (KM)	RESISTANCE (OHM)
SMOKE DETECTOR	0.000500	83	0.0415000	2*1.5 mm2	12.1	0.23	2.77
MULTI DETECTOR	0.000050	1	0.0000500				
CONTROL MODULE	0.000800	9	0.0072000				
MONITOR MODULE	0.000550	0	0.0000000				
BELL	0.000000	4	0.0000000				
STROBE LIGHT	0.000000	8	0.0000000				
DOOR HOLDER	0.000020	10	0.0002000				
RESISTANCE (OHM) x 2	TOT STAND BY CURRNT FOR ALL DEVICES	VOLTAGE DROP CALCULATIONS	OPERATING VDC	V.D%			
5.54	0.05	0.27	24.00	1.13%			

Figure FA.6: Voltage drop calculation for fire alarm Loop 1 and Loop 2.

Explanation: The voltage drop calculation verifies that the selected fire alarm loop cable can supply all connected devices without excessive voltage loss. The calculation uses the standby current of each device, the number of devices, cable resistance per kilometer, loop length, and operating voltage. The two loops shown in the sheet correspond to the two fire alarm zones/loops used in the floor shop drawing.

Calculation Step			Meaning	
1. Total standby current			Standby current per device x number of devices. This gives the total current carried by the loop during normal monitoring.	
2. Cable resistance			Cable resistance per km x loop length. For the used 2 x 1.5 mm2 cable, the sheet uses 12.1 ohm/km.	
3. Two-way loop path			The resistance value is multiplied by 2 because the current path includes outgoing and return conductors.	
4. Voltage drop percentage			$V.D\% = (2 \times \text{loop resistance} \times \text{total current} / \text{operating VDC}) \times 100$.	
5. Acceptance			The loop is accepted when the voltage drop is within the allowed limit and the final device still receives suitable 24 VDC supply.	
Loop / Zone	Cable	Length	Resistance	V.D Result
Loop 1 / Zone 1	2 x 1.5 mm2	0.19 km	2.32 ohm	0.92% Accepted
Loop 2 / Zone 2	2 x 1.5 mm2	0.23 km	2.77 ohm	1.13% Accepted

1.8 Main Fire Alarm Components

The following components represent the main devices used in the fire alarm system. They are grouped in one compact page to document the system equipment without wasting many pages.

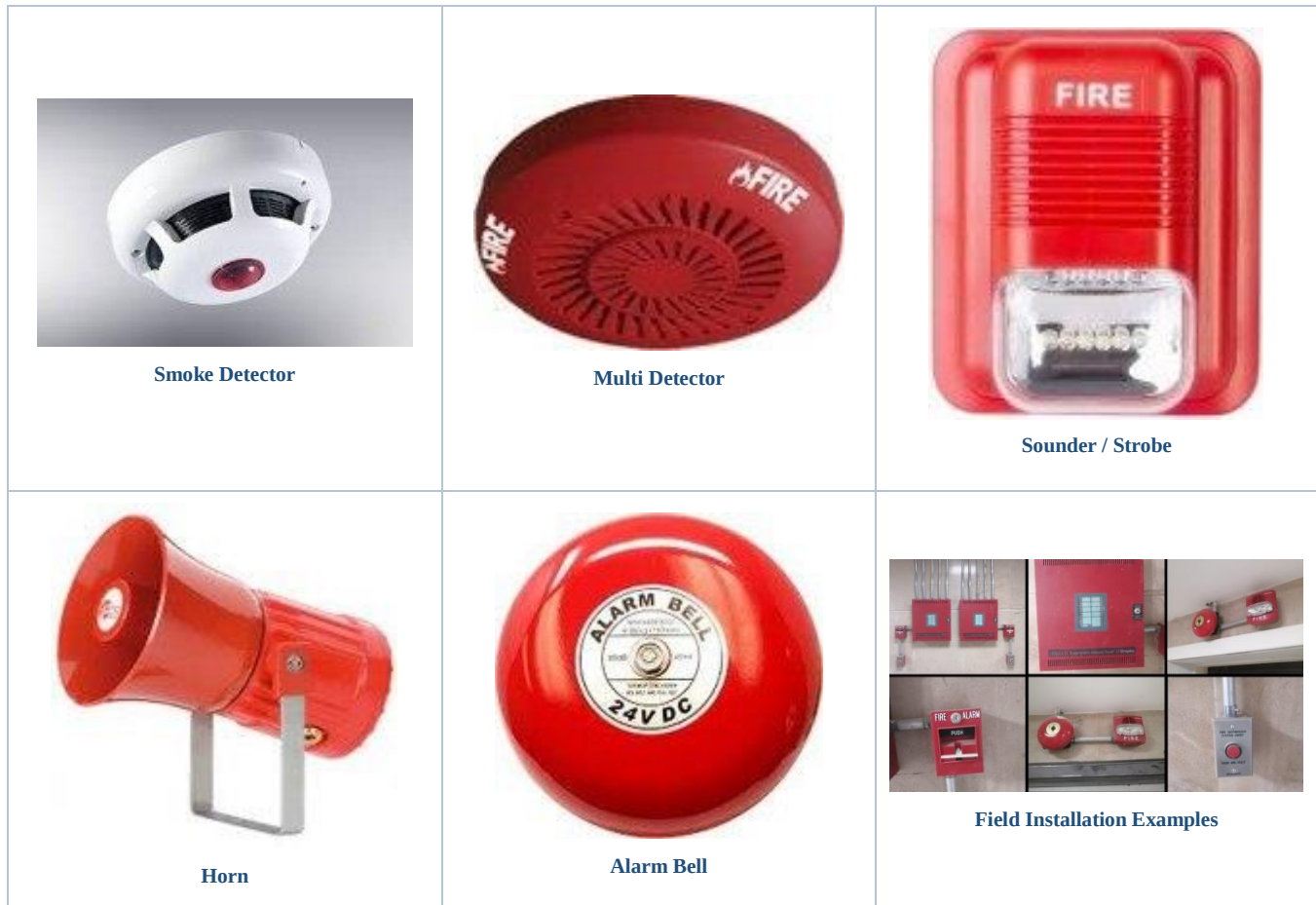


Figure FA.7: Main fire alarm system components.

Component	Function
Smoke Detector	Detects smoke particles and sends an addressable alarm signal to the FACP.
Multi Detector	Combines more than one sensing principle, usually smoke and heat, to improve detection reliability.
Sounder / Strobe	Provides audible and visual alarm indication during fire alarm condition.
Horn / Bell	Provides strong audible warning to alert occupants and support evacuation.
Manual Call Point	Allows manual activation of the fire alarm system by occupants.
Control / Monitor Modules	Control external devices and monitor auxiliary signals such as door holders or other interfaces.

Summary: The fire alarm system in this project uses an addressable architecture, two zones/loops per floor, clear shop drawing routes, and loop voltage-drop verification. This arrangement provides accurate alarm location, easier maintenance, and safer evacuation support inside the hospital.

Data and Telephone System

1. System Overview

The Data and Telephone System is part of the hospital light-current systems. It provides communication and network infrastructure for data outlets, telephone outlets, wireless access points, TV/service points, floor boxes, and bed head unit points. The design is based on a structured cabling system where all outlets are routed back to the communication rack using CAT6/RJ45 connections.

2. Main Plan

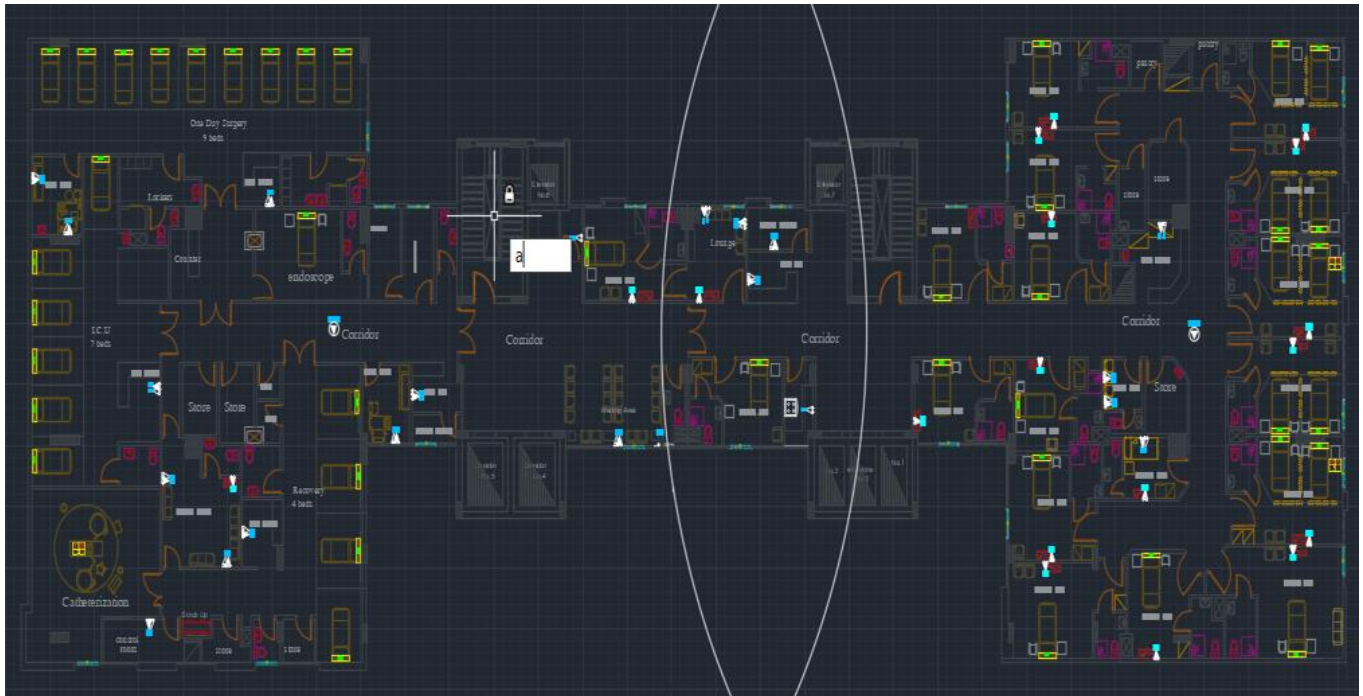


Figure LC-DT.1: Data and telephone main plan.

This drawing shows the distribution of data and telephone outlets across the hospital floor. It includes wall-mounted telephone points, single data outlets, WAP points, TV/data service points, floor boxes, and BHU communication points. The plan is the main reference before reading the legend, riser, rack sizing and shop drawing details.

3. Legend and Symbol Interpretation

ACCESS POINT (WIFI)

STRUCTURED CABLING SYSTEM SERIES LEGEND

SYMBOL	DESCRIPTION
	RJ 45 IP TELEPHONE OUTLET , WALL MOUNTED TYPE AT (1200 mm HEIGHT).
	RJ 45 SINGLE DATA OUTLET, WALL MOUNTED TYPE.
	RJ 45 DATA OUTLET, WALL MOUNTED TYPE, FOR IP TV SYSTEM IN COORDINATION WITH TV SETS.
	Service Port: 1x 10/100/1000 Mbps RJ45 (PoE) DATA OUTLET, CEILING MOUNTED TYPE, FOR WIRELESS RANGE 30 M
	COMMUNICATION STRUCTURED CABLING RACK. 800mm X 800mm FOR ALL TELECOM ROOMS. 1000mm X 800mm FOR SERVER ROOM.
	FLOOR BOX CONTAINS :- (DOUBLE SOCKET(NORMAL))+(DOUBLE SOCKET(UPS))+TEL.+DATA)

Figure LC-DT.2: Structured cabling system legend.

The legend defines RJ45 telephone outlets, RJ45 data outlets, TV/data points, ceiling-mounted WAP outlets with PoE service, communication racks and floor boxes. In this project, the data and telephone network uses CAT6 cabling with RJ45 outlets.

4. Riser Diagram

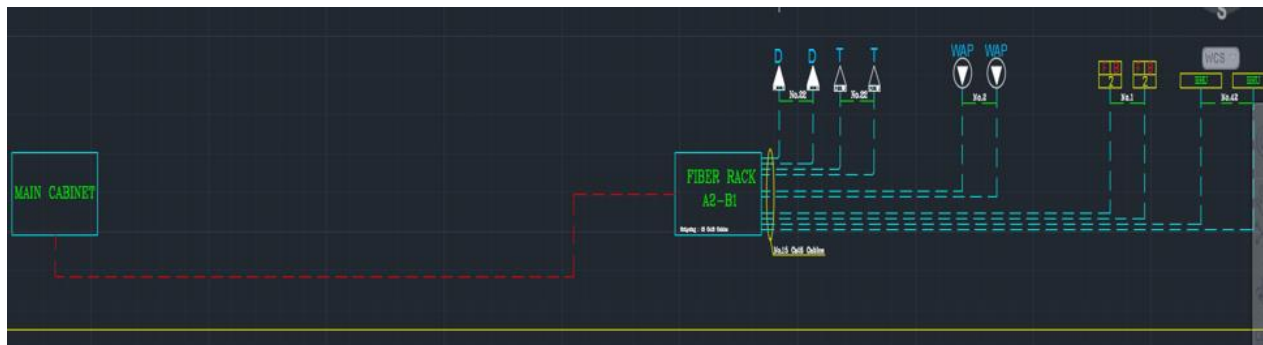


Figure LC-DT.3: Data and telephone riser diagram.

The riser diagram explains how the main cabinet is connected to the fiber rack and how the final communication points are fed from the rack. It shows the backbone relation between the rack equipment and data, telephone, WAP, TV/data points and floor boxes.

5. Shop Drawing and Installation Detail

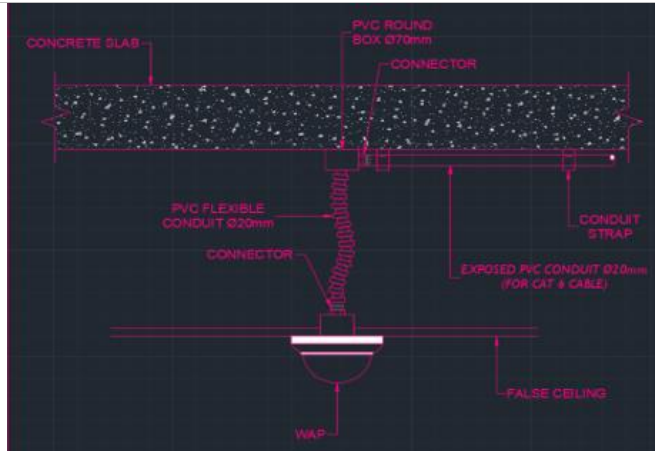


Figure LC-DT.4: WAP installation and conduit detail.



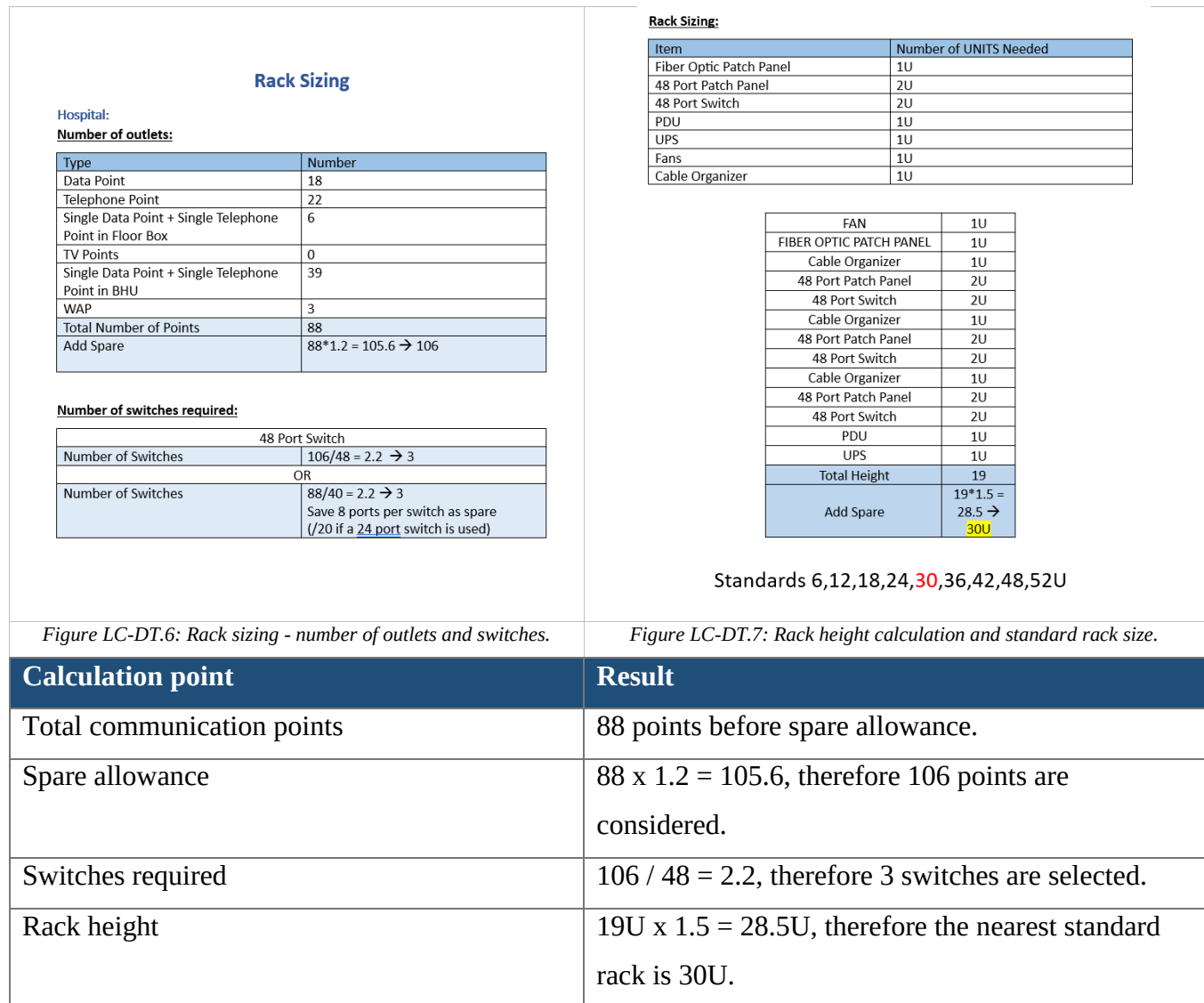
Figure LC-DT.5: Data and telephone shop drawing detail.

The shop drawing converts the design layout into practical installation routes. It shows the actual conduit paths, cable routes, cable tray coordination and outlet positions. The installation detail explains the WAP installation using PVC round box, flexible conduit, connector, exposed PVC conduit, conduit strap and false ceiling coordination.

Item	Function in shop drawing
Main route	Defines the major cable/conduit path from the rack area to the communication outlets.
Outlet drops	Show the final route down to data, telephone, TV, WAP or floor-box points.
Conduit detail	Clarifies PVC conduit, connector and ceiling coordination for site installation.
Shop output	Provides a readable site drawing for technicians before installation.

6. Rack Sizing

Rack sizing determines the number of switches and the required rack height according to the number of data, telephone, WAP, floor-box and BHU communication points.



The communication rack is similar in concept to an electrical panel, but it is used for network and telephone systems. It contains patch panels, switches, PDU, UPS, fans and cable organizers. All floor outlets are terminated and organized inside the rack for maintenance and future expansion.

7. Rack Reference and CAT6 Cabling

شكلها	Cable/Socket	مواصفات الكابل	نوع الكابل			
	CAT 3 / RJ11 sockets	كانت منتشرة في التسعينات الآن تراجعت بفضل الكابلات الأعلى كفاءة و قد تستخدم الآن في شبكات التليفون.	CAT 3		CAT6 / RJ45 sockets	الأكثر انتشارا الآن نظرا لكفاءتها.
	CAT5 / RJ45 sockets	حلت محل كابلات CAT3. لكنها الآن تقادمت نظرا لظهور الكابلات الأعلى كفاءة مثل لكابلات CAT5e.	CAT 5		CAT6a	الفرق الرئيسي بينها وبين النوع السابق هو أن CAT6a يمكن أن تعمل على تردد يصل إلى 600 ميجاهيرتز . بالإضافة إلى أنها أقل عرضة للتدخل والتداخل.
	CAT5e / RJ45 sockets	تشبه إلى حد كبير كابلات CAT5 إلا أنها تتميز عنها بأن الكابل يتكون من 4 أطراف مزدوجة على عكس CAT5 الذين يتكون من طرفين فقط، كما أن الكابلات مبرومة twisted أكثر من نظيرتها	CAT 5e			

Figure LC-DT.9: CAT cable/RJ45 references.

CAT6/RJ45 was selected for this project because it provides better performance than older CAT3, CAT5 and CAT5e references and is suitable for modern hospital data and telephone networks. CAT6 supports reliable structured cabling for data outlets, telephone points, WAP points and related communication services.

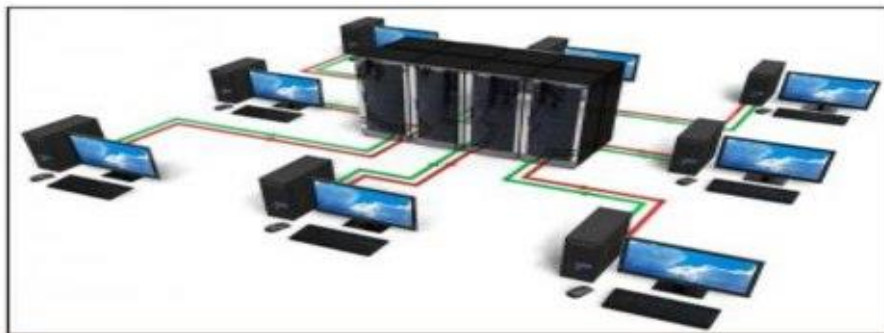


Figure LC-DT.10: Typical structured network connection concept.

8. Section Summary

Part	Design purpose
Main plan	Shows all data, telephone, WAP, TV/data and floor-box points.
Legend	Explains the symbol and service represented by each mark.
Riser	Explains connection between main cabinet, rack and final outlets.
Shop drawing	Shows real routing and installation coordination.
Rack sizing	Determines the number of switches and selected rack height.
CAT6 cabling	Defines the cable type used for structured cabling.

This section should be placed after the Fire Alarm System in the Light Current track, before CCTV, Access Control, Nurse Call and Public Address sections.

CCTV System

1. Main Plan

The CCTV system is an IP surveillance system used to monitor corridors, patient areas, entrances, lounges and important spaces. Camera locations were selected according to coverage direction, monitoring purpose and security level.

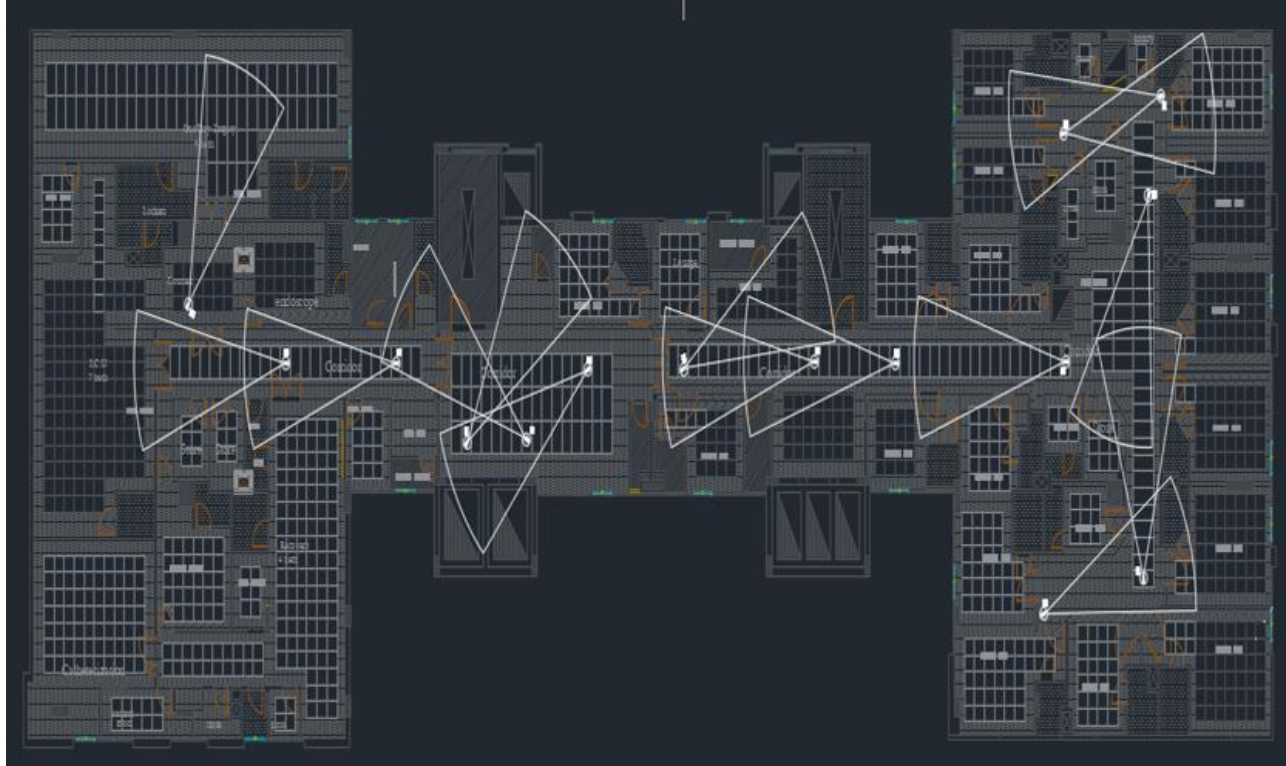


Figure CCTV-1: CCTV main plan.

Explanation: The plan shows camera locations and coverage sectors across the hospital floor. The view sectors help verify monitoring coverage before preparing the riser, shop drawing and rack sizing.

2. Legend

SECURITY SYSTEMS SERIES LEGEND	
SYMBOL	DESCRIPTION
IP . CCTV SYSTEM	
ID 	ID – Hikvision HWT-T358-Z, 5MP Turret Camera, 2.7–13.5mm Motorized Lens, 60m IR, 130dB WDR, IP66
ABBREVIATIONS	
ID	IDENTIFICATION VIEW (250 PIXEL/M)
RG	RECOGNITION VIEW (125 PIXEL/M)
DT	DETECTION or OBSERVATION VIEW (25 PIXEL/M)
SM	SURFACE MOUNTED

Figure CCTV-2: CCTV legend and abbreviations.

Explanation: The legend defines the camera symbol and abbreviations. ID = Identification View, RG = Recognition View, DT = Detection/Observation View, and SM = Surface Mounted. The selected camera is a Hikvision 5 MP turret camera with motorized lens, IR, WDR and IP66 protection

3. Riser Diagram

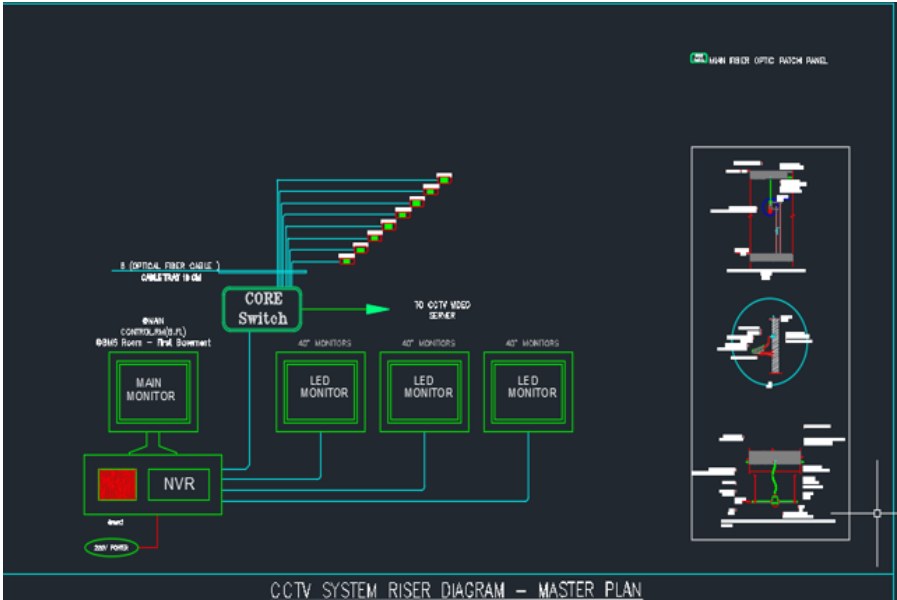


Figure CCTV-3: CCTV riser diagram.

Explanation: The riser shows camera connections to the CCTV rack, core switch, NVR and monitors. The NVR records video streams, while the monitors provide live viewing and supervision.

4. Shop Drawing Explanation

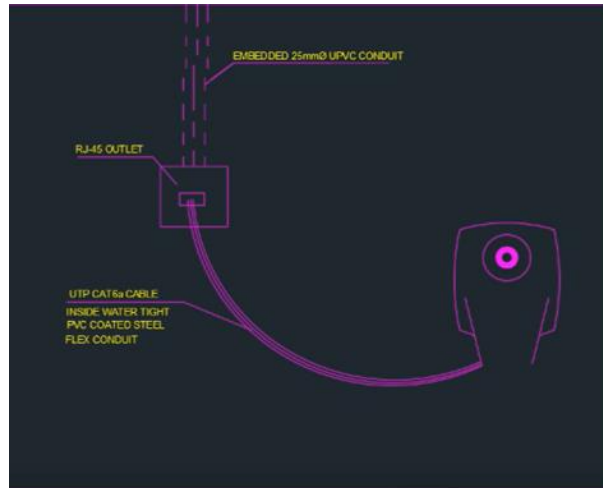


Figure CCTV-4: CCTV camera outlet installation detail.

Explanation: This detail shows the RJ45 outlet, PVC conduit, flexible conduit and camera connection. It explains the practical installation method used on site, not only the symbol location on the plan.

5. Shop Drawing



Figure CCTV-5: CCTV shop drawing routing view.

Explanation: The shop drawing converts design positions into practical cable routes. It shows camera outlet routing to the cable tray/main pathway and coordinates CCTV routes with ceiling works and other electrical systems.

6. Camera Type and Lens Check

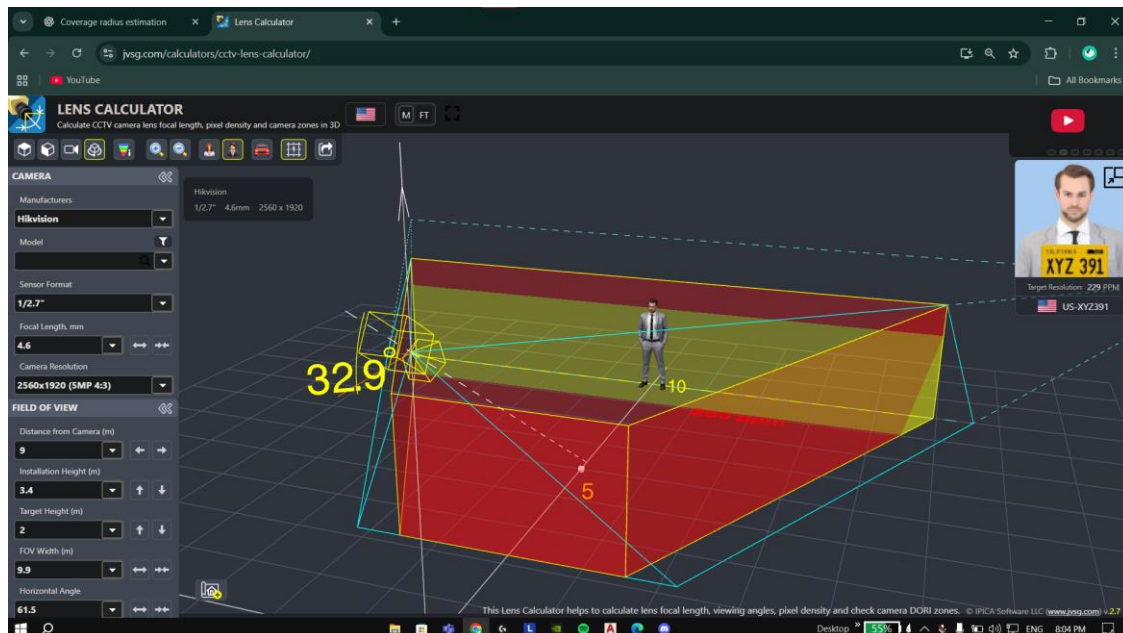


Figure CCTV-6: Camera lens calculator and ID camera reference.

Explanation: The lens calculator confirms coverage, field of view and viewing angle. In this project, the indoor camera was checked for Identification View (ID), and the shown angle reference is about 32.9 degrees.

7. Rack Sizing

◀ Rack Sizing CCTV: Number of outlets: <table> <tr> <th>Type</th><th>Number</th></tr> <tr> <td>CCTV point</td><td>37</td></tr> <tr> <td>Total Number of Points</td><td>37</td></tr> <tr> <td>Add Spare</td><td>$37 * 1.2 = 44.4 \rightarrow 45$</td></tr> </table> Number of switches required: <table> <tr> <td colspan="2">48 Port Switch</td></tr> <tr> <td>Number of Switches</td><td>$45/48 = 0.9375 \rightarrow 1$</td></tr> <tr> <td colspan="2">OR</td></tr> <tr> <td>Number of Switches</td><td>$37/40 = 0.9375 \rightarrow 1$ Save 8 ports per switch as spare (/20 if a 24 port switch is used)</td></tr> </table>		Type	Number	CCTV point	37	Total Number of Points	37	Add Spare	$37 * 1.2 = 44.4 \rightarrow 45$	48 Port Switch		Number of Switches	$45/48 = 0.9375 \rightarrow 1$	OR		Number of Switches	$37/40 = 0.9375 \rightarrow 1$ Save 8 ports per switch as spare (/20 if a 24 port switch is used)																						
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Rack Sizing: <table> <tr> <th>Item</th><th>Number of U Needed</th></tr> <tr> <td>Fiber Optic Patch Panel</td><td>1U</td></tr> <tr> <td>48 Port Patch Panel</td><td>2U</td></tr> <tr> <td>48 Port Switch</td><td>2U</td></tr> <tr> <td>PDU</td><td>1U</td></tr> <tr> <td>UPS</td><td>1U</td></tr> <tr> <td>Fans</td><td>1U</td></tr> <tr> <td>Cable Organizer</td><td>1U</td></tr> <tr> <td>NVR 48 Channel</td><td>2U</td></tr> </table> <table> <tr> <td>Fans</td><td>1U</td></tr> <tr> <td>Fiber Optic Patch Panel</td><td>1U</td></tr> <tr> <td>Cable Organizer</td><td>1U</td></tr> <tr> <td>48 Port Patch Panel</td><td>2U</td></tr> <tr> <td>48 Port Switch</td><td>2U</td></tr> <tr> <td>NVR 48 Channel</td><td>2U</td></tr> <tr> <td>UPS</td><td>1U</td></tr> <tr> <td>PDU</td><td>1U</td></tr> <tr> <td>Total Height</td><td>16</td></tr> <tr> <td>Add Spare</td><td>$16 * 1.5 = 24 \rightarrow 24$</td></tr> </table> Standards 6,12,18,24,30,36,42,48,52U		Item	Number of U Needed	Fiber Optic Patch Panel	1U	48 Port Patch Panel	2U	48 Port Switch	2U	PDU	1U	UPS	1U	Fans	1U	Cable Organizer	1U	NVR 48 Channel	2U	Fans	1U	Fiber Optic Patch Panel	1U	Cable Organizer	1U	48 Port Patch Panel	2U	48 Port Switch	2U	NVR 48 Channel	2U	UPS	1U	PDU	1U	Total Height	16	Add Spare	$16 * 1.5 = 24 \rightarrow 24$
Item	Number of U Needed																																						
Fiber Optic Patch Panel	1U																																						
48 Port Patch Panel	2U																																						
48 Port Switch	2U																																						
PDU	1U																																						
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PDU	1U																																						
Total Height	16																																						
Add Spare	$16 * 1.5 = 24 \rightarrow 24$																																						

Figure CCTV-7: CCTV rack sizing and rack height calculation.

Explanation: The project has 37 CCTV points. Adding 20% spare gives about 45 points, so one 48-port switch is sufficient. The rack contains patch panel, switch, NVR, UPS, PDU, fan and cable organizer; the selected standard rack is 24U.

8. Camera References



Figure CCTV-8: CCTV camera reference images.

Explanation: The reference images show bullet and dome/turret camera forms. Selection depends on view quality, mounting location, lens angle, IR range, protection level and monitoring purpose.

CCTV Output Summary

Final CCTV design output: main plan, legend, riser diagram, shop drawing explanation, shop drawing routing, lens check, rack sizing, and selected CCTV camera references. The rack design uses 37 points + 20% spare, one 48-port switch, and a 24U rack.

Access Control System

The access control system controls entry to selected hospital rooms using an IP controller, card reader, door contact/sensor, electric strike lock, exit push button and break-glass door release. The system is used to improve security, restrict unauthorized access, and monitor door status through a central control point.

Main Plan and Legend



Figure AC-1: Access control main plan.

Explanation: The main plan shows the access-controlled doors and equipment locations inside the hospital floor. Each controlled door includes the required field devices and is coordinated with the architectural room layout and circulation paths.

IP ACCESS CONTROL		
TYPE	SYMBOLS	DESCRIPTION
1	IP CONTROLLER	IP CONTROLLER
2	CR	ACCESS CONTROL CARD READER
3	DC	DOOR CONTACTS/SENSOR
4	DL	ELECTRIC STRIKE DOOR LOCK
5	PS	EXIT PUSH BUTTON
6	BR	BREAK GLASS DOOR RELEASE

Figure AC-2: IP access control legend.

The legend identifies the IP controller, card reader, door contact/sensor, electric strike lock, exit push button, and break glass door release used in the access control drawings.

Figure AC-3: Access control riser diagram.

The riser shows the vertical/system connection between the main control equipment and access-controlled doors through the structured cabling path.

Access Control System - Shop Drawing and Components

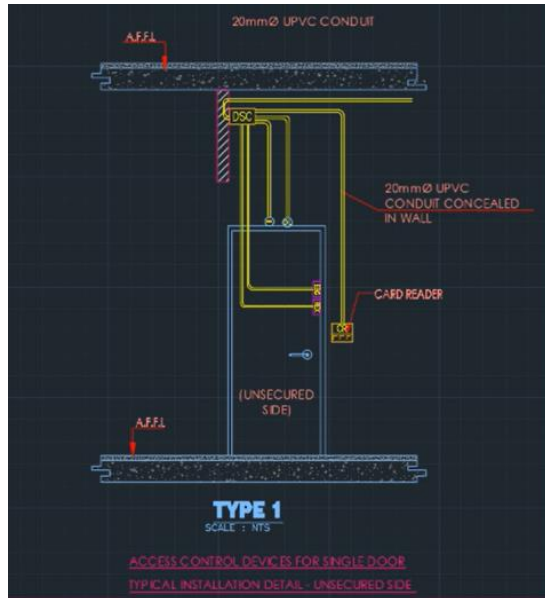


Figure AC-4: Installation explanation detail.

This detail explains the single-door installation method. The card reader is installed on the unsecured side, the door contact monitors door position, and conduits are routed inside the wall/ceiling according to the shop drawing.

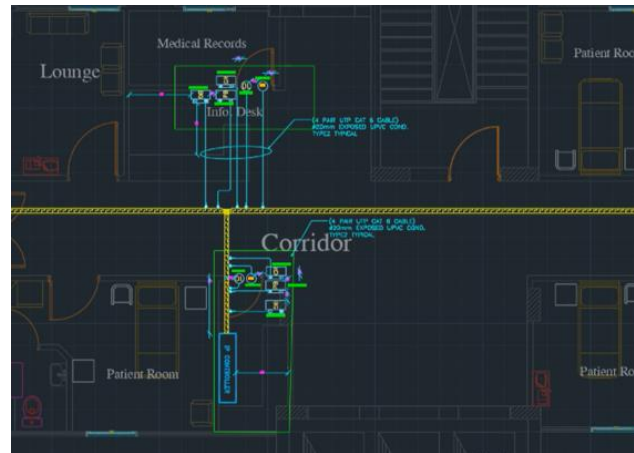


Figure AC-5: Access control shop drawing detail.

The shop drawing converts the design concept into installation routing. It shows the practical cable paths, conduit routes, and device positions required for site execution.

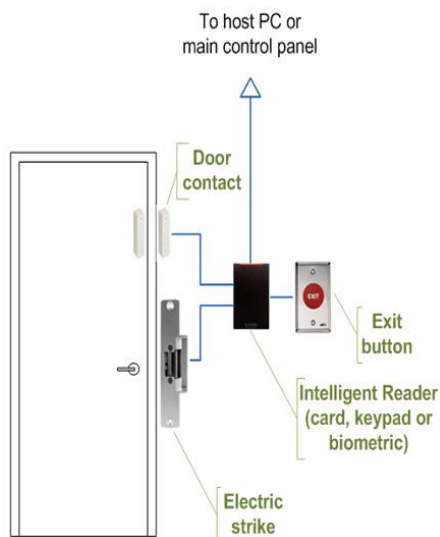


Figure AC-6: Typical access door system concept.

The system consists of a reader, door contact, exit button and electric strike connected to the controller or host PC/main panel. Access is granted when an authorized card, keypad, or biometric input is accepted.

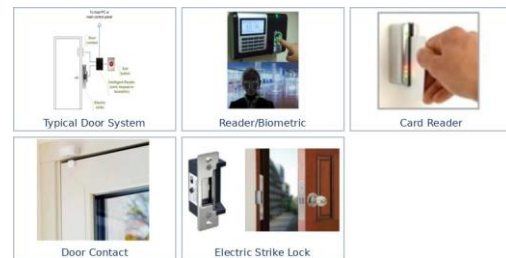


Figure AC-7: Access control components.

- Main components: the IP controller processes access decisions, the card reader reads authorized credentials, the door contact monitors door status, the electric strike locks/unlocks the door, and the exit button or break-glass unit provides safe release during exit or emergency conditions.

Nurse Call System

System Overview

The Nurse Call System (NCS) allows patients to call nursing staff from patient rooms, beds, toilets, ICU rooms and medical spaces. Calls are sent to the nurse station and are also indicated by corridor overhead lights, which helps the staff identify the calling room quickly.

1. Main Plan

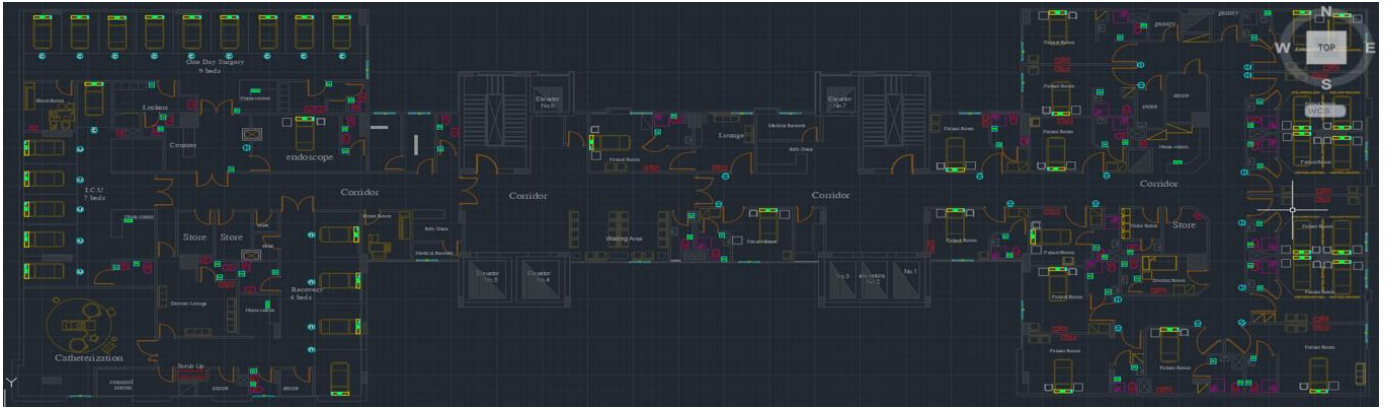


Figure NCS-1: Nurse Call System main plan.

Explanation: This plan shows the location of the nurse call points, patient room units, bathroom call units, corridor overhead lights and nurse station related devices on the hospital floor.

2. Legend and Riser Diagram

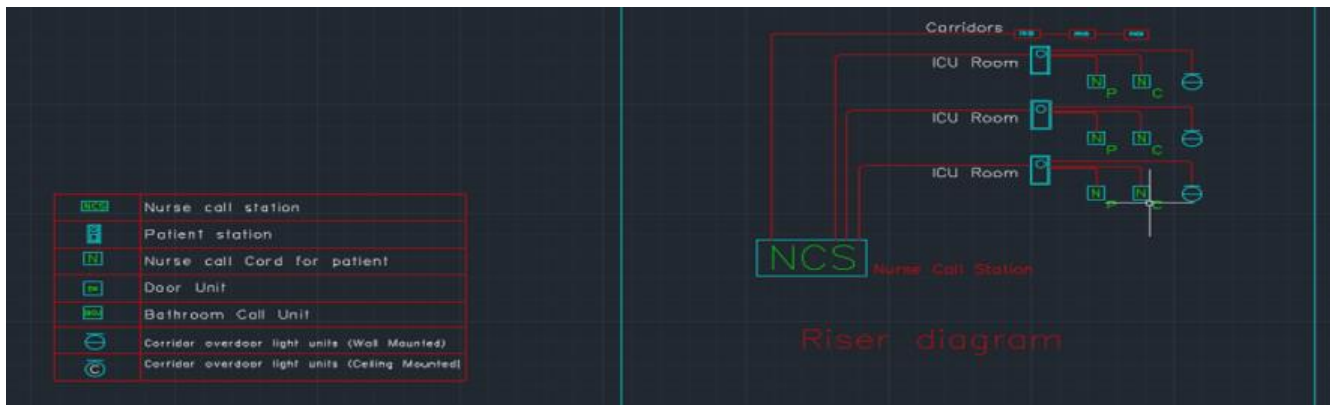


Figure NCS-2: Nurse call legend and riser diagram.

Explanation: The legend identifies the symbols used in the NCS drawing. The riser shows how the nurse call station is connected with room devices and corridor indication units.

Nurse Call Equipment and Sizing

Equipment List						
SR	Item	Required Quantity	Used Quantity	Category	Power (Watt)	Total Power
Nurses' station and management equipment						
1	Display unit	4.0000	4	No. 0 766 60	9.2	36.8
1.1	Installation Components		4	No. 0 800 53		
1.2			4	No. 0 802 53		
1.3			4	No. 0 766 60		
2	Management module	7.6667	8	No. 0 782 12	1	8
Corridor equipment						
3	Corridor overdoor light units	42.0000	42	No. 0 766 72	0.9	37.8
Room/bathroom equipment						
4	Door unit	31.0000	31	No. 0 782 04	0.7	21.7
4.1	Installation Components		31	No. 0 801 41		
5	Bathroom call unit	28.0000	28	No. 0 766 64	0.7	19.6
5.1	Installation Components		28	No. 0 801 41		
Bed equipment						
6	Socket for corded push-button	42	42	No. 0 766 63		
7	Corded push-button	42	42	No. 0 783 62		
Needed Power Supply Rating						123.9
8	Power Supply	0.63888889	3	No. 1 467 23	60 W	

No. of Nurse Stations		4
A	Number of separate rooms	0
B	Number of rooms with WC	20
C	Number of separate WCs	3
D	Number of beds	42
E	Number of directions	23

Figure NCS-3: Equipment list and power supply calculation.

Explanation: The equipment list defines the required and used quantities for each NCS component, including display units, management modules, corridor overdoor lights, door units, bathroom call units, corded push buttons and power supplies. The calculation confirms the required power supply rating and the selected number of power supplies.

Key equipment values

Item	Value	Meaning	Design note
Nurse stations	4	Main reception points	Receives patient calls.
Beds	42	Bed call points	Used for corded push-buttons.
Rooms with WC	20	Bathroom coverage	Affects bathroom call unit quantity.
Overdoor lights	42	Corridor indication	Shows the calling room location.
Required power	123.9 W	Total load	Used to size power supplies.
Power supplies	3 x 60 W	Selected source	Gives margin above demand.

Final placement note: In the final report, this system should stay compact because it does not include a separate shop drawing page. The required order is: main plan, legend/riser, equipment list, then equipment and power supply explanation.

Public Address System

1.0 Overview

The Public Address (PA) / Voice Alarm system distributes announcements and emergency voice messages through loudspeakers installed in corridors, rooms and public areas. The system supports daily paging and can also help guide occupants during emergency conditions. In this project, the PA design includes ceiling loudspeakers, cabinet loudspeakers, amplifiers, riser connections and voltage drop checks.

1.1 Main Plan and Legend



Figure PA.1: Public Address main plan.

Explanation: The main plan shows the distribution of PA loudspeakers and public address circuits across the hospital floor. The loudspeaker positions are selected to cover corridors, waiting zones, rooms and service areas with clear voice announcement coverage.

SYMBOL	DESCRIPTION
PUBLIC ADDRESS SYSTEM	
	IP Ceiling Loudspeaker, 30W (tapped at 15W), 1-way, Built-in MIC, 88 dB @1m, 24V DC/POE.
	CABINET LOUDSPEAKER L-PWP10A, 10 W (5 W tapped), 92 dB @1m, 150 Hz–17 kHz, 170° dispersion, White, Indoor.
	PUBLIC ADDRESS / VOICE ALARM SYSTEM AMPLIFIER.

Figure PA.2: Public Address system legend.

Explanation: The legend defines the symbols used in the PA drawings, including IP ceiling loudspeaker, cabinet loudspeaker and PA/voice alarm system amplifier. The ceiling loudspeaker is rated 30 W and tapped at 15 W, while the cabinet loudspeaker is rated 10 W and tapped at 5 W.

Public Address System

1.2 Riser Diagram

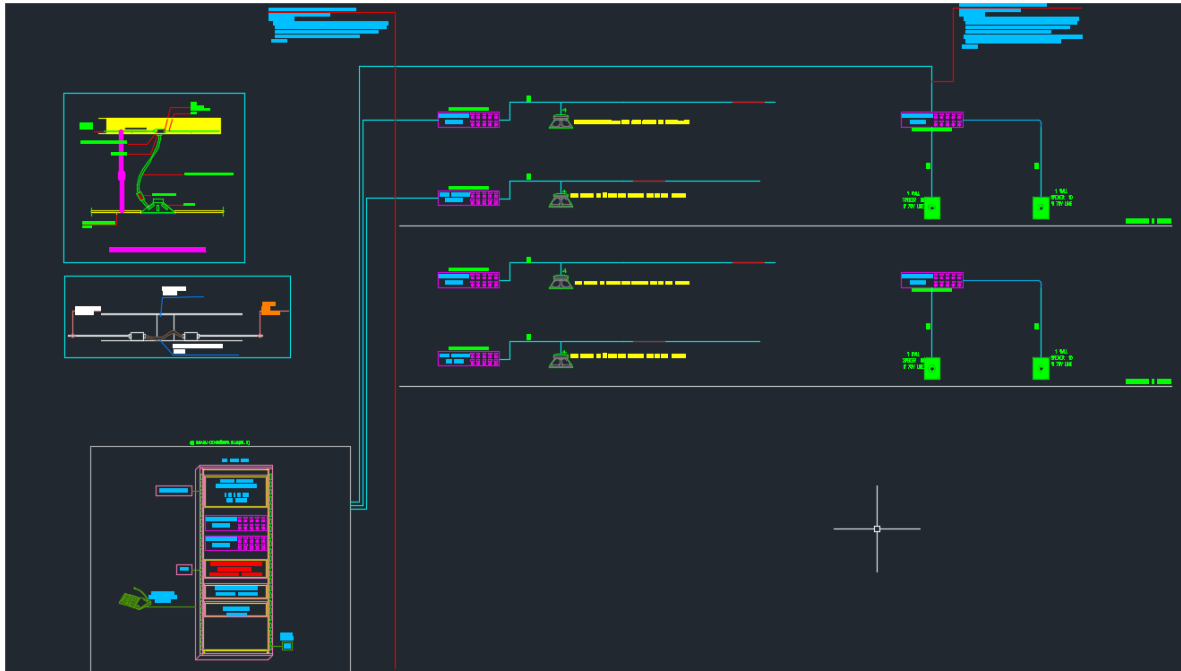
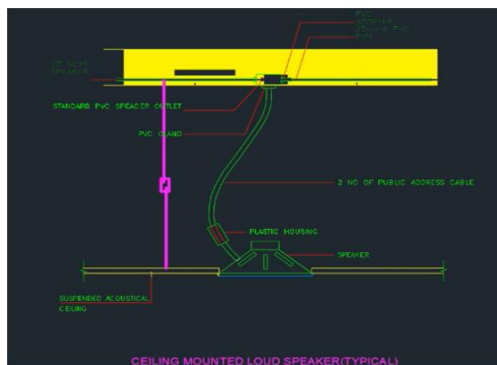


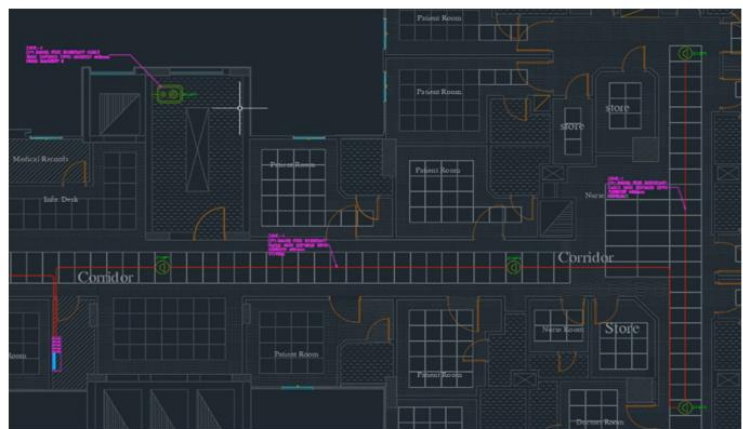
Figure PA.3: Public Address riser diagram.

Explanation: The riser diagram explains the vertical system connection between the PA amplifier/rack and the loudspeaker zones. It shows how speakers are grouped in zones and connected back to the PA equipment so each area can be controlled and monitored clearly.

1.3 Shop Drawing and Installation Detail



Typical ceiling-mounted loudspeaker installation detail



PA shop drawing detail

Figure PA.4: PA shop drawing explanation and shop drawing reference.

Explanation: The shop drawing converts the design into practical installation routes. It shows the speaker locations, cable routes, conduit paths and connection to the PA rack/amplifier. The installation detail clarifies how the speaker is fixed in the suspended ceiling and connected using PA cable through conduit.

Public Address System

1.4 Amplifier Sizing and Voltage Drop Calculation

Item	Public Address Zone	Level	No. of Loudspeakers			Volume Control with override relay watts as indicated in SLD	Wattage / Zone	Wattage / Zone + 20 % Spare	Power Amplifier (PA)		Cables	Estimated Distance meter	Voltage Drop %		P=VI I=P/V
			WALL MOUNTED	CELLING MOUNTED	Horn				Connection	Rate Power Amplifier (Wattage + 20% spare)			with 2x1.5mm2	with 2x2.5mm2	
			6/3/1.5 Watts Tapped @ 3W	15/10 Watts Tapped @ 10W	20/15 Watts Tapped @ 15W										
1	Zone. 01	Right Ground Floor	0	11	0	0	110	132	PA1	180	Fire Resistance Cable with (minimum cross section 2 X 1.5mm2).	43.7	2.85%	1.74%	
2	Zone. 02	Left Ground Floor	0	6	0	0	60	72	PA2	120		43.3	1.54%	0.94%	
3	Zone. 03	Stair 1		1	0	0	10	12	PA3	60		25	1.26%	0.77%	
4	Zone. 04	Stair 2		1	0	0	10	12	PA4	60		25	1.26%	0.77%	
Total			0	19	0	0	No.s Power Amplifiers								

Amplifier standard sizes:
60, 120, 180, 240, 360,
400, 480, 500, 640 W

Figure PA.5: PA amplifier sizing and voltage drop calculation.

Explanation: The PA calculation table divides the system into public address zones. For each zone, the number of wall-mounted, ceiling-mounted and horn loudspeakers is counted, then the total wattage is calculated. A spare margin is added before selecting a standard amplifier size. The table also checks voltage drop using different cable sizes, such as 2x1.5 mm2 and 2x2.5 mm2, to confirm that the speaker line losses remain within the acceptable range.

Zone	Loudspeaker Count	Selected Amplifier	VD check
Zone 01	11 ceiling speakers	180 W	Accepted
Zone 02	6 ceiling speakers	120 W	Accepted
Zone 03	1 ceiling speaker	60 W	Accepted
Zone 04	1 ceiling speaker	60 W	Accepted

1.5 PA Components

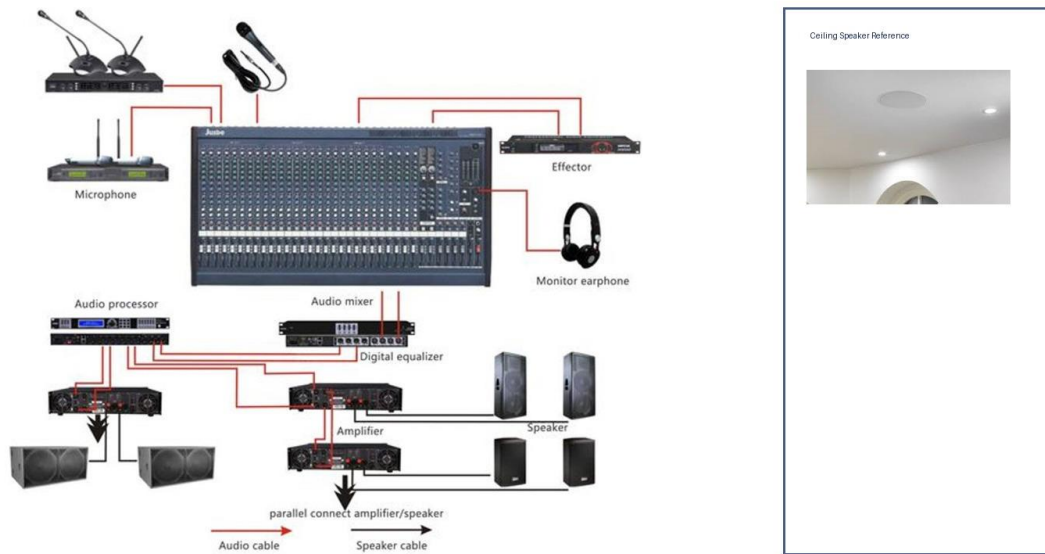


Figure PA.6: Public Address components and ceiling speaker reference.

Explanation: The PA system includes microphones, audio processing/mixing equipment, amplifiers and loudspeakers. The amplifier feeds speaker circuits using speaker cable, while the ceiling loudspeaker is installed flush with the suspended ceiling to provide clean appearance and uniform sound coverage

Final Coordination Drawings

Ceiling Coordination

After finishing the main design stages, ceiling coordination is prepared to combine all ceiling-related services in one coordinated drawing. This stage is important because the false ceiling contains many systems at the same time, such as lighting fixtures, emergency lighting, fire alarm detectors, CCTV cameras, public address speakers, HVAC diffusers, cable trays and conduits. The purpose is to make sure that no device, route, or outlet conflicts with another one before installation.



Figure LC.1: Ceiling coordination layout after completing the design stage.

Explanation: The drawing overlays the final ceiling devices and routes from different disciplines. It is checked before site execution to confirm clear spacing, correct device positions, accessible maintenance zones, and safe installation paths. Any conflict between services can be solved in the drawing stage instead of being discovered on site.

Check point	What is verified
Device coordination	Lighting fixtures, speakers, smoke detectors, CCTV cameras and HVAC diffusers do not overlap.
Route coordination	Cable trays, conduits and ceiling routes have enough clearance and do not block other services.
Site execution	The final ceiling layout becomes easier for installation teams to follow accurately.
Main benefit	Reduces rework, prevents clashes and keeps the hospital ceiling layout organized.

Chapter 4: Mechanical Fire Fighting System

4.1 Fire Fighting System Overview

This chapter presents the mechanical fire fighting design prepared for the hospital building. This chapter is separated from the electrical fire alarm system and light current systems. It focuses only on the mechanical fire protection scope, including sprinkler distribution, pre-action protection for operation rooms, portable extinguishers, fire hose cabinets, zone control valves, pipe sizing, hydraulic calculations, fire pump selection, fire water tank, FM200 clean agent protection, and Fire Search protection for electrical panels.

The design concept is based on matching the protection method with the risk of each area. General hospital spaces are protected by a wet type sprinkler system. Operation rooms are protected by a pre-action sprinkler system. UPS rooms and medical record rooms are protected using FM200 gas suppression, while small electrical rooms and electrical panels are protected locally using Fire Search. Manual response is supported by fire hose cabinets and portable extinguishers located near the required areas.

متطلبات الحريق للمستشفيات		
اسم الغرفة	الاطفاء العام داخل الغرفة	الاطفاء داخل الوحدات
R M U		لا يوجد
M D P	FM200	ان ترم FIRE SEARCH
DISTRIBUTER	FM200	لا يوجد
TRANSFORMER	FM200	لا يوجد
GENERATOR	FM200	FIRE SEARCH IN ATS
IT ROOM	FM200	لا يوجد
CONTROL ROOM	FM200	لا يوجد
COMMUNICATION ROOM	FM200	لا يوجد
UPS ROOM	FM200	لا يوجد
غرف الكهرباء داخل المبني والتي تحتوي على لوحات كهرباء	لا يوجد	في اللوحات الكبيرة FIRSEARCH أو وفقًا للحماية المدنية
محطة الغازات الطبية وغرفة التخزينات	رشاشات مياه	
غرف العمليات	رشاشات مياه مع Pre Action	
غرف الإنعشة	FM200	
الموقع العام	توزيع صناعي حريق ٢.٥" بالتوزيع العام وتركيب Slamese Connection في المدخل الرئيسي	
بالقرب فراغات المستشفى	رشاشات مياه لا تزيد المسافة بين الرشاشات عن ١.٢ متر وبين الرشاش والحائط عن ٢ متر ويتم تركيب رشاشات في حمامات المرضى	
تركيب صندوق مزودج ٢.٥" و ١" بغطاء على السلام مع محاولة تركيب ZCV عند نقطة الدور في السلم ولا يزيد المسافة بين الصندوقين عن ٥٠ متر وفي حالة زيادة المسافة عن ذلك يتم تركيب صندوق ١ بوصة في منتصف المسافة		
المسافة بين الطفايات البودرة في الدور لا تتجاوز ال ٢٠ متر + طفلة CO2 امام غرف الكهرباء والشارب الخفيف والإنعشة لظف وطفلة فوم ١٥ لتر في غرفة التخزينات وامام غرفة المولد الكهربائي		
تقديم لوحة لفرقة الطفايات والغازان موضح بها جميع المتناسيب لمياه الحريق والترب وخط الحريق وال Test وال OverFlow وال PRV وبعد الغازان وغرفة التخزينات وموقعه بالموقع العام وربطه مع المبني		
عمل خط هيدر لخط الاختيار منفصل عن الهيكل الرئيسي للحريق وعن خط ال PRV وذلك لاختيار كل أنظمة تون فصل النظامية الأخرى عن إطفاء المبني مع محاولة ربط خط ال OverFlow الى خارج الغازان بالقرب منقوش لو سمحت المتناسيب		

Figure 4.1: Egyptian code requirement reference used for sprinkler spacing and room coverage checks.

The Egyptian code requirement reference was placed at the beginning of the fire fighting chapter because it is used as a general check for sprinkler spacing and coverage. This reference supports the review of sprinkler distribution inside rooms before moving to the detailed plans and calculations.

4.2 Fire Fighting General Layout and Legend

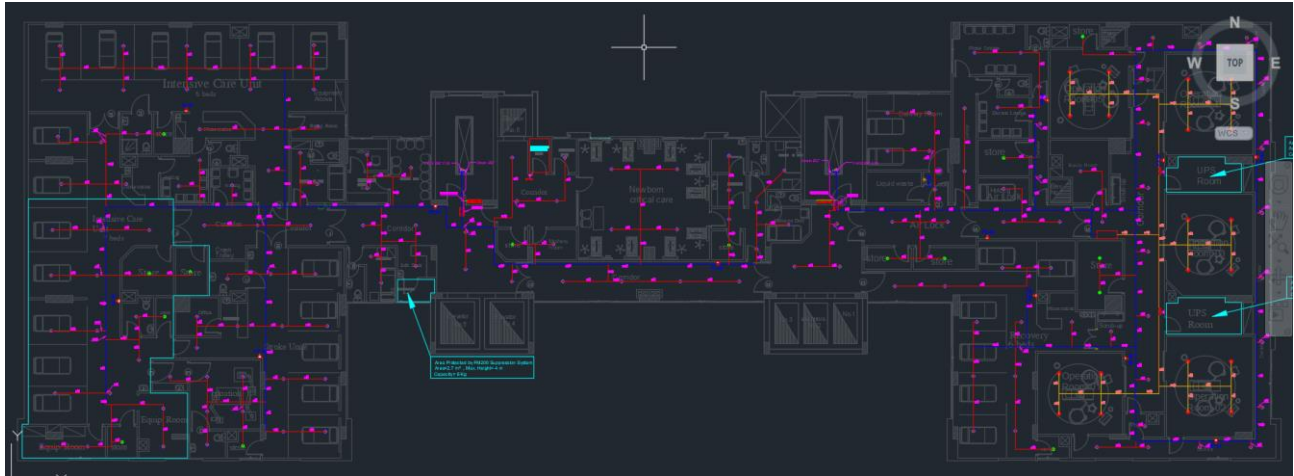


Figure 4.2: Complete fire fighting layout for the hospital floor.

The complete fire fighting layout shows the distribution of sprinklers, pipe routes, fire hose cabinets, zone control valves, flow switches, fire risers, extinguishers, pre-action areas, FM200 protected rooms and Fire Search protected areas. This plan is the main coordination drawing for the mechanical fire fighting design.

LEGEND	
	Pendent sprinkler K=5.6 68C
	Upright Sprinkler K=5.6
	Pre Action Sprinkler K=5.6
FE-01 	PORTABLE DRY POWDER FIRE EXTINGUISHER 6K.g
FE-02 	PORTABLE Co2 FIRE EXTINGUISHER 6K.g
FE-04 	PORTABLE FOAM FIRE EXTINGUISHER
F/B& T/B	FROM BELOW & TO BELOW
	FIRE HOSE CABINET, CLASS III(2.5"+1")
	FIRE RISER
	ZONE CONTROL VALVE
	Flow Switch (F.S)

Figure 4.3: Fire fighting legend used in the mechanical fire fighting drawings.

BLOCKS COUNT		
PREVIEW	NAME	NUMBER
	FIRE HOSE CABINET CLASS III	2
	Flow Switch (F.S)	2
	Pendent sprinkler K5.6	146
FE-02 	PORTABLE Co2 FIRE EXTINGUISHER	3
FE-01 	PORTABLE DRY POWDER FIRE EXTINGUISHER	12
FE-04 	PORTABLE FOAM FIRE EXTINGUISHER	1
	Pre Action Spr	20
	Upright Sprinkler K5.6	15
	ZONE CONTROL VALVE	2

Figure 4.4: Fire fighting blocks count used to check the number of installed fire protection elements.

Element	Quantity / note
Fire Hose Cabinet Class III	2
Flow Switch	2
Pendant Sprinkler K=5.6	146
CO2 portable extinguisher	3
Dry powder portable extinguisher	12
Foam portable extinguisher	1
Pre-action sprinkler	20
Upright sprinkler K=5.6	15
Zone Control Valve	2

4.3 Sprinkler Distribution and Wet Type Sprinkler System

The main sprinkler network is a wet type sprinkler system for the general hospital areas. The pipes are normally filled with water under pressure. When a sprinkler reaches its activation temperature, the sprinkler opens and discharges water directly over the fire area. This system is reliable and suitable for corridors, rooms and general hospital spaces.

Wet Type Water Sprinkler System

Components:

- Sprinklers.
- Pipe network.
- Valves.
- Pump station.
- Tank.

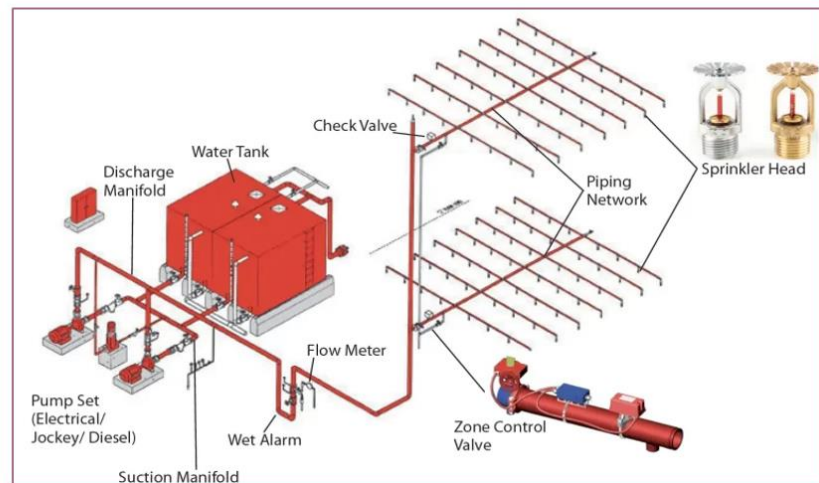


Figure 4.5: Wet type water sprinkler system concept and main components.

Sprinkler Types



Figure 4.6: Sprinkler types used as references in the design.

Pendant sprinklers were used in most areas with suspended ceilings. Upright sprinklers were used in store rooms and areas without false ceiling because they can be installed above the branch pipe and are less exposed to damage during storage activities. Pre-action sprinklers were used in operation rooms because those areas are sensitive to accidental water discharge.

Upright Sprinklers Used in Store Areas



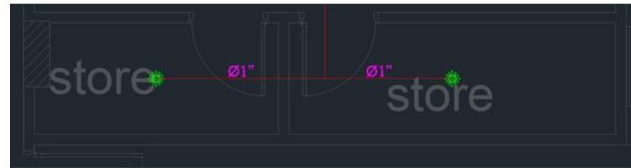
Store area with upright sprinklers on the branch line



Upright sprinkler inside a store room



Store room sprinkler branch and upright sprinkler locations



Two store rooms protected using upright sprinklers

Figure 4.7: Upright sprinkler distribution in store areas.

Figure 4.7 shows the store room sprinkler distribution. The store areas use upright sprinklers because these rooms do not depend on a normal suspended ceiling layout. This arrangement also helps keep the sprinkler heads safer from accidental contact with stored items or workers movement.

4.4 Most Remote Area and Hydraulic Calculation Reference

The most remote area is the hydraulically most demanding part of the sprinkler network. It is selected because it normally has the highest pressure losses due to distance, fittings and elevation. If the pump and pipe network can satisfy this area, the closer zones will also have enough pressure and flow.

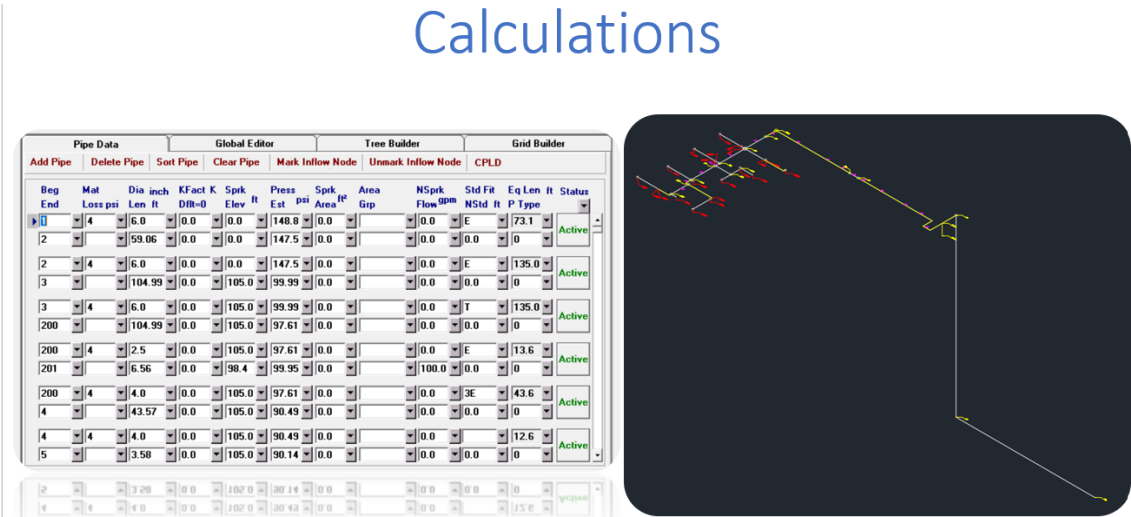
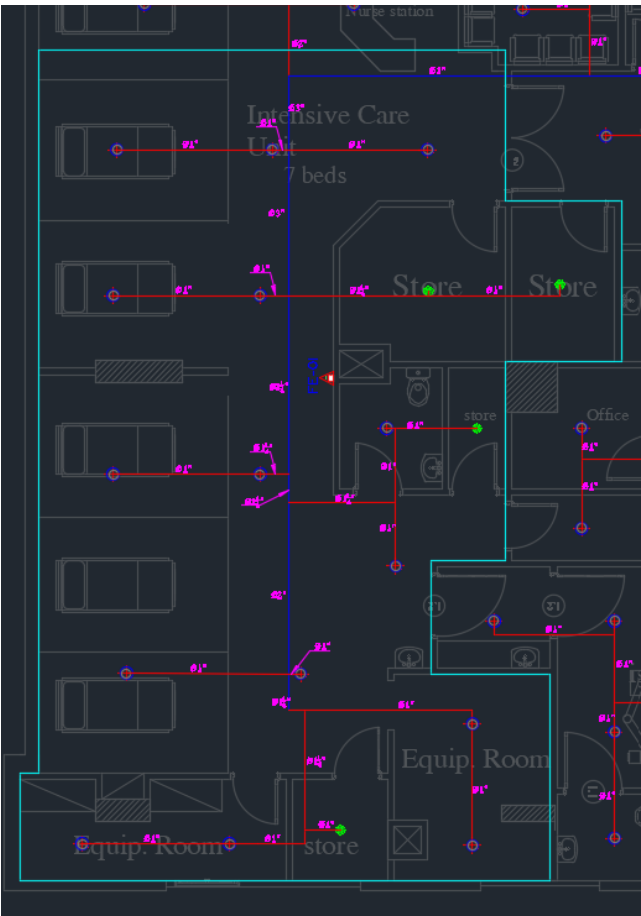


Figure 4.8: Hydraulic calculation model and most remote area reference.



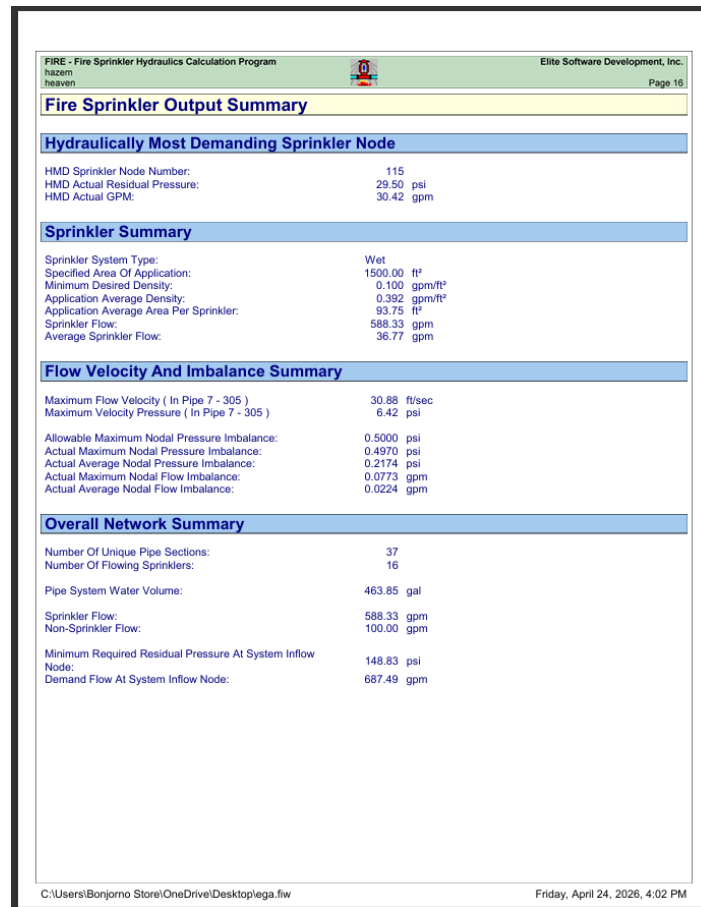


Figure 4.9: Fire sprinkler hydraulic calculation output summary.

Hydraulic result / design value	Project value
Specified design area	1500 ft²
Minimum design density	0.100 gpm/ft²
Hydraulically most demanding sprinkler node	Node 115
HMD actual residual pressure	29.50 psi
HMD actual flow	30.42 gpm
Flowing sprinklers	16
Sprinkler flow	588.33 gpm
Non-sprinkler hose allowance	100 gpm
Demand flow at system inflow node	687.49 gpm
Minimum required residual pressure at system inflow node	148.8 psi

The hydraulic calculation output was placed beside the sprinkler distribution and most remote area section because it verifies the sprinkler system performance. The calculation confirms that the sprinkler network, pipe sizes and pump selection can provide the required flow and pressure at the most difficult point in the system.

4.5 Pre-action System for Operation Rooms

Operation rooms are sensitive medical areas. Direct water discharge by mistake can damage medical equipment and interrupt room operation. Therefore, a pre-action sprinkler system was selected for operation rooms instead of a normal wet pipe sprinkler system. In a pre-action system, the pipe network is not normally filled with water until a confirmed fire condition allows water to enter the system.

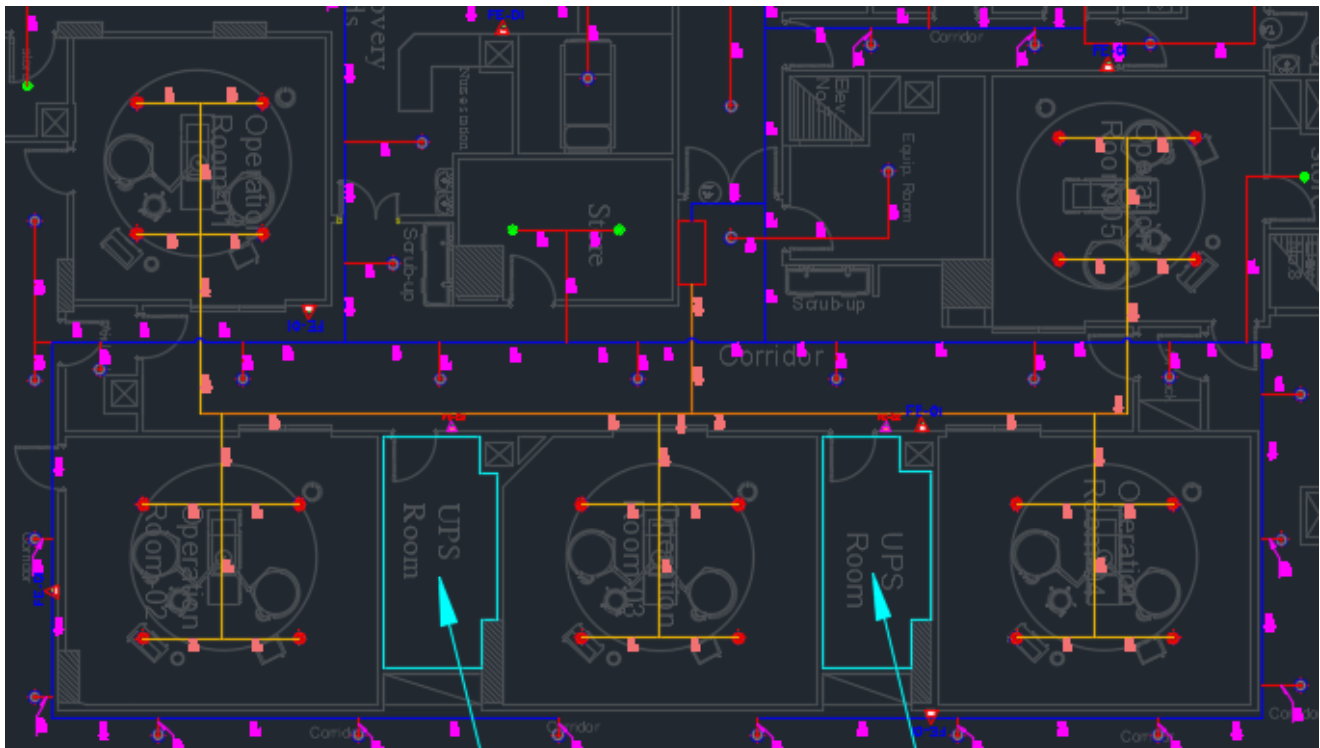
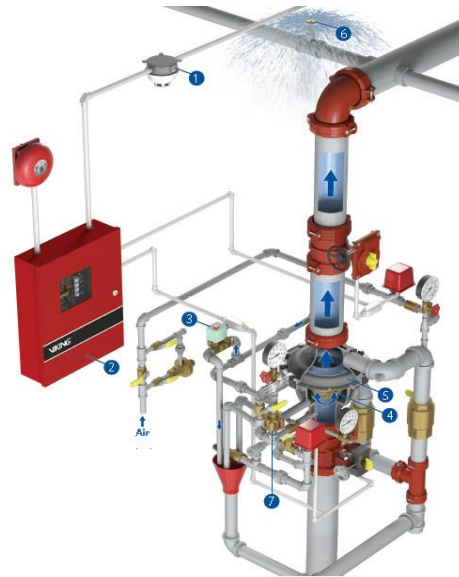


Figure 4.10: Operation rooms protected by pre-action sprinklers with adjacent UPS room references.

Figure 4.10 shows the operation room area where pre-action sprinklers are distributed. The important point in this section is the operation room protection, not the electrical room protection. The nearby UPS rooms are shown on the same drawing but are explained separately in the FM200 section.

Area	Protection method	Reason
Operation rooms	Pre-action sprinkler system	Avoid accidental water discharge in sensitive surgical spaces.
General rooms and corridors	Wet sprinkler system	Fast automatic fire suppression.
UPS / records / electrical rooms	FM200 or Fire Search	Waterless suppression for sensitive electrical or document areas.

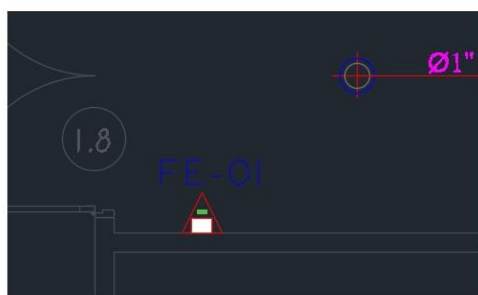
4.6 Portable Fire Extinguishers

Portable fire extinguishers were distributed to support manual first response. The extinguisher type was selected according to the expected fire risk in each area. Dry powder extinguishers are suitable for general areas, CO2 extinguishers are used near electrical risks, and foam extinguishers are used where liquid fire risk exists. Extinguishers were located so that they remain visible, accessible, and within the allowed travel distance from protected rooms.

Portable Fire Extinguisher Locations

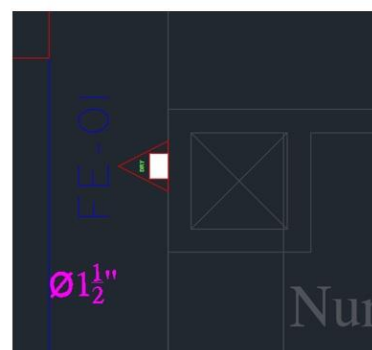


Foam extinguisher / liquid waste area reference



Dry powder extinguisher location example

Dry powder extinguisher location near stroke unit



Extinguisher point near nurse/service area

Figure 4.11: Portable fire extinguisher location examples on the fire fighting drawings.

Symbol / type	Quantity	Typical location / use
FE-01 Dry powder extinguisher 6 kg	12	General rooms, corridors and mixed fire risk areas.
FE-02 CO2 extinguisher 6 kg	3	Electrical rooms, UPS rooms and equipment areas.
FE-04 Foam extinguisher	1	Liquid waste / possible flammable liquid risk area.

The extinguisher section is placed before the riser and pump sections because it belongs to the local manual response layer of the design. The locations shown in the drawings indicate how each extinguisher type was placed beside the proper room or risk area, instead of using one extinguisher type randomly for all spaces.

Choosing a Fire Extinguisher				ABC DRY CHEMICAL	BC DRY CHEMICAL	DRY POWDER	WATER	FOAM	WET CHEMICAL	HALOGENATED	CARBON DIOXIDE
Class	Symbol	Type of Fire	Examples								
A		Common combustibles	Wood, paper, cloth etc.								
B		Flammable liquids and gases	Gasoline, propane and solvents								
C		Live electrical equipment	Computers, fax machines (see note)								
D		Combustible metals	Magnesium, lithium, titanium								
K		Cooking media	Cooking oils and fats								

4.7 Fire Hose Cabinet and Standpipe System

The Fire Hose Cabinet (FHC) and standpipe system provide a manual firefighting water point for trained users and fire brigade use. In the project, Class III fire hose cabinets were used. A Class III cabinet can include hose equipment and a larger connection for fire brigade use. FHCs are located near accessible circulation routes and stairs to support manual fire response.

Fire Hose Cabinet and Standpipe System

Standpipe System

Fire Hose Cabinet (FHCs)

Components:

1. **Cabinet:** It is located at a height of 90 cm to 150 cm from the ground.
2. **Hose:** It has a diameter of 1 in, 1.5 in or 2.5 in and has a length of 15 m, 20 m or 30 m.
3. **Landing Valve:** It Connects the hose to the riser.
4. **Discharge Nozzle:** Converts pressure into velocity.
5. **Fire Extinguisher (Optional).**



Fire hose cabinet / standpipe system components

FHCs Classes

Class1

Class 2

Class 3

Item	Class 1	Class 2	Class 3
Hose Size	2.5 "	1 " or 1.5 "	2.5 " or 1 "
Min. Press.	6.9 bar	4.5 bar	6.9 bar for 2.5 " and 4.5 bar for 1.5 "
Min. Flow Rate	250 gpm	100 gpm	250 gpm
Pmax	12 bar	6.9 bar	6.9 bar
Diam. Of Pipe	2.5 "	1 " or 1.5 "	2.5 " or 3 " for 2.5 " 1 " for 1.5 "
Description			
Travel Distance	45.7 m	39.7 m for 1.5 " 36.6 m for 1 "	30 m

FHC classes and design data comparison

Figure 4.12: Fire Hose Cabinet and standpipe system references.

The FHC image and the FHC classes table are kept together in this section to avoid mixing them with the Zone Control Valve section. The cabinet includes the hose, landing valve, discharge nozzle and optional extinguisher, while the FHC class table explains the basic design requirements for the cabinet class.

4.8 Zone Control Valve and Flow Switch

The Zone Control Valve (ZCV) assembly is used to control each sprinkler zone. It allows isolation, testing, draining and monitoring of the zone. The flow switch indicates water movement inside the pipe network, which happens when sprinkler water is flowing in the zone.

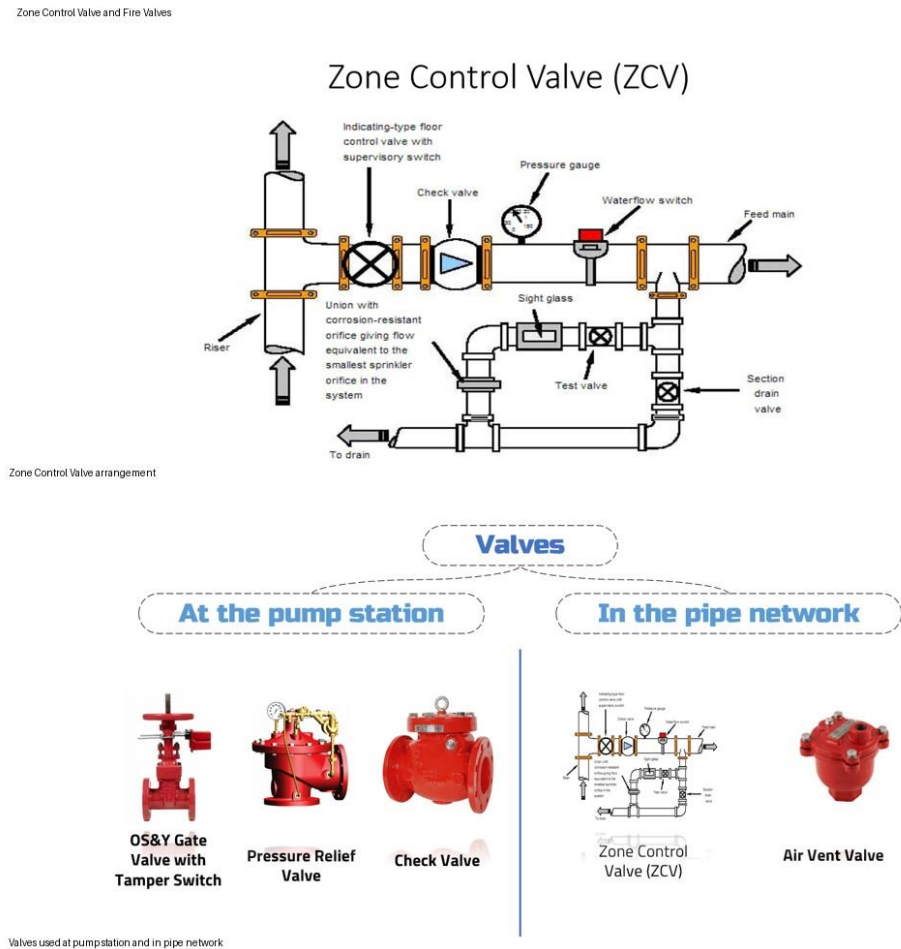


Figure 4.13: Zone Control Valve arrangement and valve references.

ZCV component	Function
Indicating control valve	Isolates or opens the protected sprinkler zone.
Check valve	Prevents reverse flow.
Flow switch	Detects water flow in the zone.
Pressure gauge	Shows zone pressure.
Test valve	Used to test the zone.
Section drain valve	Used to drain the zone during maintenance.
Sight glass	Helps visual inspection of flow/drain operation.

The ZCV images were separated from the FHC images because they represent different parts of the system. The FHC is a manual water outlet, while the ZCV is a control and monitoring assembly for the sprinkler zone.

4.9 Pipe Network and Pipe Sizing

The pipe network transfers water from the fire pump room and tank to all protected zones. The design includes main risers, branch pipes, sprinkler connections, drain points, flow switches and zone control valves. Pipe sizing was selected according to the number of sprinklers served by each pipe and the hydraulic calculation results.

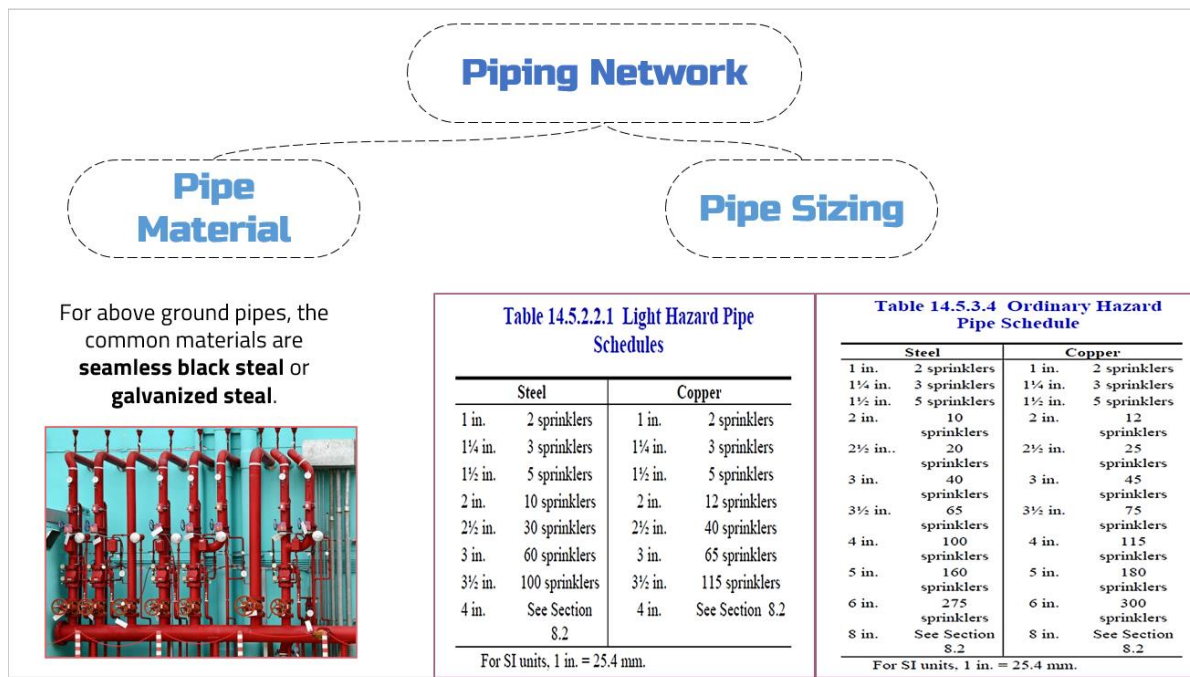


Figure 4.14: Pipe material and pipe sizing reference used in the sprinkler network.

Riser or Main Size	Size of Drain Connection
UP to 2 in	3/4 in or Larger
2.5 in, 3 in, 3.5 in	1 1/4 in or Larger
4 in or Larger	2 in Only

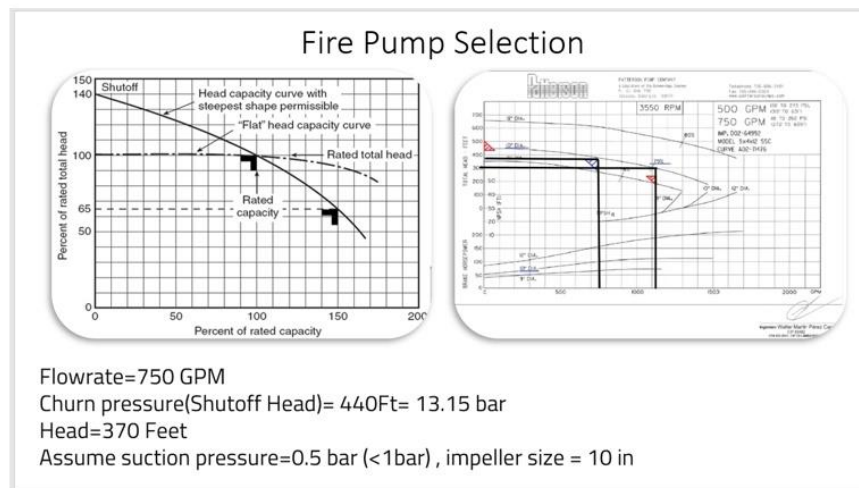
Figure 4.15: Drain connection size reference according to riser or main size.

Above-ground sprinkler piping commonly uses seamless black steel or galvanized steel. Main risers carry the main system flow, while branch pipes supply smaller sprinkler groups. The drain size was selected according to the riser or main size as shown in the drain connection reference.

4.10 Fire Pump Selection and Suction / Discharge Sizing

The fire pump was selected after checking the hydraulic demand of the most remote area. The pump capacity must be greater than the calculated demand and must provide enough head to overcome pipe losses and supply the required residual pressure. Suction and discharge sizes were selected according to the pump capacity and standard fire pump data.

Fire Pump, Suction and Discharge Selection



Fire pump selection curve and selected duty point

Suction and discharge Selection :

Suction and discharge line = 6 in
Relief valve(s) = 4 in
Relief valve(d)= 6 in

TABLE 4.26(a) Summary of Centrifugal Fire Pump Data (U.S. Customary)

Minimum Pipe Sizes (Nominal) (in.)

Pump Rating (gpm)	المنبع Suction ^{a,c}	المخرج Discharge ^d	Relief Valve	Relief Valve Discharge	Meter Device	Number and Size of Hose Valves	خطوط تجميع Hose Header Supply
25	1	1	¾	1	1½	1 — 1½	1
50	1½	1½	1½	1½	2	1 — 1½	1½
100	2	2	1½	2	2½	1 — 2½	2½
150	2½	2½	2	2½	3	1 — 2½	2½
200	3	3	2	2½	3	1 — 2½	2½
250	3½	3	2	2½	3½	1 — 2½	3
300	4	4	2½	3½	3½	1 — 2½	3
400	4	4	3	5	4	2 — 2½	4
450	5	5	3	5	4	2 — 2½	4
500	5	5	3	5	5	2 — 2½	4
750	6	6	4	6	5	3 — 2½	6
1000	8	6	4	8	6	4 — 2½	6
1250	8	8	6	8	6	6 — 2½	8
1500	8	8	6	8	8	6 — 2½	8
2000	10	10	6	10	8	6 — 2½	8
2500	10	10	6	10	8	8 — 2½	10
3000	12	12	8	12	8	12 — 2½	10
3500	12	12	8	12	10	12 — 2½	12

Suction and discharge selection table

Figure 4.16: Fire pump selection with suction and discharge sizing.

Pump / pipe selection item	Selected value
Selected pump flowrate	750 GPM
Pump head	370 ft
Churn / shut-off head	440 ft
Approximate shut-off pressure	13.15 bar
Assumed suction pressure	0.5 bar
Impeller size	10 in
Suction line	6 in
Discharge line	6 in
Relief valve suction side	4 in
Relief valve discharge side	6 in

The suction and discharge reference is placed in this pump section because it belongs directly to pump selection. For the selected 750 GPM pump, the suction and discharge lines are 6 in, with relief valve sizes selected as shown in the reference table.

4.11 Pump Room Components

The pump room contains the main equipment required to supply firefighting water to the network. It includes an electric fire pump, diesel fire pump and jockey pump. The electric and diesel pumps provide the main firefighting flow, while the jockey pump maintains pressure during standby conditions.

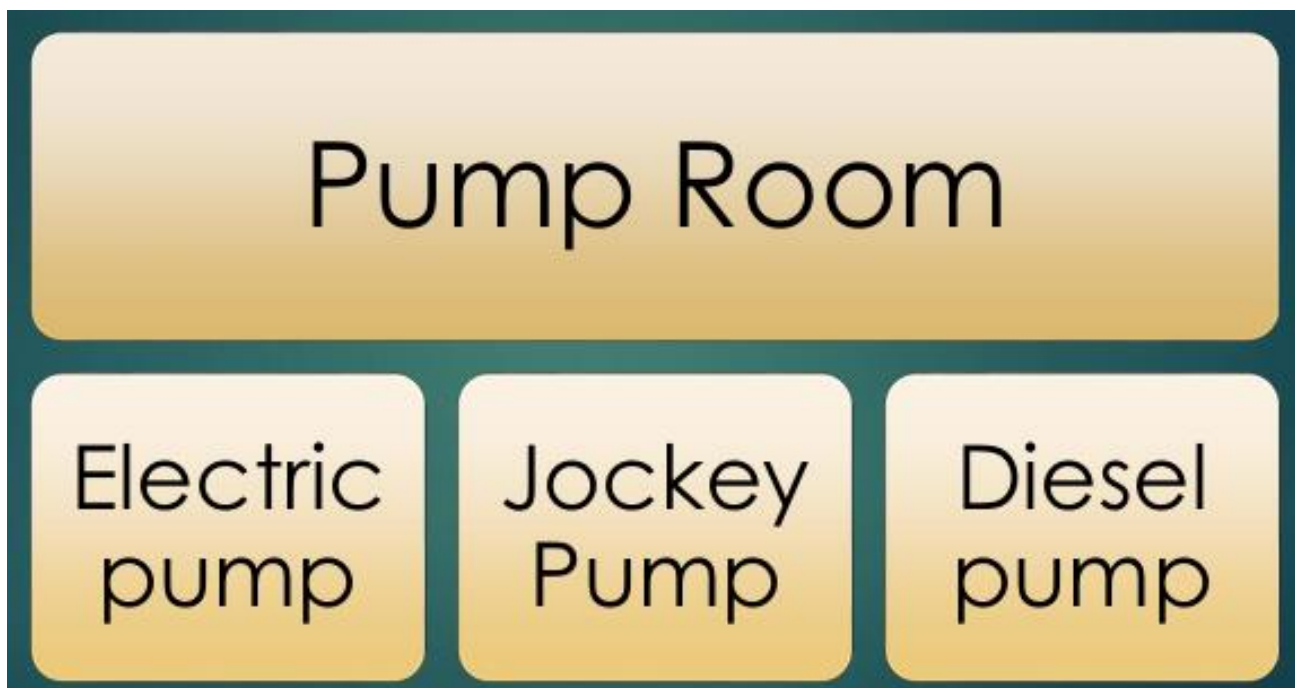


Figure 4.17: Pump room main equipment arrangement.

Pump Room Components

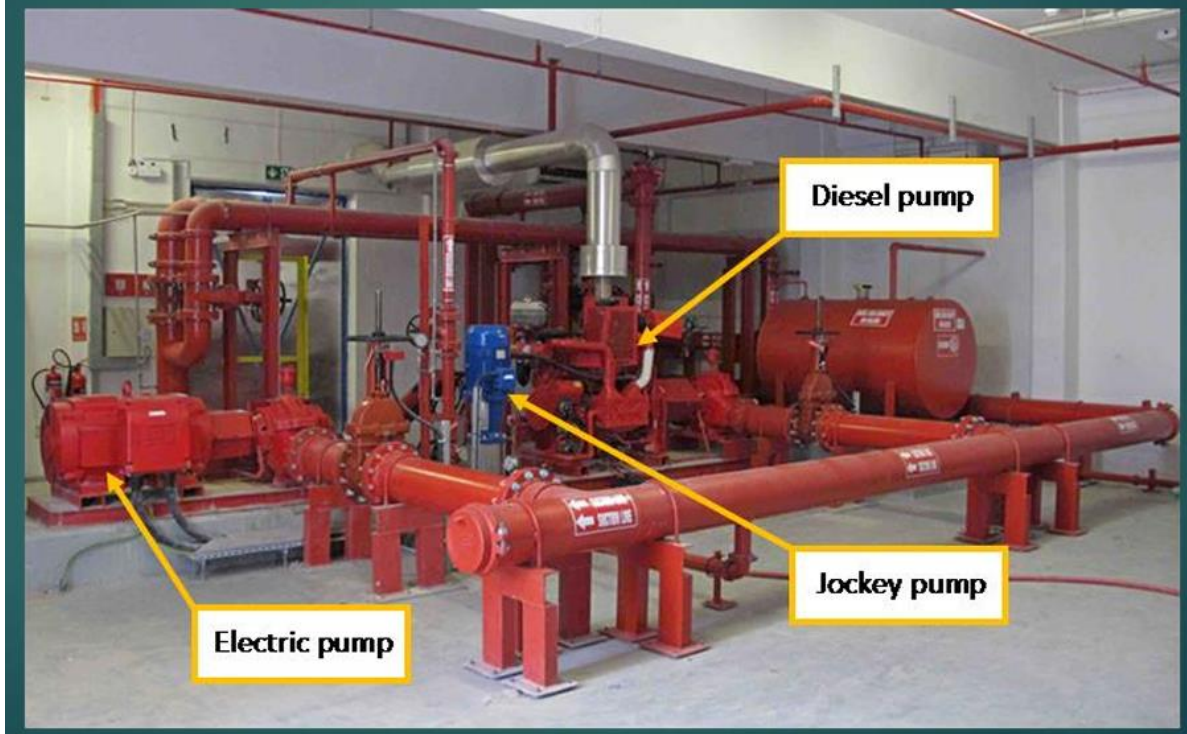


Figure 4.18: Pump room components showing electric, diesel and jockey pumps.

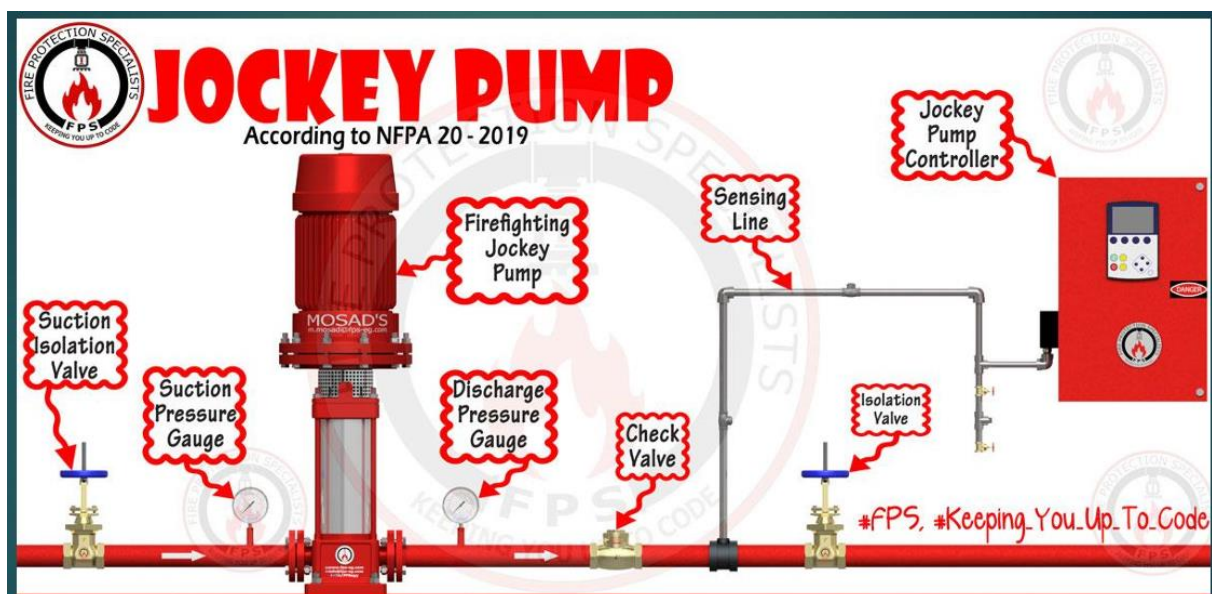


Figure 4.19: Jockey pump components and sensing line reference.

Pump room component	Purpose
Electric fire pump	Main duty pump for supplying firefighting flow and pressure.
Diesel fire pump	Backup pump used when electrical power is unavailable.
Jockey pump	Maintains system pressure and prevents unnecessary start of main pumps.
Diesel fuel tank	Supplies fuel to diesel pump during emergency operation.
Sensing line and gauges	Monitor pressure and control jockey/main pump operation.
Check and isolation valves	Control direction of flow and allow maintenance.

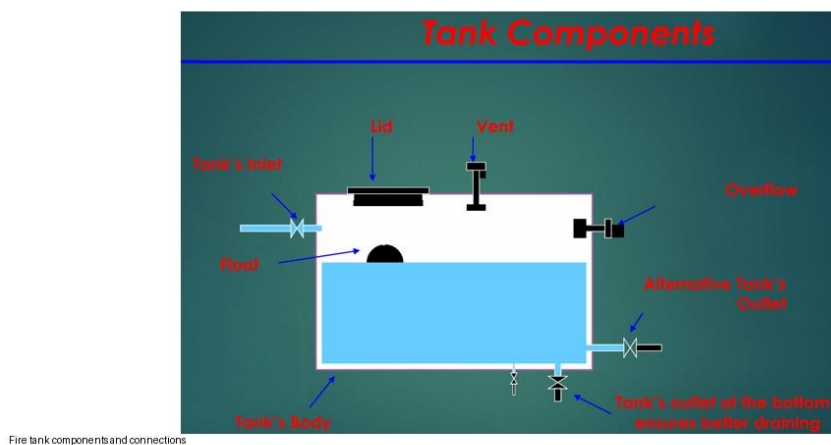
4.12 Fire Water Tank

The fire water tank stores the water required for firefighting operation. In this project, the tank was considered a concrete fire tank. The tank must include the necessary components such as inlet, outlet, overflow, vent, lid, float and drain. The pump suction line should be connected properly to ensure reliable pump operation.

Fire Water Tank and Components



Concrete fire water tank reference



Fire tank components and connections

Figure 4.20: Concrete fire water tank and tank components.

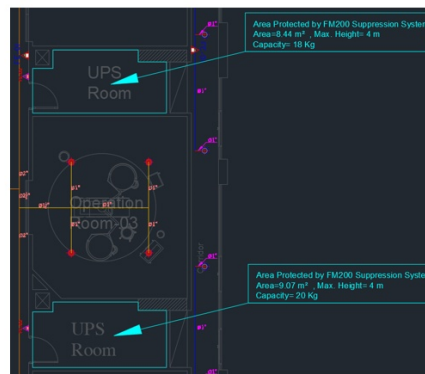
Tank component	Function
Tank inlet	Fills the tank from the water supply.
Lid / access opening	Allows maintenance access.
Vent	Allows air movement during filling and discharge.
Overflow	Protects the tank from overfilling.
Float	Supports water level control.
Bottom outlet / drain	Allows draining and cleaning of the tank.
Fire outlet / suction connection	Feeds the fire pump suction line.

The tank images are placed in this section only, because they belong to the fire water storage part of the system. The tank outlet and pump suction connection must be arranged to reduce losses and maintain a stable water supply to the fire pumps.

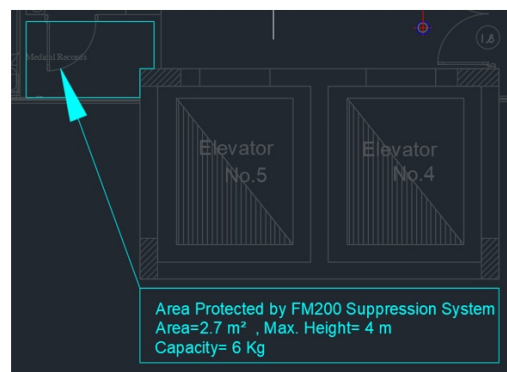
4.13 FM200 Clean Agent System

FM200 was used for rooms where water-based suppression may damage equipment or documents. This includes UPS rooms and medical records rooms. The system suppresses fire using a clean gaseous agent, which is suitable for sensitive electrical and document storage areas.

FM200 Protected Rooms



UPS rooms protected by FM200 suppression system



Medical records room protected by FM200 suppression system

Figure 4.21: UPS rooms and medical records room protected by FM200.

Design calculation sheet			
Project no: D0271	Date: 9/3/2003	Sheet no: 1 of 1	Computed by: R.G.H
Subject: Dubai Royal Airwing		Checked by: M.A	
Administration Archive - FM200 Total Flooding Quantity		Approved by:	
Basic: NFPA 2001, Standard on Clean Agent Fire Extinguishing Systems, 2000 Edition The amount of Halocarbon agent required to achieve the design concentration shall be calculated from the following Formula $W = \frac{V \left(\frac{C}{100 - C} \right)}{S}$			
Hazard Class	=	Class A	
W	=	Weight of FM200 (kg)	
V	=	Net Volume of the Hazard	12.48 (m ³)
C	=	FM 200 Design Concentration	6 %
sf	=	Safety factor	1.2
	=	Required Concentration	7.2 %
t	=	Minimum anticipated temp. of the protected volume	40 °C
S	=	Specific Volume of superheated agent vapor at 1 atmosphere and the temperature t	0.1476 (m ³ /kg)
W	=	$\frac{12.48}{0.1476} \times \left[\frac{7.2}{(100 - 7.2)} \right]$	
W	=	7	kg of FM-200 agent

FM200 design calculation sheet

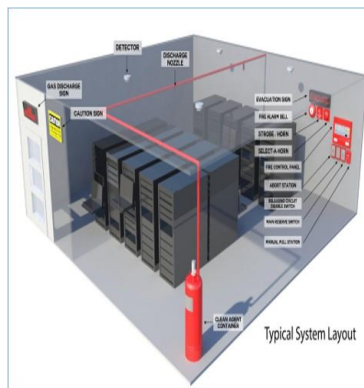
W	=	7	kg of FM-200 agent
Class A Fires : Fire in ordinary combustible materials, such as wood, cloth, paper, rubber, and many plastics. Class B Fires : Fire in flammable liquids, oils, greases, tars, oil-base paints, lacquers, and flammable gases. Class C Fires : Fire that involves energized electrical equipment where the electrical resistivity of the the extinguishing media is of importance.			
3-4 Design Concentration Requirements. 3-4.2.1 The flame extinguishing concentration for Class B fuels shall be determined by the cup burner method described in Appendix B. 3-4.2.2 The flame extinguishing concentration for Class A fuels shall be determined by test as part of a listing program. As a minimum, the listing program shall conform to UL 2127, Standard for Inert Gas Clean Agent Extinguishing System Units, or UL 2166, Standard for Halocarbon Clean Agent Extinguishing System Units, or equivalent. 3-4.2.3 The minimum design concentration for a Class B fuel hazard or an only manually actuated system shall be the extinguishing concentration, as determined in 3-4.2.1, times a safety factor of 1.3. 3-4.2.4 The minimum design concentration for a Class A surface fire hazard shall be the extinguishing concentration as determined in 3-4.2.2, times a safety factor of 1.2. 3-4.2.5 Minimum design concentration for Class C hazards shall be at least that for Class A surface fire.			

FM200 design concentration requirements

Figure 4.22: FM200 gas quantity calculation and design concentration requirements.

Protected area	Area	Maximum height	Selected FM200 capacity
UPS room / operation area 1	8.44 m ²	4 m	18 kg
UPS room / operation area 2	9.07 m ²	4 m	20 kg
Medical records room	2.7 m ²	4 m	6 kg

The FM200 room images are placed beside the FM200 calculation images because the room area and height are directly used to determine the gas quantity. The calculation uses room volume, design concentration and safety factor to select the required FM200 cylinder capacity.



4.14 Fire Search System for Electrical Panels

Fire Search was used as a local protection system for electrical panels and small electrical rooms. The system uses a cylinder connected to a pressurized detection tube installed around or inside the protected panel. When the temperature becomes high, the tube ruptures at the hottest point and releases the agent directly into the hazard area.

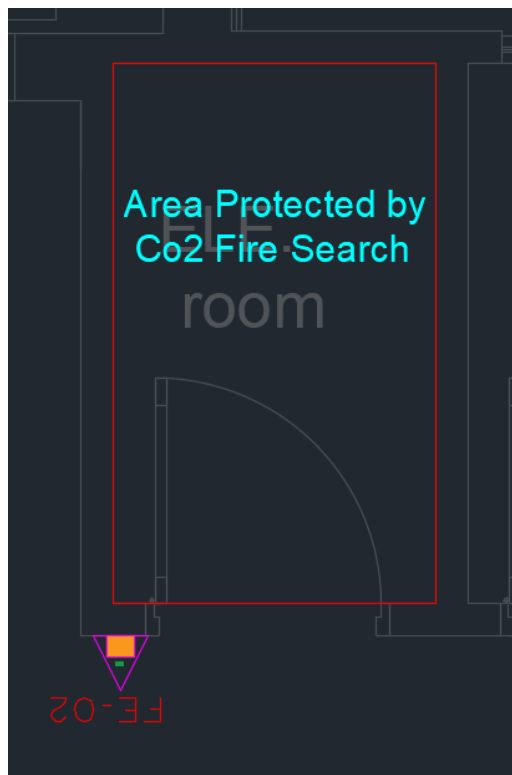


Figure 4.23: Area protected by CO2 Fire Search system.



Figure 4.24: Fire Search panel protection reference.

Fire Search item	Description
Cylinder	Contains CO2 or clean agent under pressure.
Detection tube	Flexible tube routed around the electrical panel.
Activation method	Tube ruptures at high temperature and releases the agent.
Typical application	Electrical panels, UPS panels and small electrical rooms.
Purpose	Fast local suppression without waiting for full room activation.

Fire Search is separated from FM200 in the report because FM200 protects a whole room volume, while Fire Search protects a localized electrical panel or small electrical enclosure. This makes it suitable for panel-level automatic suppression.

4.15 Mechanical Fire Fighting System Summary

The mechanical fire fighting system was organized as a complete protection strategy. General hospital areas use a wet sprinkler network, stores use upright sprinklers, operation rooms use a pre-action system, UPS and medical records rooms use FM200, electrical panels use Fire Search, and manual response is supported using fire hose cabinets and portable extinguishers.

The fire water tank, pump room, suction and discharge sizing, pipe network, ZCVs and hydraulic calculations ensure that the system can deliver the required water flow and pressure to the most remote area. This chapter represents the mechanical fire fighting work only and does not include the electrical fire alarm system.

Chapter 5: Final Integration

5.1 Integration Between Project Tracks

The main value of the project comes from combining building services design with smart hardware automation. The electrical distribution track provides reliable power, the fire fighting track provides building protection, and the hardware track adds monitoring, automatic response, and prototype-level smart hospital functions.

The three project tracks complete each other. The hardware system adds monitoring and automation, the electrical distribution system provides the power infrastructure, and the fire fighting system provides life safety protection. Together, they form an integrated hospital engineering solution.

Track	Integration Role
Hardware System	Smart monitoring, emergency detection, access response, patient data, and prototype automation.
Electrical Distribution	Reliable power supply, UPS/generator backup, low current systems, and practical shop drawings.
Fire Fighting	Fire protection, emergency firefighting systems, and special suppression for sensitive rooms.

5.2 Safety and Reliability Improvements

Safety is improved through gas and fire detection, automatic access door opening during emergencies, fire alarm systems, sprinkler and suppression systems, and reliable electrical supply. Reliability is improved through proper cable and breaker selection, backup power, local controller operation, and offline operation when Wi-Fi is unavailable.

5.3 Project Limitations

- The hardware system is implemented as a prototype and not a full-scale commercial hospital installation.
- Some sensor thresholds were calibrated experimentally according to the prototype environment.
- The ETAP work was used as a brief verification reference, while the main electrical calculations were manual.
- The fire fighting section is summarized to keep the report focused and within the required page limit.

5.4 Future Work

- Add a full dashboard for live hardware monitoring.
- Improve robot navigation using mapping and more advanced path planning.
- Add more rooms and multiple sensor nodes connected through a network.
- Integrate all alarms with a centralized hospital monitoring station.
- Use industrial-grade controllers and protected enclosures for real implementation.

Conclusion

This project presented an integrated engineering solution for a smart hospital environment. It combined embedded hardware systems, electrical distribution design, and fire fighting design into one complete project. The hardware prototype demonstrated monitoring, safety, access control, climate control, lighting control, and robot delivery. The electrical distribution design covered the main power infrastructure, equipment sizing, calculations, shop drawings, and light current systems. The fire fighting design covered sprinkler protection, special systems, pump room, tank, and suppression systems. The final result is a practical and well-documented engineering solution that supports safety, reliability, and smart hospital operation.

References / Bibliography

NFPA standards for fire protection system design and sprinkler system concepts.

Egyptian electrical code references for lighting and power requirements.

DIALux software documentation and lighting calculation outputs.

Electrical cable and circuit breaker manufacturer catalogues used for cable selection.

ESP32, Arduino Uno, MAX30102, DHT22, MQ5, MFRC522, TCS3200, and HC-05 module datasheets.

Project drawings, Excel sheets, screenshots, and prototype photos prepared by the project team.

Appendix A: Source Code Documentation

This appendix includes the complete final source codes used in the hardware prototype. The main chapters explain the function of each controller, while this appendix documents the full implementation for the ESP32 and the three Arduino Uno boards.

Status	Meaning
0	Normal
1	Gas Leak
2	Fire
3	Gas + Fire
Pulse Width	Status
100 ms	Normal
200 ms	Gas Leak
300 ms	Fire
400 ms	Gas + Fire
Command	Function
1	Go to red station
2	Go to white station
3	Go to station / black crossing line
S	Stop

Appendix A: Project Team Members

Student Name	ID	Department	Role
Mohamed Ahmed Salem Abul-ola	21-0-6145	EME	Hardware System & Team Leader
Omar Mohamed Arafat Hussein	21-0-6327	ECE	Hardware System
Ahmed Mohamed Ahmed Abdelsattar	22-1-11147	EME	Hardware System
Kareem Ibrahim Abdel moaneim Ibrahim	21-0-6267	EME	Electrical Distribution
Kareem Mohamed Farag Abu Al-Nasr	21-0-6353	EME	Electrical Distribution
Ahmed Khaled Mohamed Elkhoully	22-1-11115	EME	Electrical Distribution
Mazen Abd Elmonsef Khamis Ahmed	21-0-6170	ECE	Hardware System
Moamen Ayman Mabrouk Galhoum	21-0-7725	EME	Fire Fighting System

Appendix B:: ESP32 Main Controller Full Code

Full source code:

```
#include <Wire.h>
#include "MAX30105.h"
#include "spo2_algorithm.h"
#include <LiquidCrystal_I2C.h>
#include "DHT.h"
#include "BluetoothSerial.h"

#include <WiFi.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>

BluetoothSerial SerialBT;
MAX30105 particleSensor;

// ===== WiFi =====
const char* ssid = "test";
const char* password = "";

// API URL
const char* serverName = "https://lqum8loiu7.execute-api.us-east-1.amazonaws.com/prod/data";

// Patient Data
String patient_id = "P001";

// WiFi non-blocking variables
bool wifiWasConnected = false;
unsigned long lastWiFiCheckTime = 0;
unsigned long lastWiFiReconnectTime = 0;
unsigned long wifiMessageUntil = 0;

#define WIFI_CHECK_INTERVAL 3000
#define WIFI_RECONNECT_INTERVAL 15000

// Bluetooth status
bool bluetoothStarted = false;
```

```
// ===== I2C Pins =====
#define SDA_PIN 21
#define SCL_PIN 22

// Arduino 1 Sensors I2C Address
#define SENSOR_SLAVE_ADDRESS 0x08

// Signal to Arduino 2 Access
#define ACCESS_SIGNAL_PIN 17

// ===== Emergency Status =====
// 0 = Normal
// 1 = Gas Leak
// 2 = Fire
// 3 = Gas + Fire
byte emergencyStatus = 0;
byte lastEmergencyStatus = 255;

bool emergencyMode = false;
bool alertVisible = true;

unsigned long lastEmergencyReadTime = 0;
unsigned long lastAccessSendTime = 0;
unsigned long lastBlinkTime = 0;

#define EMERGENCY_READ_INTERVAL 250
#define ACCESS_SEND_INTERVAL 500
#define BLINK_INTERVAL 500
```

```

// ===== MAX30102 =====
#define BUFFER_SIZE 100

uint32_t irBuffer[BUFFER_SIZE];
uint32_t redBuffer[BUFFER_SIZE];

int32_t spo2;
int8_t validSP02;

int32_t heartRate;
int8_t validHeartRate;

int fakeBPM = 0;
int finalBPM = 0;
int finalSpO2 = 0;
bool maxReadingValid = false;

// ===== PCA9548A =====
#define PCA_ADDR 0x70
#define LCD_ADDR 0x27

LiquidCrystal_I2C lcdMAX(LCD_ADDR, 16, 2); // PCA Channel 0
LiquidCrystal_I2C lcdDHT(LCD_ADDR, 16, 2); // PCA Channel 1

// ===== DHT22 =====
#define DHTPIN 13
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

```

```

// ===== Relays =====
#define FAN_RELAY_PIN 26
#define MIST_RELAY_PIN 27
#define MIST_FAN_RELAY_PIN 16

#define TEMP_LIMIT 28.0

#define RELAY_ON LOW
#define RELAY_OFF HIGH

bool mistState = false;

// ===== LDR + LEDs =====
#define LDR_PIN 35
#define LDR_LED_1 18
#define LDR_LED_2 19

// ESP32 ADC range: 0 to 4095
// عند LDR القيمة دي حسب قراءة الـ LDR عندك
int LDR_STRONG_LIGHT = 3000; //OFF الليدات → ضوء قوي
int LDR_WEAK_LIGHT = 1200; //إضاءة الليدات لمر إضاءة ضوء ضعيف

// ===== Bluetooth =====
unsigned long lastToggleTime = 0;
const unsigned long toggleDelay = 150;

```

```
// ===== DHT Stored Values =====
float lastTemperature = 37.0;
float lastHumidity = 0;
bool lastFanState = false;
bool dhtDataValid = false;

unsigned long lastDHTTime = 0;
#define DHT_INTERVAL 1000

// ===== Cloud Timing =====
unsigned long lastCloudSendTime = 0;
#define CLOUD_SEND_INTERVAL 30000 //كل 30 ثانية

// ===== PCA =====
void selectPCAChannel(uint8_t channel) {
    if (channel > 7) return;

    Wire.beginTransmission(PCA_ADDR);
    Wire.write(1 << channel);
    Wire.endTransmission();
}
```

```
// ===== LDR Control =====
void controlLDRLeds() {
    int ldrValue = analogRead(LDR_PIN);
    int ledBrightness = 0;

    /*
    OFF  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة
    إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة
    إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة  إضاءة
    */

    if (ldrValue >= LDR_STRONG_LIGHT) {
        ledBrightness = 0;
    }
    else if (ldrValue >= LDR_WEAK_LIGHT && ldrValue < LDR_STRONG_LIGHT) {
        ledBrightness = 128;
    }
    else {
        ledBrightness = 255;
    }

    analogWrite(LDR_LED_1, ledBrightness);
    analogWrite(LDR_LED_2, ledBrightness);
}

// ===== Bluetooth Control =====
void startBluetooth() {
    if (!bluetoothStarted) {
        SerialBT.begin("ESP32_Mist_Maker");
        bluetoothStarted = true;

        Serial.println("Bluetooth Started");
        SerialBT.println("ESP32 Mist Maker Ready");
        SerialBT.println("Send R once to toggle ON/OFF");
    }
}
```



```

void stopBluetooth() {
  if (bluetoothStarted) {
    Serial.println("Stopping Bluetooth once because WiFi Cloud is active...");
    SerialBT.end();
    bluetoothStarted = false;
    delay(10);
    Serial.println("Bluetooth Stopped");
  }
}

// ===== WiFi Non Blocking =====
void startWiFi() {
  WiFi.mode(WIFI_STA);
  WiFi.setSleep(false);
  WiFi.begin(ssid, password);

  Serial.print("Trying WiFi: ");
  Serial.println(ssid);
}

void showWiFiConnectedOnMAX() {
  if (emergencyMode) return;

  selectPCAChannel(0);
  lcdMAX.clear();
  lcdMAX.setCursor(0, 0);
  lcdMAX.print("WiFi Connected");
  lcdMAX.setCursor(0, 1);
  lcdMAX.print(WiFi.localIP());

  wifiMessageUntil = millis() + 2500;
}

```

```

void checkWiFi() {
  if (millis() - lastWiFiCheckTime < WIFI_CHECK_INTERVAL) {
    return;
  }

  lastWiFiCheckTime = millis();

  if (WiFi.status() == WL_CONNECTED) {
    if (!wifiWasConnected) {
      wifiWasConnected = true;

      Serial.println();
      Serial.println("WiFi Connected!");
      Serial.print("IP: ");
      Serial.println(WiFi.localIP());
      Serial.print("RSSI: ");
      Serial.println(WiFi.RSSI());

      showWiFiConnectedOnMAX();

      stopBluetooth();
    }
  }
  else {
    if (wifiWasConnected) {
      Serial.println("WiFi Disconnected!");
    }

    wifiWasConnected = false;

    startBluetooth();

    if (millis() - lastWiFiReconnectTime >= WIFI_RECONNECT_INTERVAL) {
      lastWiFiReconnectTime = millis();
    }
  }
}

```

```

        Serial.println("WiFi not connected, system still running...");
        Serial.println("Trying reconnect...");

        WiFi.disconnect(false);
        WiFi.begin(ssid, password);
    }
}

// ===== Build JSON =====
String buildJson() {
    float temperatureToSend;

    if (dhtDataValid) {
        temperatureToSend = lastTemperature;
    } else {
        temperatureToSend = 37.0;
    }

    String jsonData = "{";
    jsonData += "\"patient_id\": \"" + patient_id + "\", ";
    jsonData += "\"temperature\": " + String(temperatureToSend, 1) + ", ";
    jsonData += "\"heart_rate\": " + String(finalBPM) + ", ";
    jsonData += "\"oxygen\": " + String(finalSpO2);
    jsonData += "}";

    return jsonData;
}

```

```

// ===== Send To Cloud =====
void sendToCloud(String jsonData) {
    if (WiFi.status() != WL_CONNECTED) {
        Serial.println("WiFi not connected, data not sent.");
        return;
    }

    Serial.println("Starting HTTPS request...");

    WiFiClientSecure client;
    client.setInsecure();

    HTTPClient http;
    http.setTimeout(12000);
    http.setReuse(false);

    if (!http.begin(client, serverName)) {
        Serial.println("HTTP begin failed!");
        http.end();
        return;
    }

    http.addHeader("Content-Type", "application/json");

    int httpResponseCode = http.POST(jsonData);

    Serial.print("HTTP Response code: ");
    Serial.println(httpResponseCode);

    if (httpResponseCode > 0) {
        String response = http.getString();
        Serial.println("Response from server:");
        Serial.println(response);
    }
}

```

```

} else {
    Serial.print("HTTP Error: ");
    Serial.println(http.errorToString(httpResponseCode));
}

http.end();

Serial.println("HTTPS request finished. System still running.");
}

```

```

// ===== Emergency Messages =====
void printEmergencyOnLCDs() {
    const char* line1 = "";

    if (emergencyStatus == 1) {
        line1 = "GAS LEAK!";
    }
    else if (emergencyStatus == 2) {
        line1 = "FIRE ALERT!";
    }
    else if (emergencyStatus == 3) {
        line1 = "GAS + FIRE!";
    }

    selectPCAChannel(0);
    lcdMAX.clear();
    lcdMAX.setCursor(0, 0);
    lcdMAX.print(line1);
    lcdMAX.setCursor(0, 1);
    lcdMAX.print("Room Number 1");

    selectPCAChannel(1);
    lcdDHT.clear();
    lcdDHT.setCursor(0, 0);
    lcdDHT.print(line1);
    lcdDHT.setCursor(0, 1);
    lcdDHT.print("Room Number 1");
}

void clearESP32LCDs() {
    selectPCAChannel(0);
    lcdMAX.clear();

    selectPCAChannel(1);
    lcdDHT.clear();
}

```

```

void handleEmergencyBlink() {
    if (!emergencyMode) return;

    if (millis() - lastBlinkTime >= BLINK_INTERVAL) {
        lastBlinkTime = millis();
        alertVisible = !alertVisible;

        if (alertVisible) {
            printEmergencyOnLCDs();
        } else {
            clearESP32LCDs();
        }
    }
}

```

```

// ===== I2C Receive From Arduino 1 =====
byte readArduino1Status() {
    byte status = 0;

    Wire.requestFrom(SENSOR_SLAVE_ADDRESS, 1);

    if (Wire.available()) {
        status = Wire.read();
    }

    return status;
}

void checkArduino1Emergency() {
    if (millis() - lastEmergencyReadTime < EMERGENCY_READ_INTERVAL) {
        return;
    }

    lastEmergencyReadTime = millis();

    byte newStatus = readArduino1Status();

    if (newStatus != emergencyStatus) {
        emergencyStatus = newStatus;

        Serial.print("Received From Arduino 1: ");
        Serial.println(emergencyStatus);

        if (emergencyStatus == 0) {
            emergencyMode = false;
            alertVisible = true;

            updateDHTLCD();
        }
    }
}

```

```

    selectPCChannel(0);
    lcdMAX.clear();
    lcdMAX.setCursor(0, 0);
    lcdMAX.print("Emergency Clear");
    lcdMAX.setCursor(0, 1);
    lcdMAX.print("System Normal");
    delay(20);
}
else {
    emergencyMode = true;
    alertVisible = true;
    printEmergencyOnLCDs();
}

```

```

// ===== Send Status To Access Arduino =====
// 0 = 100ms HIGH
// 1 = 200ms HIGH
// 2 = 300ms HIGH
// 3 = 400ms HIGH
void sendStatusToAccess() {
    if (millis() - lastAccessSendTime < ACCESS_SEND_INTERVAL) {
        return;
    }

    lastAccessSendTime = millis();

    int pulseTime = 100;

    if (emergencyStatus == 0) pulseTime = 100;
    else if (emergencyStatus == 1) pulseTime = 200;
    else if (emergencyStatus == 2) pulseTime = 300;
    else if (emergencyStatus == 3) pulseTime = 400;

    digitalWrite(ACCESS_SIGNAL_PIN, HIGH);
    delay(pulseTime);
    digitalWrite(ACCESS_SIGNAL_PIN, LOW);

    if (emergencyStatus != lastEmergencyStatus) {
        lastEmergencyStatus = emergencyStatus;

        Serial.print("Sent To Access: ");
        Serial.println(emergencyStatus);
    }
}

```

```

// ===== DHT LCD =====
void updateDHTLCD() {
    if (emergencyMode) return;

    selectPCChannel(1);
    lcdDHT.clear();

    if (!dhtDataValid) {
        lcdDHT.setCursor(0, 0);
        lcdDHT.print("DHT22 Reading");
        lcdDHT.setCursor(0, 1);
        lcdDHT.print("Waiting Data");
        return;
    }

    lcdDHT.setCursor(0, 0);
    lcdDHT.print("T:");
    lcdDHT.print(lastTemperature, 1);
    lcdDHT.print((char)223);
    lcdDHT.print("C ");

    if (lastFanState) {
        lcdDHT.print("F:ON");
    } else {
        lcdDHT.print("F:OFF");
    }

    lcdDHT.setCursor(0, 1);
    lcdDHT.print("H:");
    lcdDHT.print(lastHumidity, 0);
    lcdDHT.print("%");
}

```